POLI

ESCOLA SUPERIOR
TECNOLOGIA
GESTÃO

TÉCNICO GUARDA

KMEANS ALGORITHM

8 DATASET'S ANALYSIS

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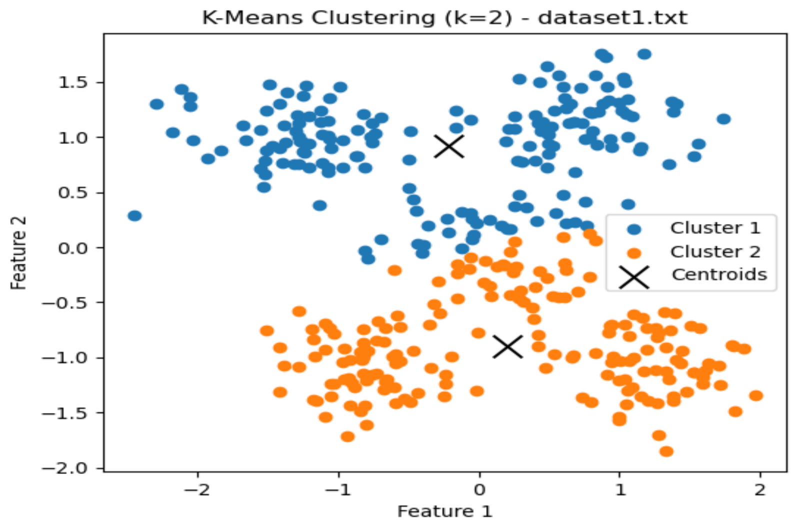
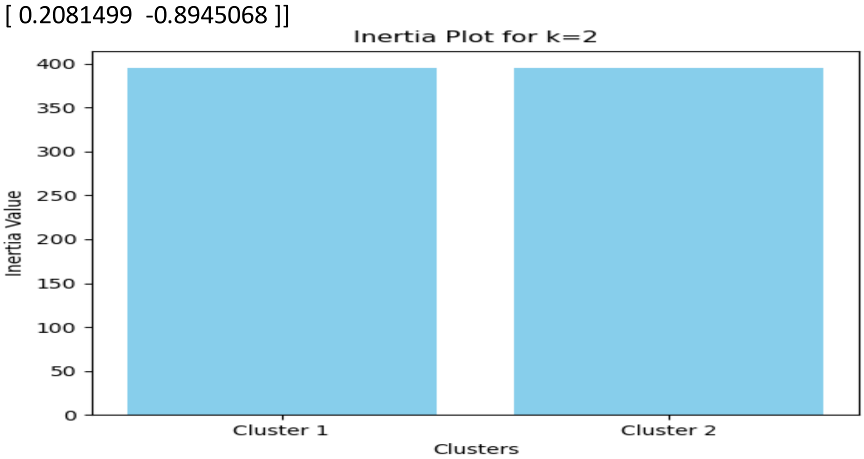
Curso	Machine Learning 2
Unidade curricular	Clique para introduzir a designação da unidade curricular
Ano letivo	2024/2025
Docente(s)	Clique para introduzir o(s) nome(s) do(s) docente(s) da unidade curricular
Data	31/03/2025

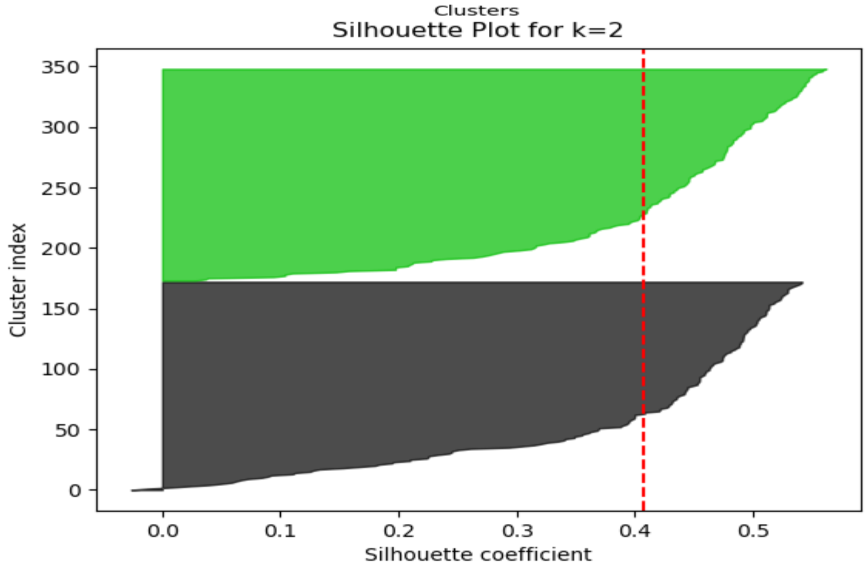
In the following sections, I present the clustering evaluation results for each dataset. The results are illustrated using graphical representations, where I analyze different values of kkk (number of clusters) ranging from **k=2 to k=10**. Specifically, I include the **Silhouette Score** and **Elbow Method** plots to determine the optimal number of clusters for each dataset. These methods help evaluate the quality of clustering by assessing both cohesion and separation.

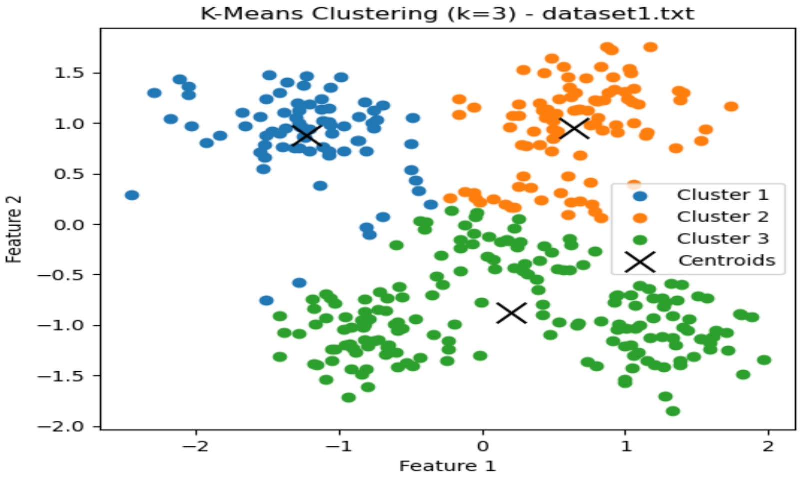
DATASET 1



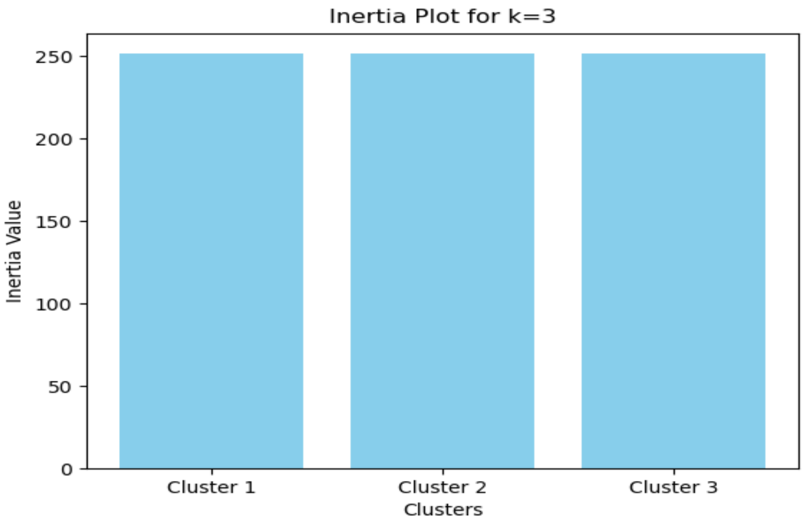
Centroids for k=2 in dataset1.txt:
[[-0.21420077 0.9205099]
 [0.2081499 -0.8945068]]

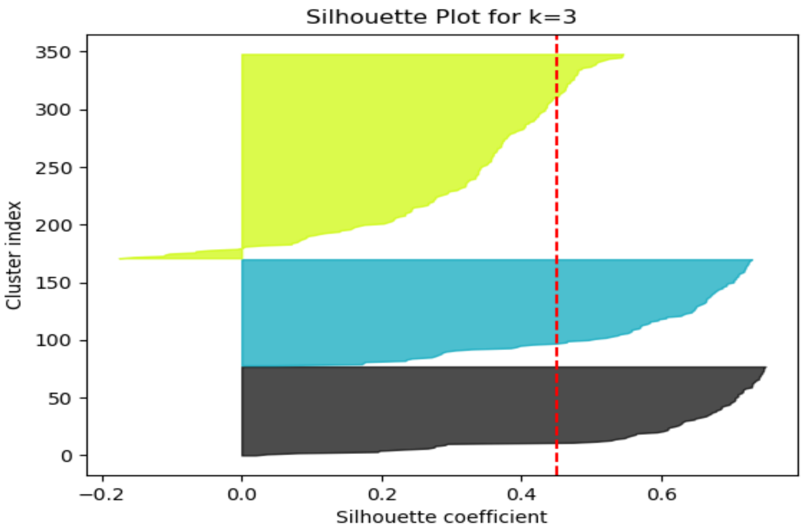


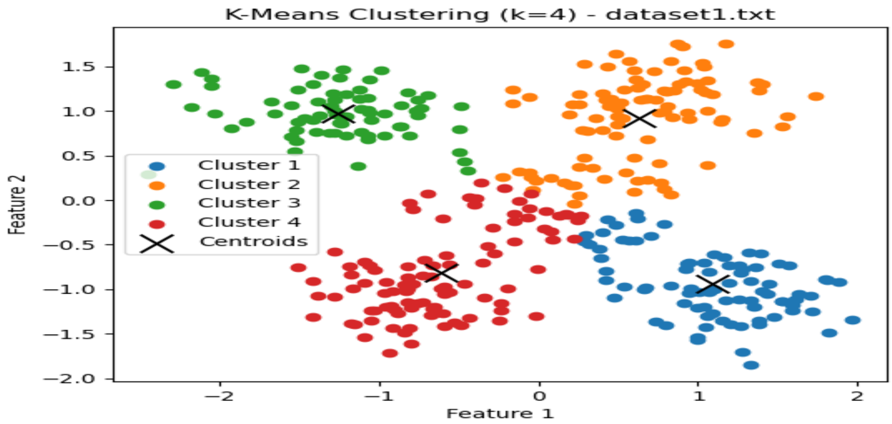




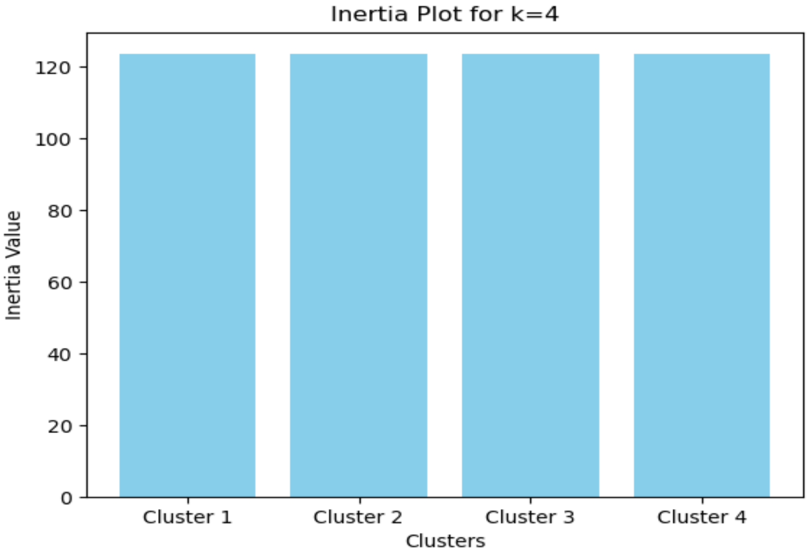
Centroids for k=3 in dataset1.txt:
[[-1.22967889 0.88219878]
 [0.64550574 0.94768581]
 [0.20158944 -0.8817207]]

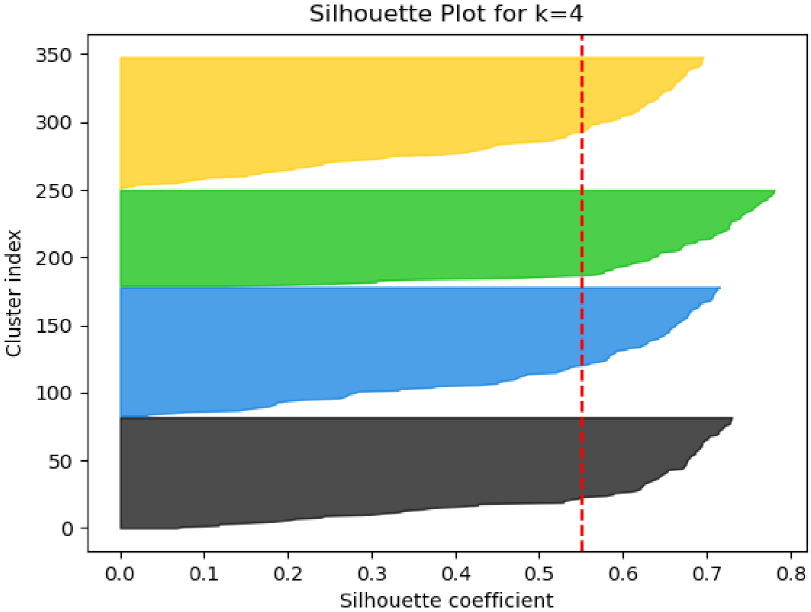


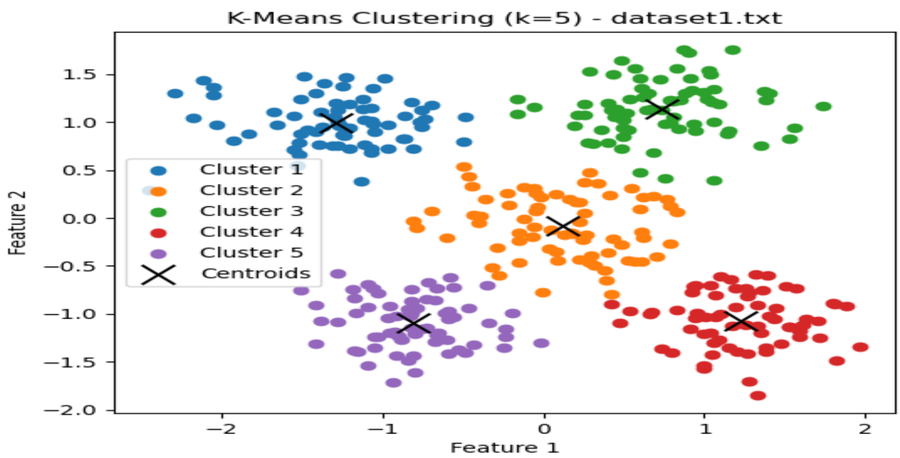




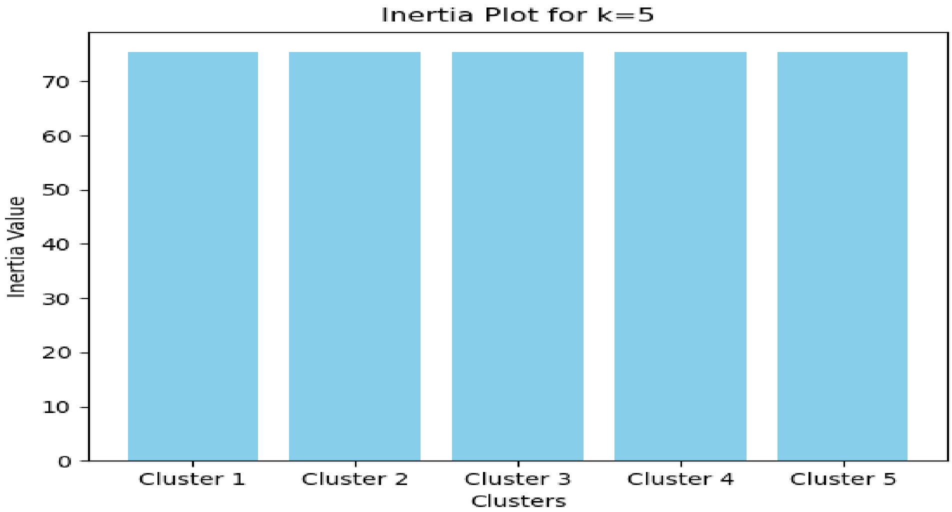
Centroids for k=4 in dataset1.txt:
[[1.08416582 -0.94434389]
 [0.62990946 0.91936424]
 [-1.25692842 0.97229786]
 [-0.61181863 -0.81514153]]

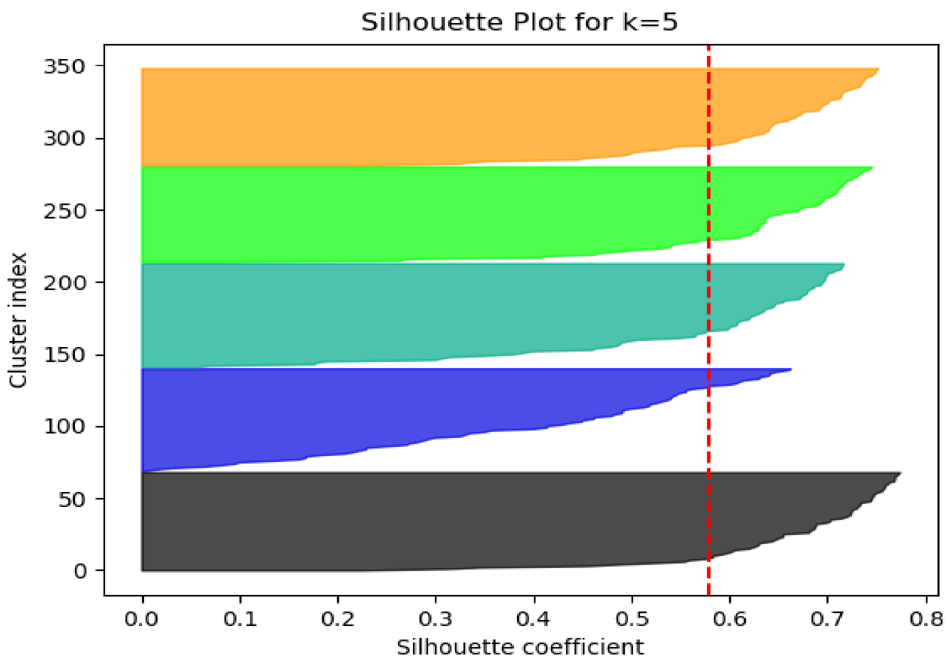




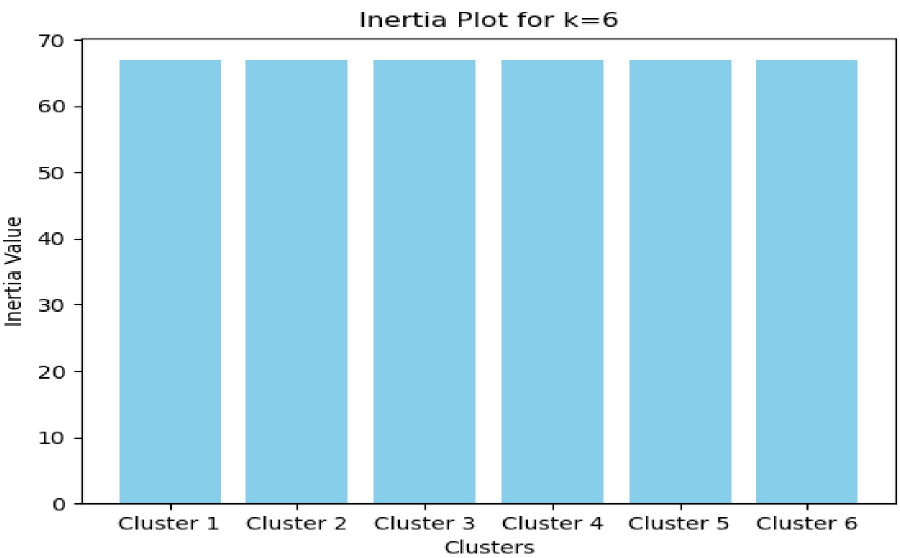


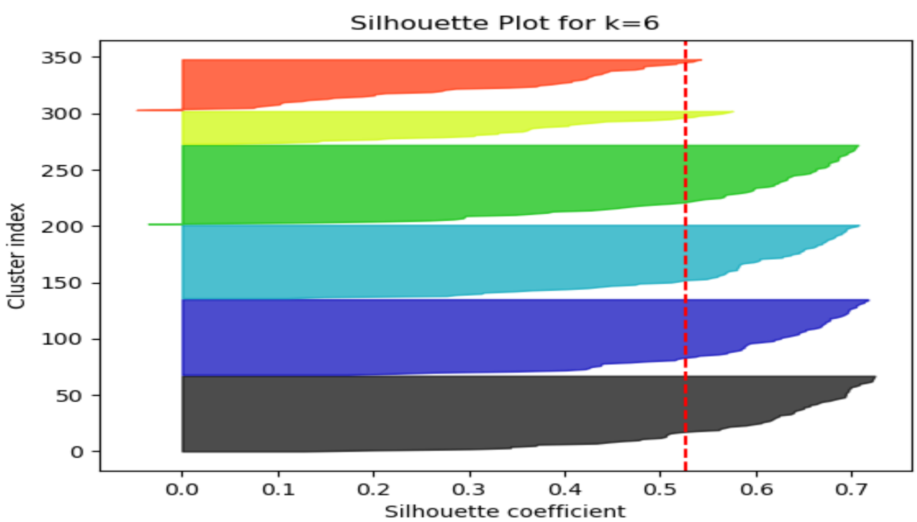
Centroids for k=5 in dataset1.txt:
[[-1.29114145 0.995575]
 [0.11524954 -0.08419612]
 [0.73580633 1.14091514]
 [1.22651068 -1.07170183]
 [-0.81028359 -1.08993143]]

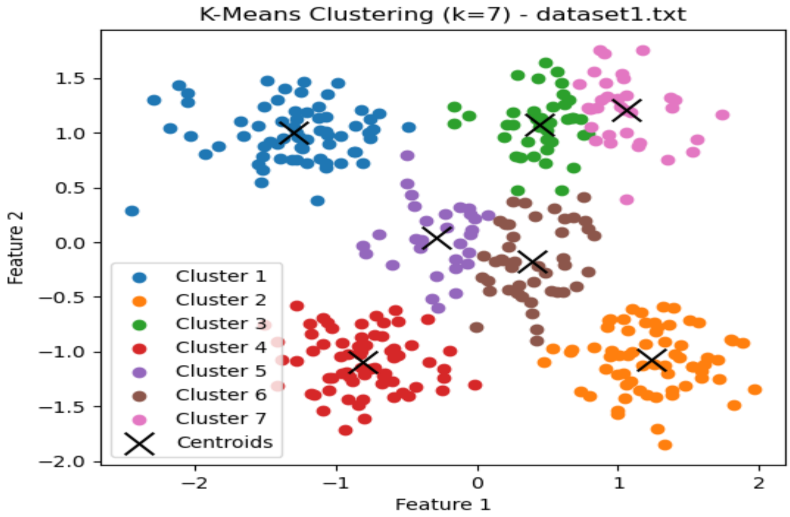




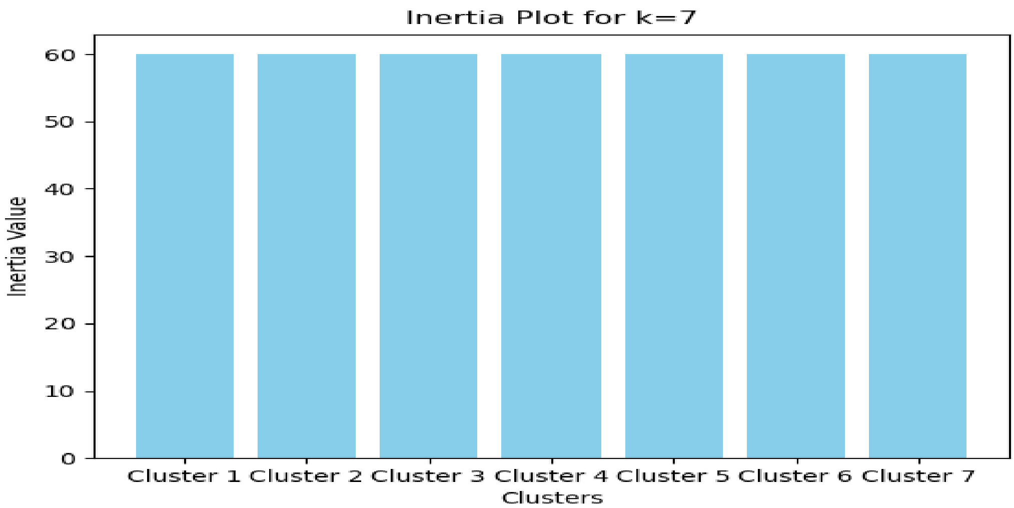
Centroids for k=6 in dataset1.txt:
[[-0.81028359 -1.08993143]
 [-1.30290568 0.99852545]
 [1.2386368 -1.07425642]
 [0.73738044 1.16050577]
 [-0.28834577 0.046851]
 [0.39659138 -0.1453241]]

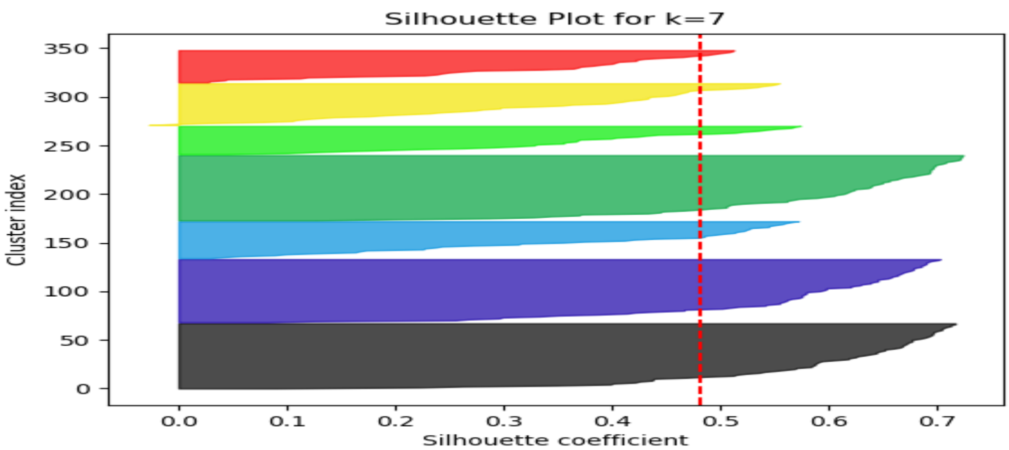


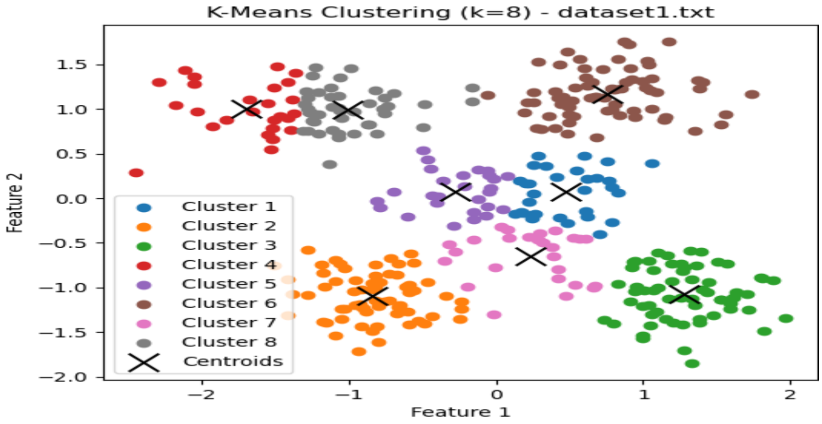




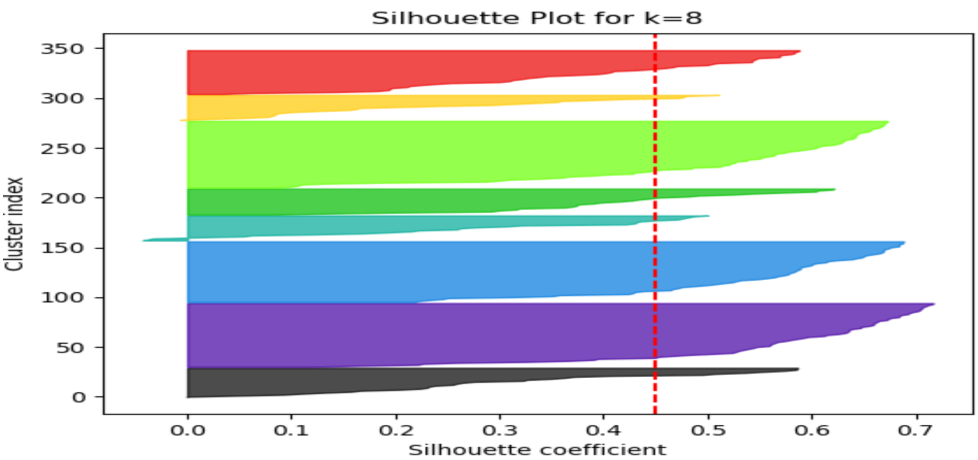
Centroids for k=7 in dataset1.txt:
[[-1.30290568 0.99852545]
 [1.2386368 -1.07425642]
 [0.44245234 1.0804   ]
 [-0.81028359 -1.08993143]
 [-0.28834577 0.046851]
 [0.39435893 -0.17357226]
 [1.05852297 1.21213472]]

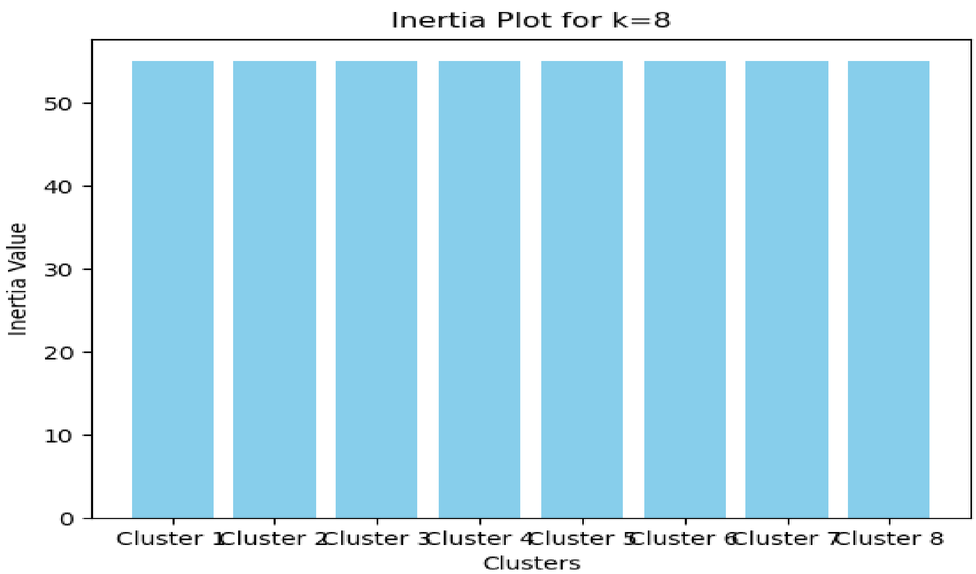


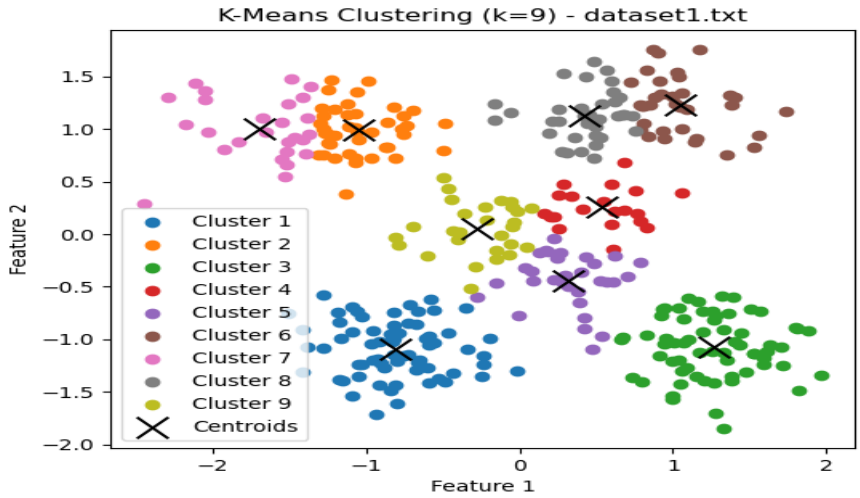




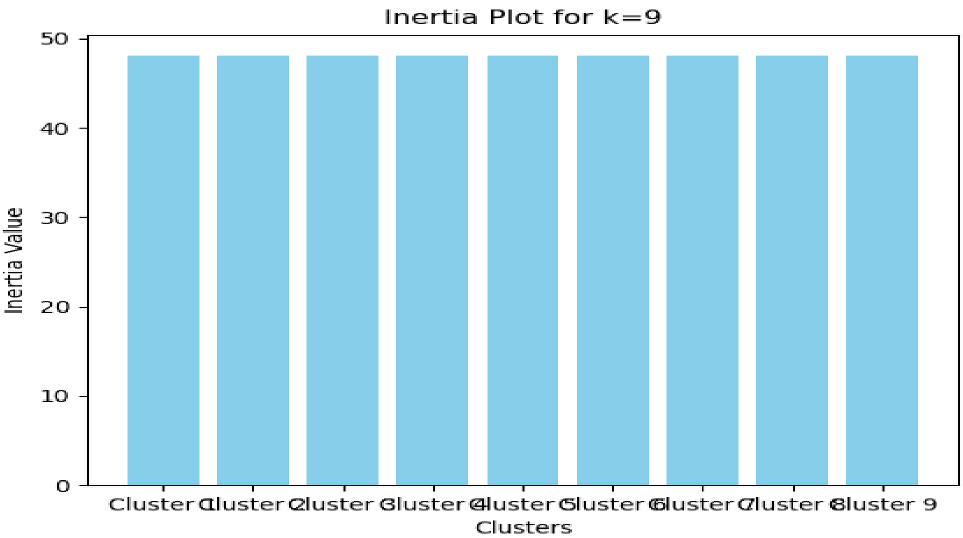
Centroids for k=8 in dataset1.txt:
[[0.4741696   0.07094639]
 [-0.83904992 -1.09409562]
 [1.28090024 -1.07825943]
 [-1.69886907 1.00687233]
 [-0.27303835 0.07637669]
 [0.75907476 1.17171944]
 [0.23306841 -0.65077884]
 [-1.00526419 0.99649744]]

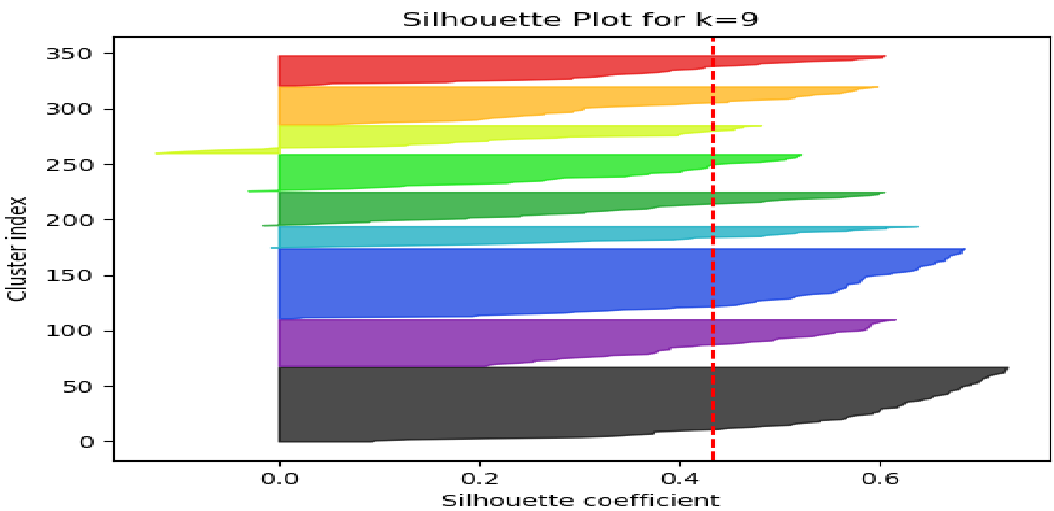


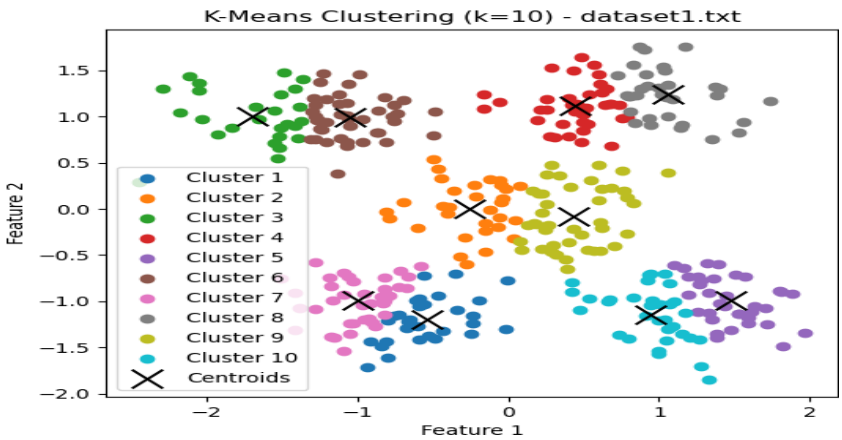




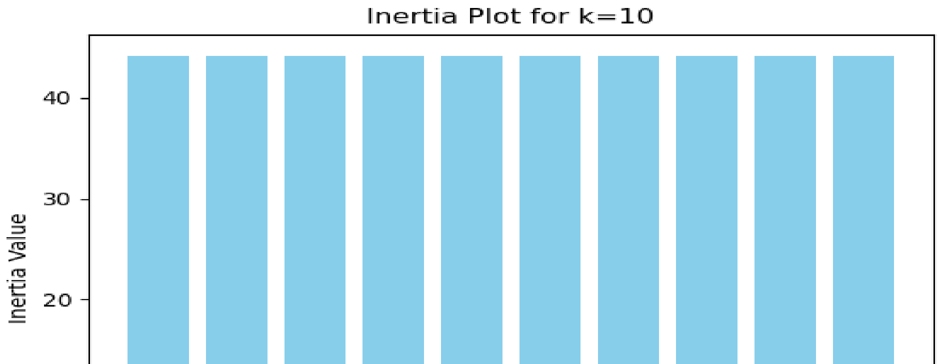
Centroids for k=9 in dataset1.txt:
[[-0.81028359 -1.08993143]
 [-1.04460847 0.98874406]
 [1.26166231 -1.07555606]
 [0.53329382 0.25501474]
 [0.31757713 -0.44234562]
 [1.05048602 1.22978506]
 [-1.69886907 1.00687233]
 [0.42576883 1.12890303]
 [-0.27455515 0.05515611]]

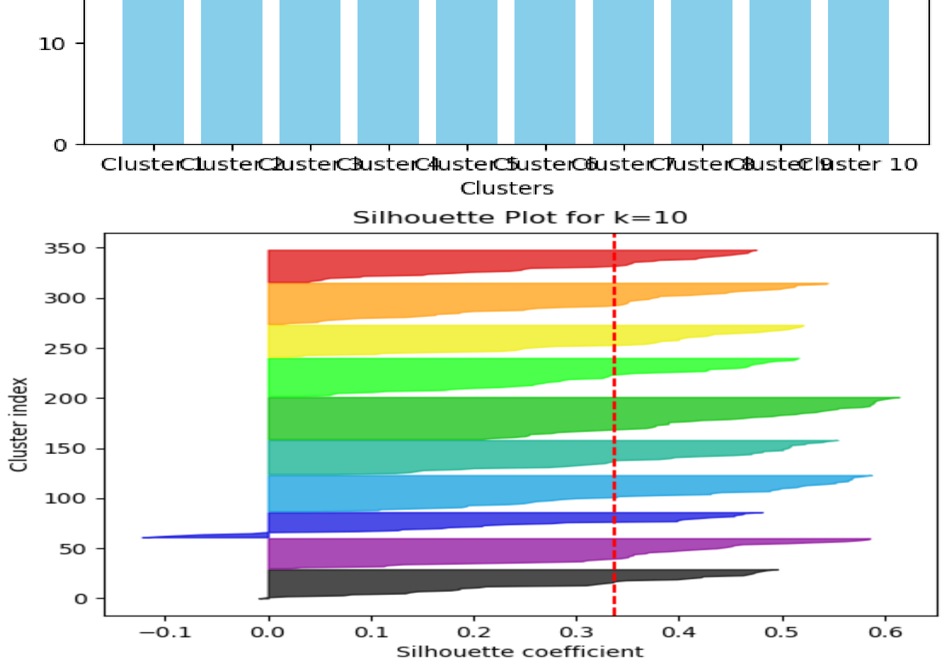


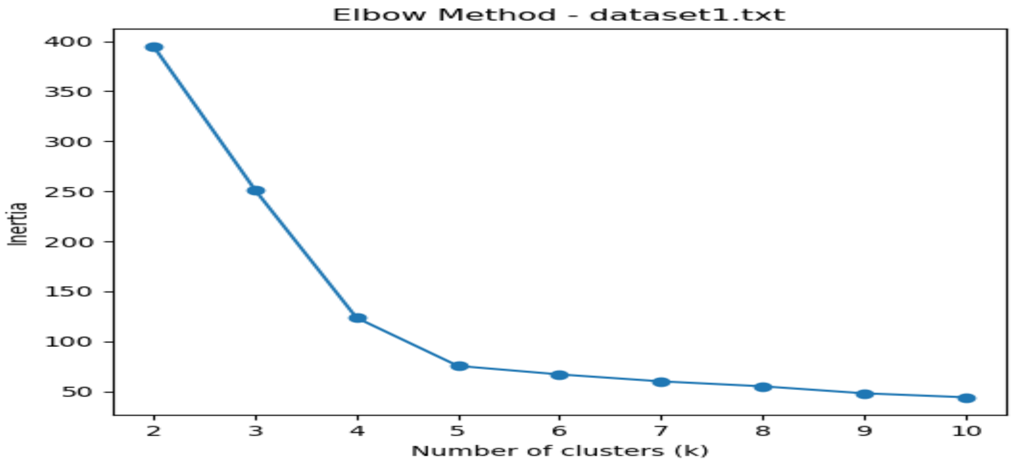




Centroids for k=10 in dataset1.txt:
[[-0.53915607 -1.20088123]
 [-0.26041674 0.00531834]
 [-1.69886907 1.00687233]
 [0.4422765   1.11306295]
 [1.47264741 -0.99379723]
 [-1.04460847 0.98874406]
 [-0.9982743 -0.99647255]
 [1.05860995 1.23696919]
 [0.43745659 -0.08029929]
 [0.94105648 -1.14601564]]







Based on the previous diagrams as we can see in the Elbow Method the optimal value is at k=5 and we confirm that at the Silhouette Score diagram. The optimal K value is first identified using the elbow graph, where we observe a distinct elbow point. At this point, the decrease in error or intra-cluster distance slows down significantly, indicating that adding more clusters no longer provides substantial improvement. We then confirm this choice using the silhouette score graph, which evaluates cluster quality, ensuring that the selected value offers the best balance between cohesion and separation.

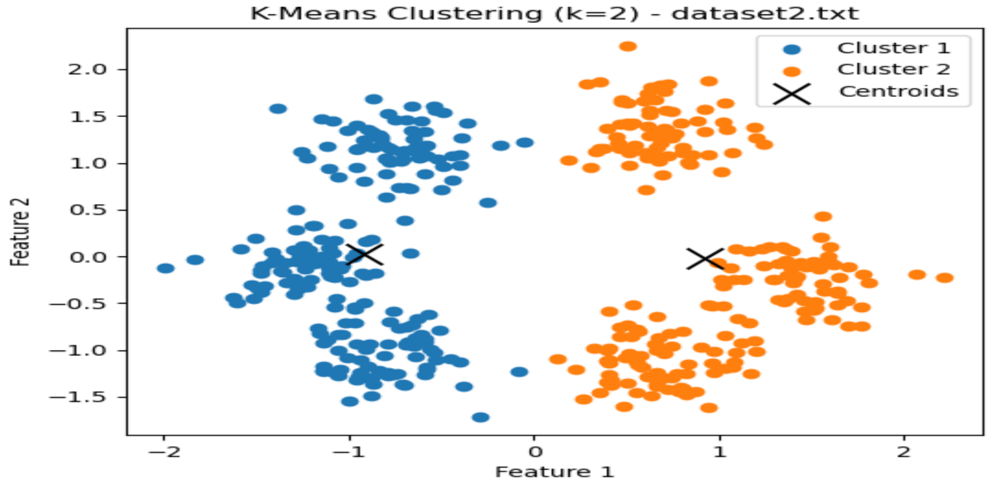
Optimal k for dataset1.txt based on silhouette score: 5

Best Silhouette Score: 0.5782869045664402

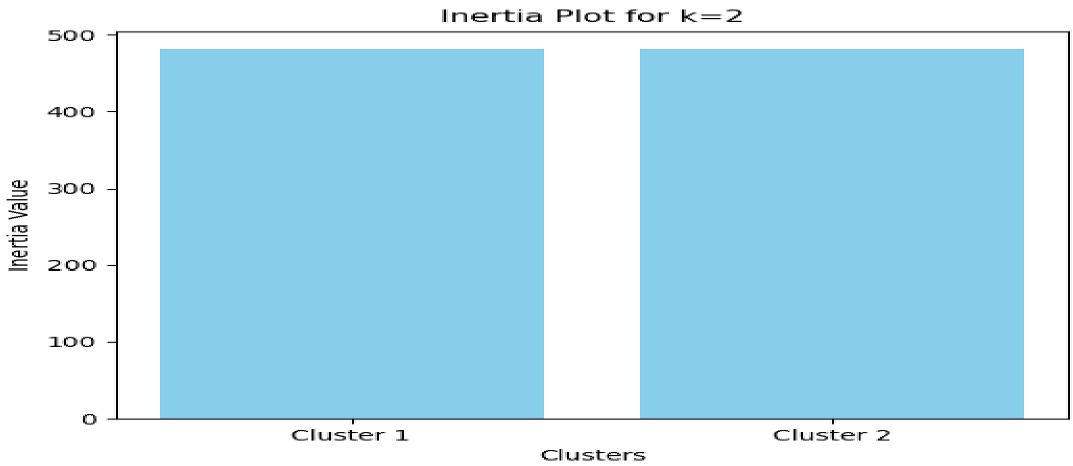
Inertia values: [395.0556146715863, 251.4584697336636, 123.57180143213473, 75.40099673630887, 66.92256662867665, 59.97714746356724, 55.03476529549739, 48.03745261023843, 44.06739321945993]

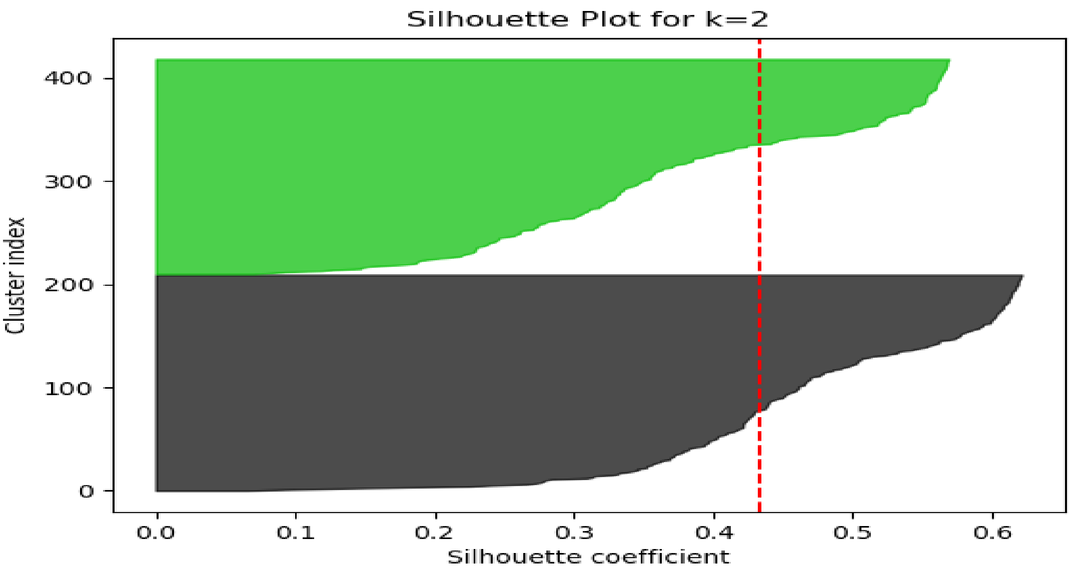
Silhouette scores: [0.40700554191831184, 0.45126793472690385, 0.5511880981182673, 0.5782869045664402, 0.526149660238036, 0.48254906584195656, 0.45051036226894575, 0.4336486918905106, 0.3377795808826339]

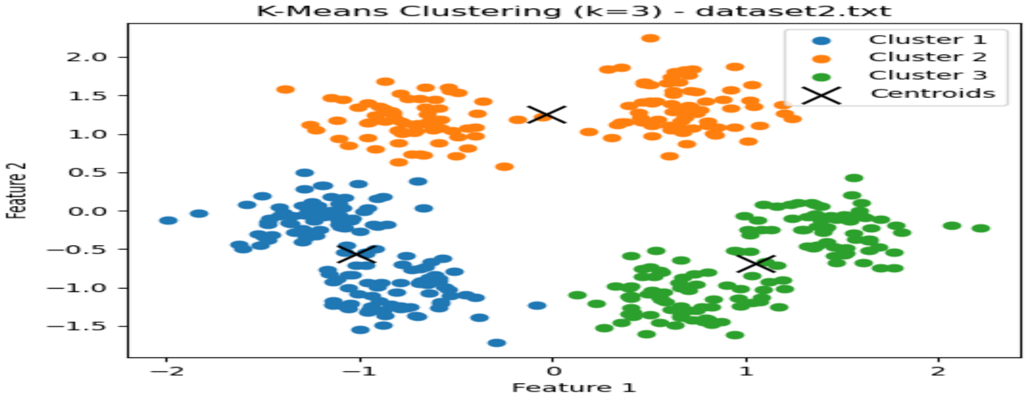
DATASET 2



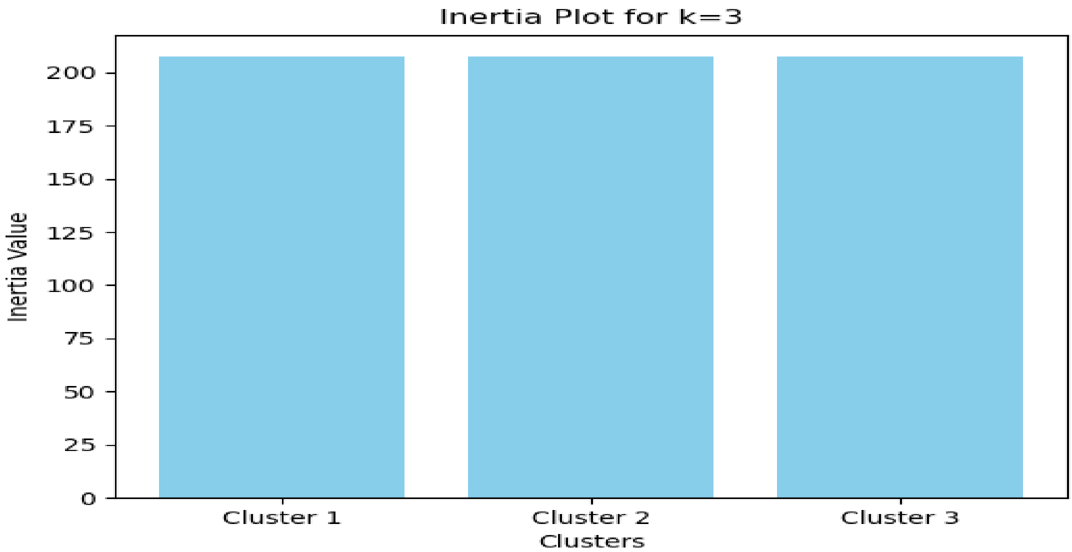
Centroids for k=2 in dataset2.txt:
[[-0.92106796 0.01863951]
 [0.92547498 -0.0187287]]

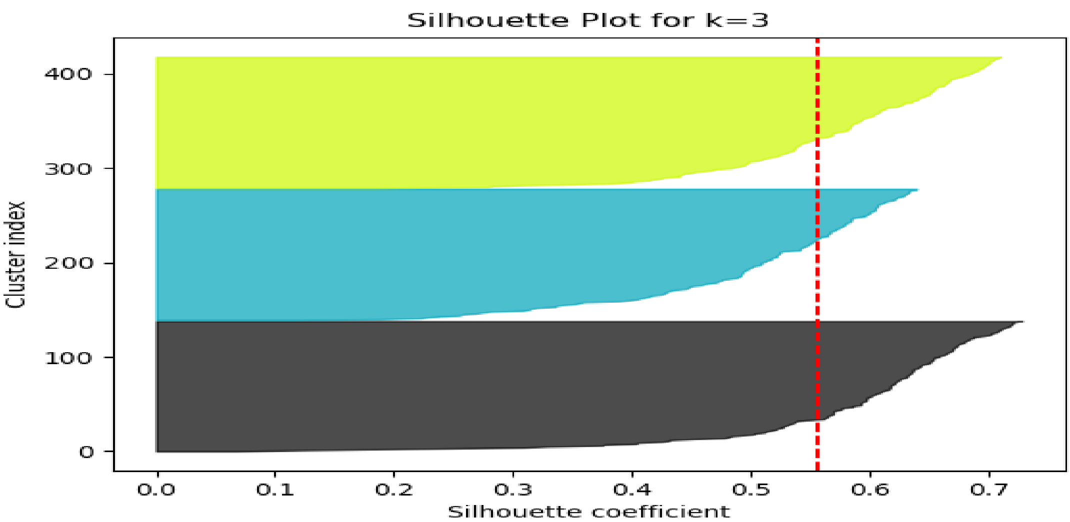


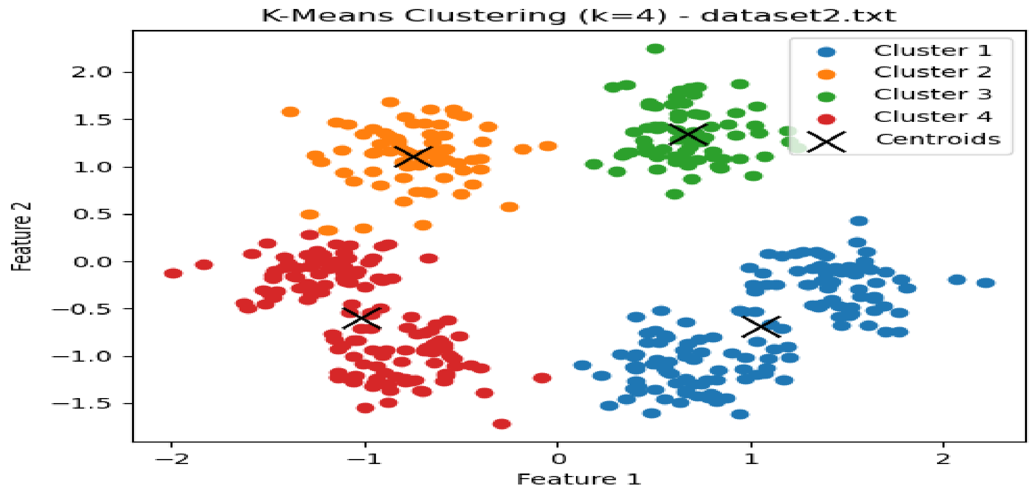




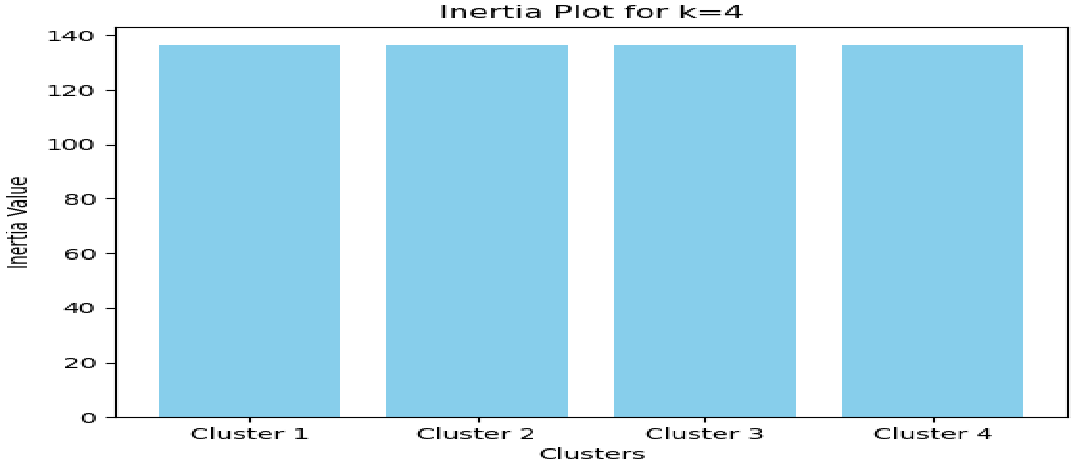
Centroids for k=3 in dataset2.txt:
[[-1.01915069 -0.5660919]
 [-0.03800076 1.25398882]
 [1.0498718 -0.69194043]]

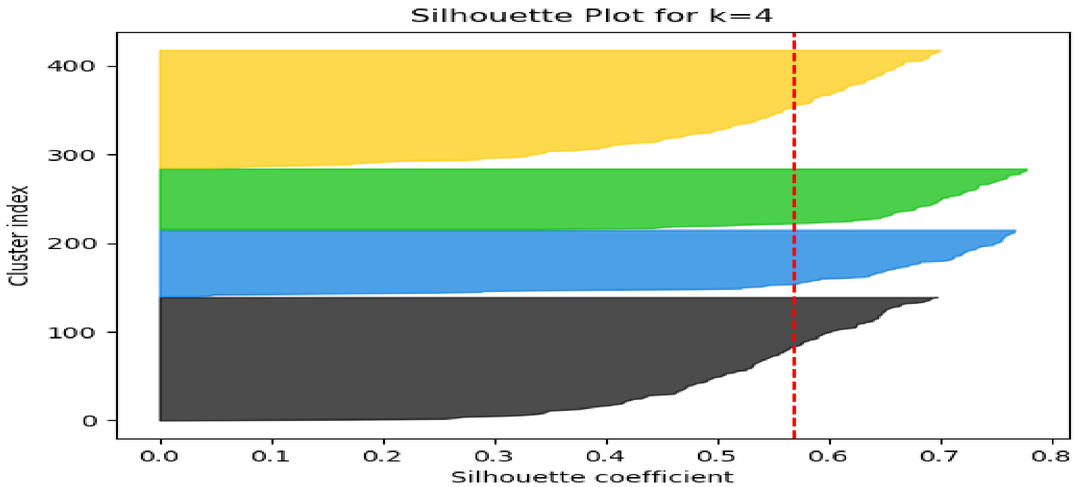


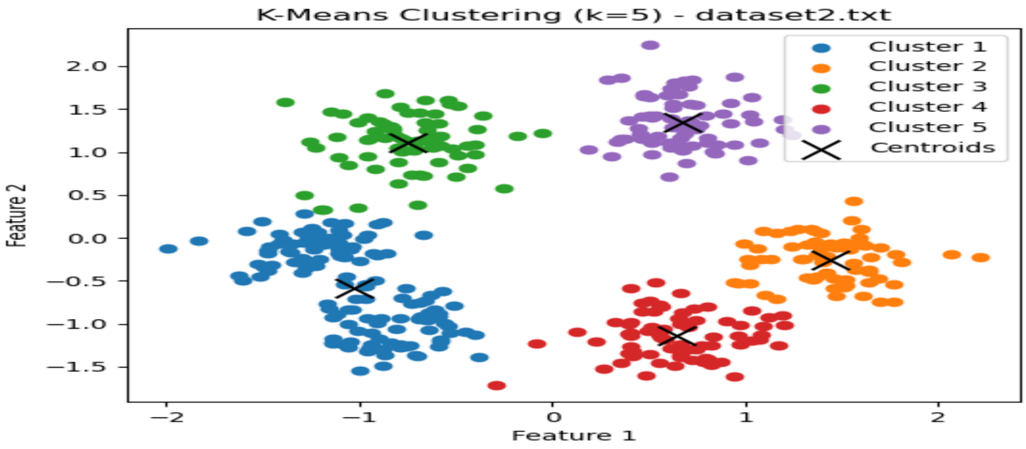




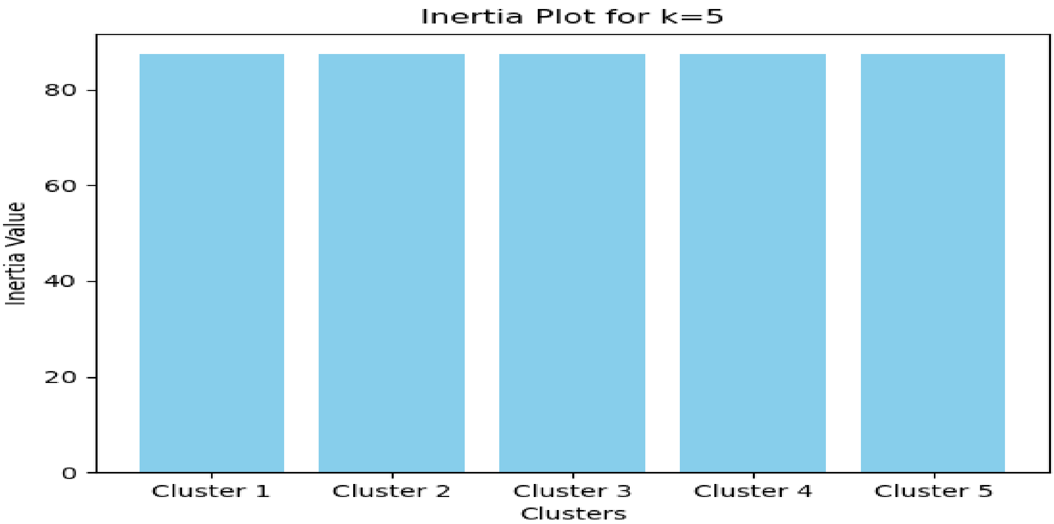
Centroids for k=4 in dataset2.txt:
[[1.0498718 -0.69194043]
[-0.75208259 1.11172171]
[0.67307564 1.34720815]
[-1.0169104 -0.60131756]]

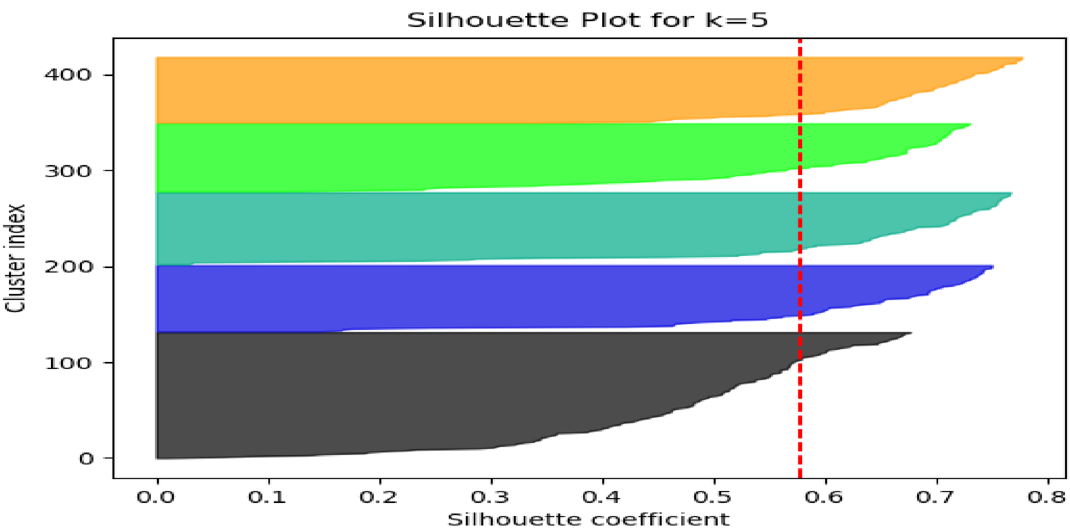


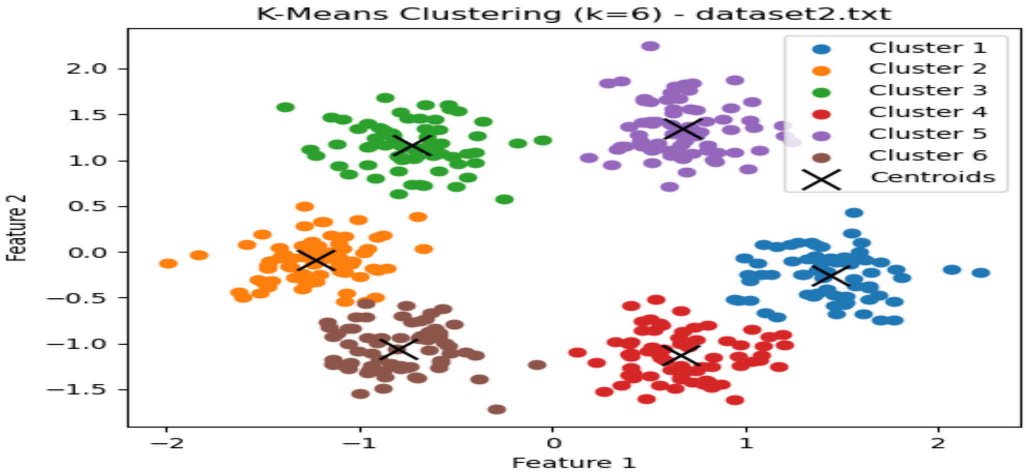




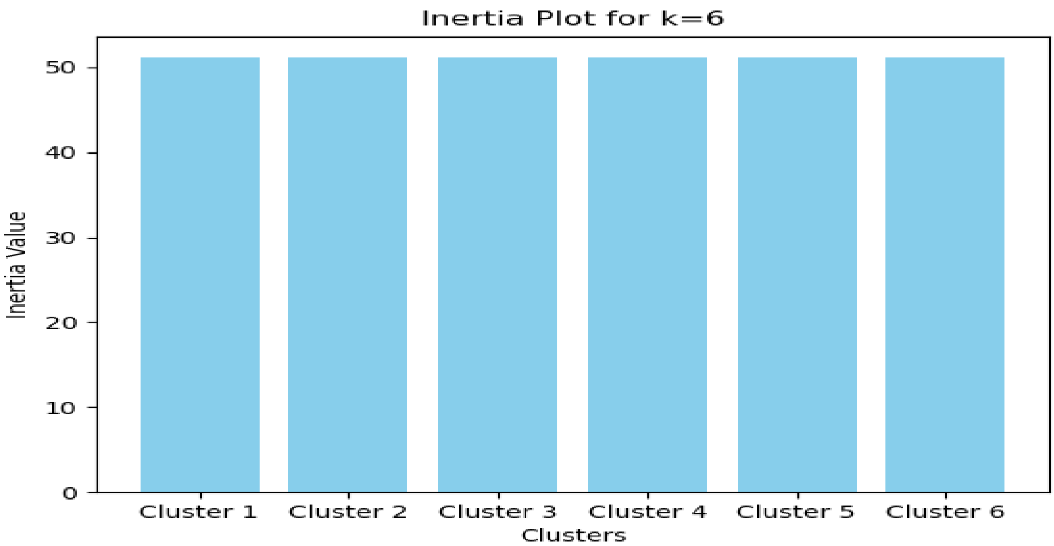
Centroids for k=5 in dataset2.txt:
[[-1.02947404 -0.58814083]
[1.43552563 -0.25442151] [-0.75208259 1.11172171]
[0.64055329 -1.13894608]
[0.67307564 1.34720815]]

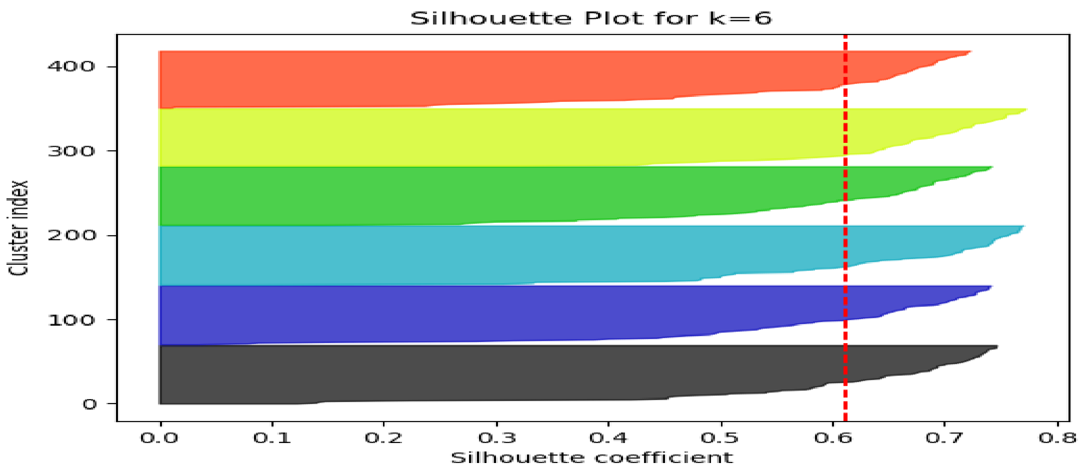


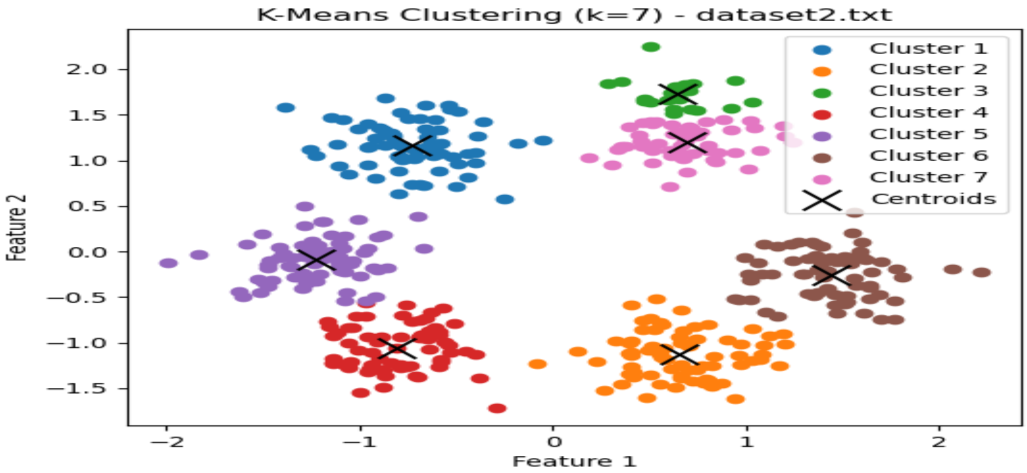




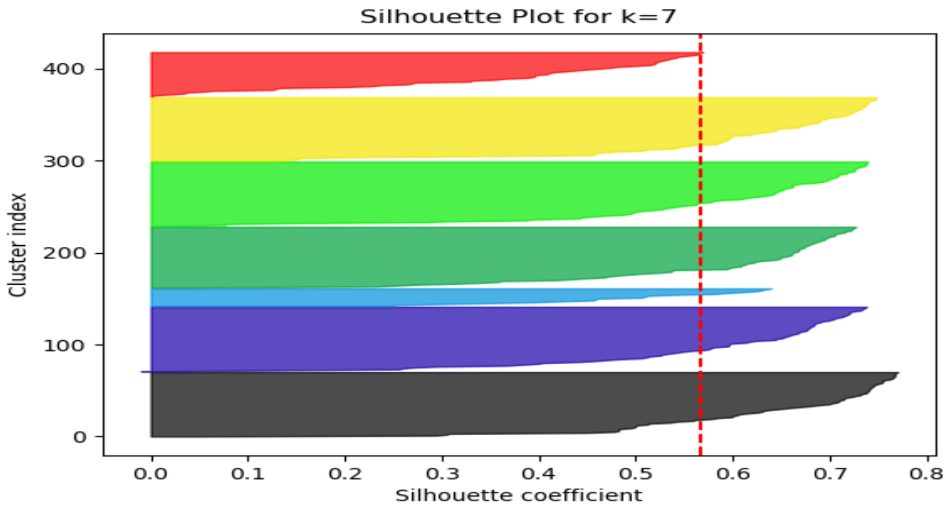
Centroids for k=6 in dataset2.txt:
[[1.43552563 -0.25442151]
[-1.22869606 -0.09413111]
[-0.72904684 1.16339538]
[0.66421797 -1.12945935]
[0.67307564 1.34720815]
[-0.80036066 -1.0588745]]

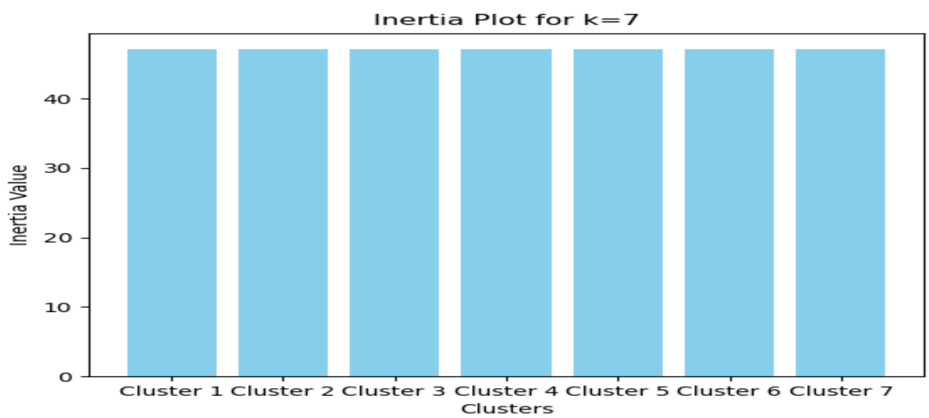


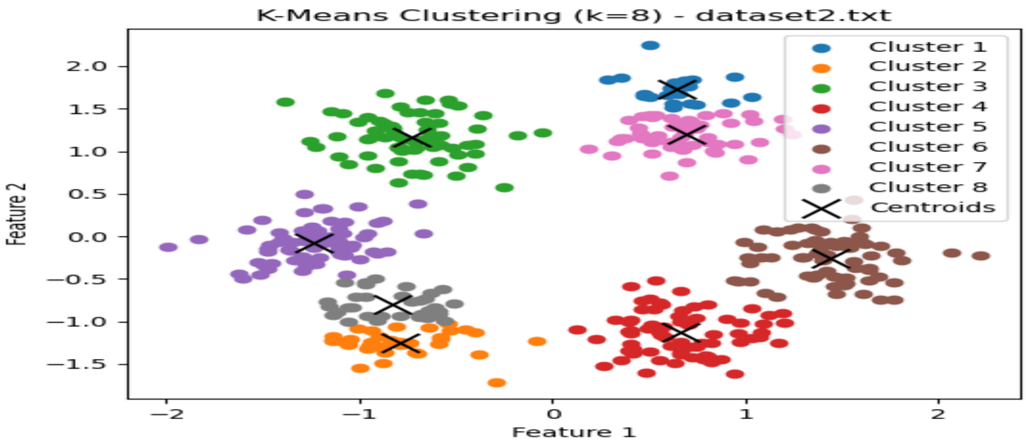




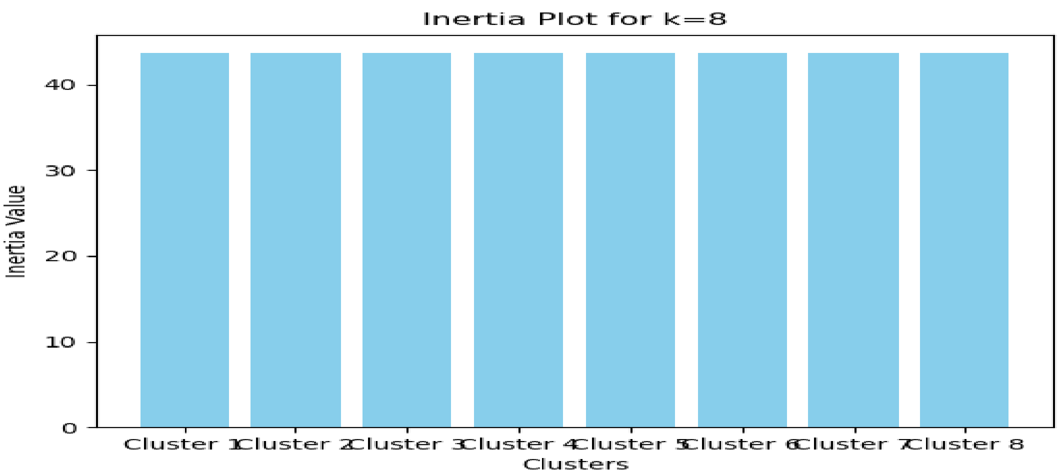
Centroids for k=7 in dataset2.txt:
[[-0.72904684 1.16339538]
 [0.65366507 -1.13090953]
 [0.63791764 1.72423042]
 [-0.81103712 -1.05628422]
 [-1.22869606 -0.09413111]
 [1.43552563 -0.25442151]
 [0.68742584 1.19332151]]

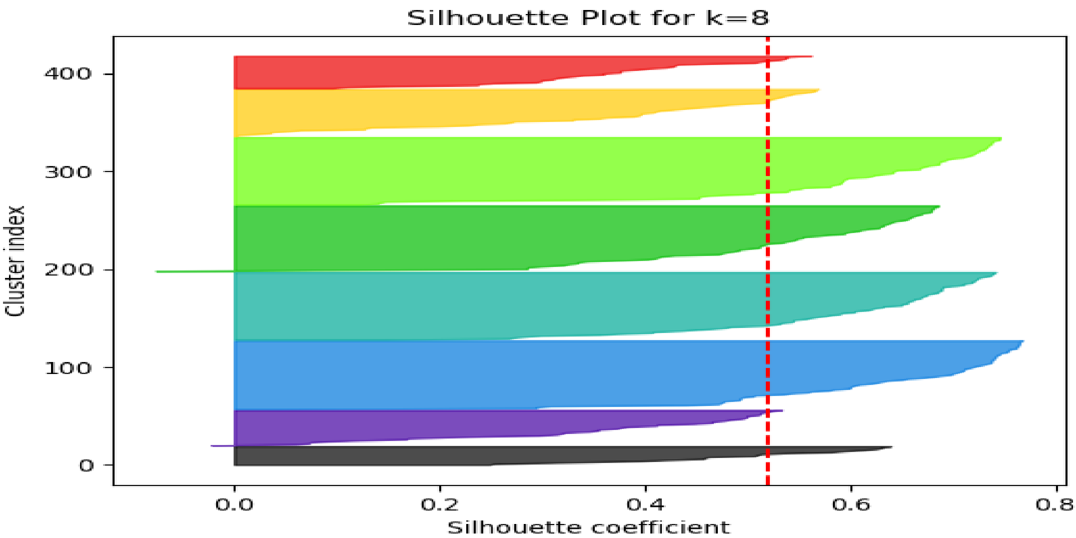


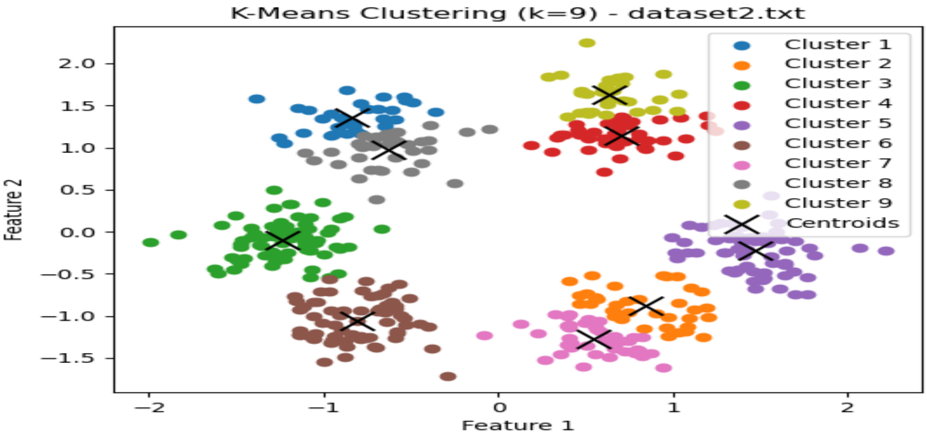




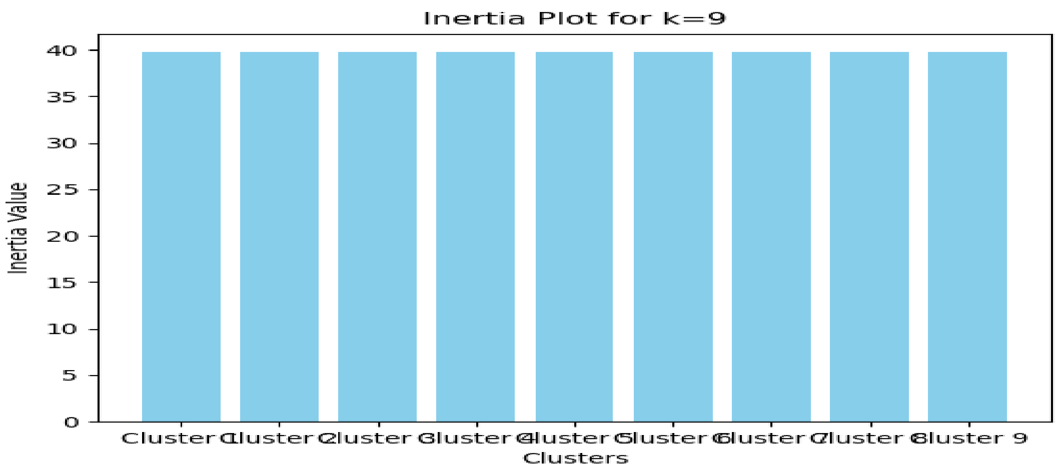
Centroids for k=8 in dataset2.txt:
[[0.63791764 1.72423042]
 [-0.78680825 -1.25033888]
 [-0.72904684 1.16339538]
 [0.66421797 -1.12945935]
 [-1.23921762 -0.07520692]
 [1.43552563 -0.25442151]
 [0.68742584 1.19332151]
 [-0.83186006 -0.80324014]]

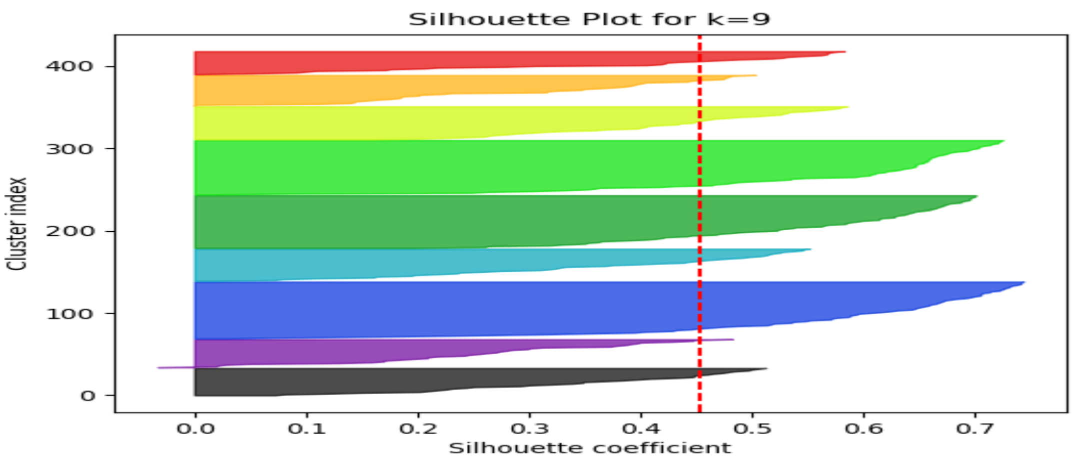


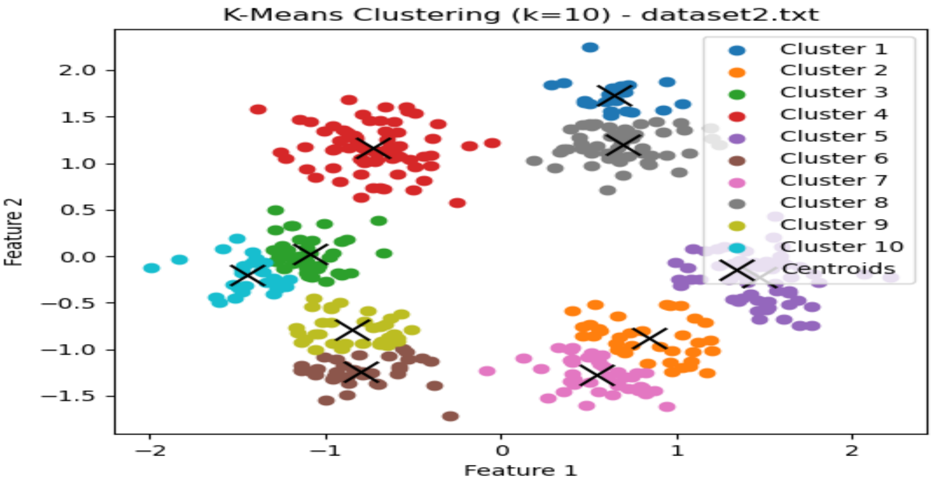




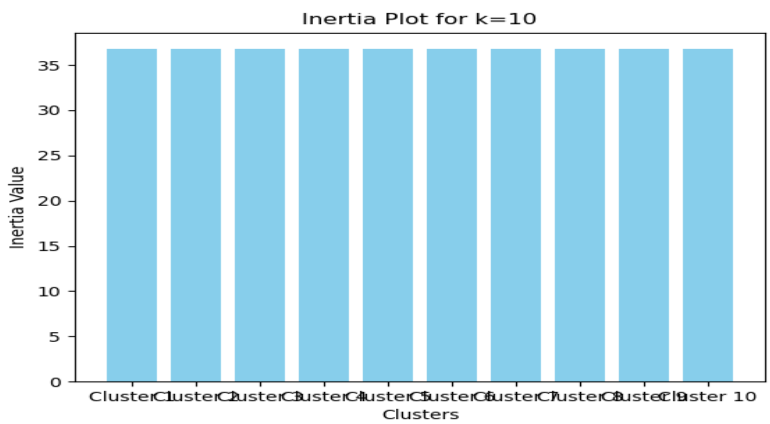
Centroids for k=9 in dataset2.txt:
[[-0.83779549 1.35360097]
 [0.84043016 -0.88330029]
 [-1.23623354 -0.10097171]
 [0.69925557 1.14356106]
 [1.466269   -0.22855887]
 [-0.81103712 -1.05628422]
 [0.54084081 -1.27639624]
 [-0.63100925 0.97271974]
 [0.63696539 1.6281007]]

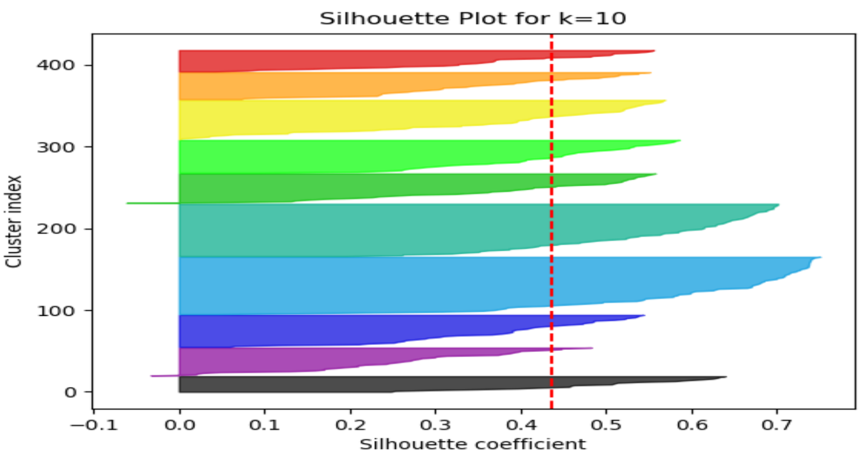


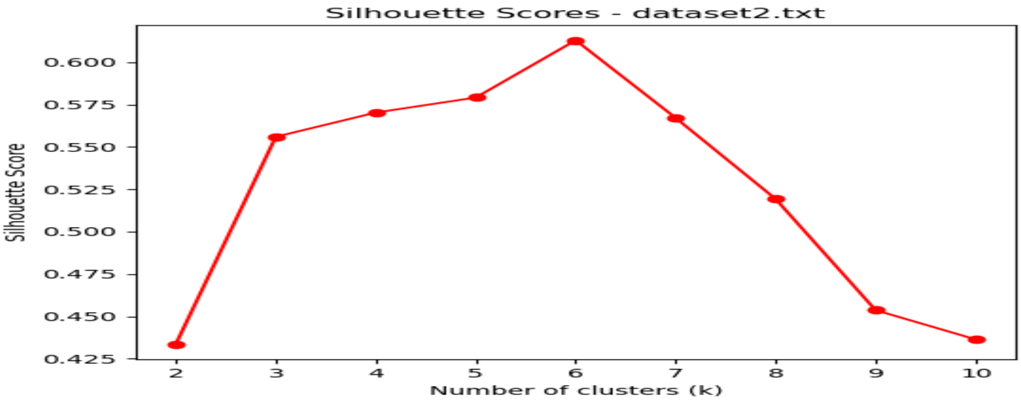


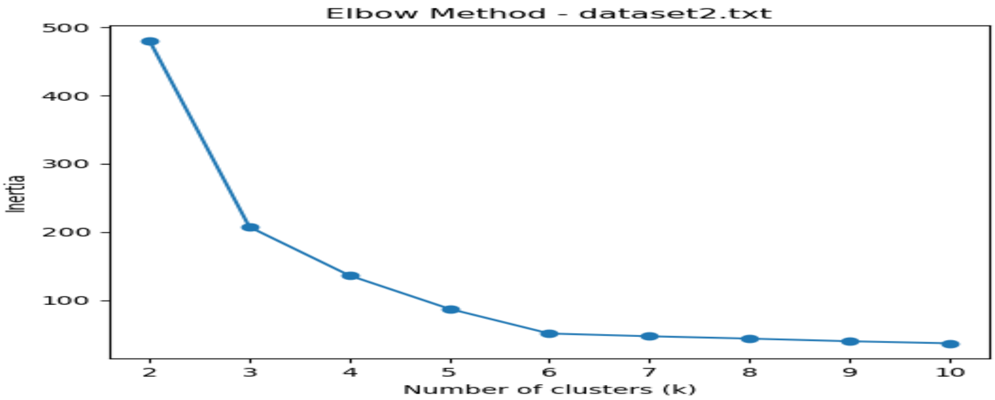


Centroids for k=10 in dataset2.txt:
[[0.63791764 1.72423042]
 [0.84043016 -0.88330029]
 [-1.09342458 0.02581739]
 [-0.72904684 1.16339538]
 [1.466269   -0.22855887]
 [-0.79982172 -1.24383429]
 [0.54084081 -1.27639624]
 [0.68742584 1.19332151]
 [-0.84654243 -0.78746868]
 [-1.44848224 -0.20233664]]









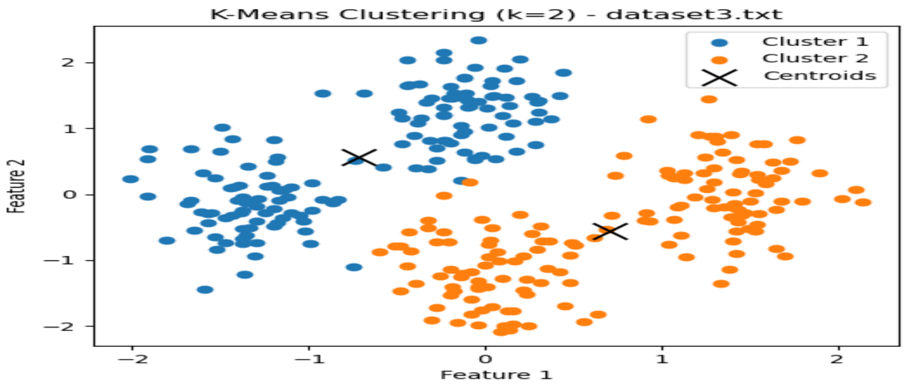
The optimal K value was first identified using the elbow graph, where we observe a clear stabilization or elbow point in the metric. We then confirmed this choice using the silhouette score graph, ensuring the best balance between cluster compactness and separation.

Optimal k for dataset2.txt based on silhouette score: 6

Best Silhouette Score: 0.6129497302185308

Inertia values: [480.6875088355889, 207.38897532019138, 136.2257672516503, 87.31103836125902, 51.0908639894117, 47.0667995601096, 43.64885382169108, 39.77056806424032, 36.76285284005939]

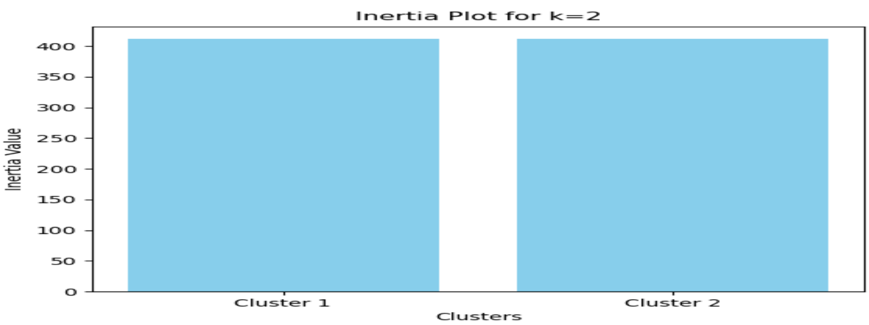
Silhouette scores: [0.43362841855607276, 0.5559763914873764, 0.5702750673522017, 0.5791354312623921, 0.6129497302185308, 0.5671042674108933, 0.5192529196679174, 0.453622325564437, 0.4363941525073924]

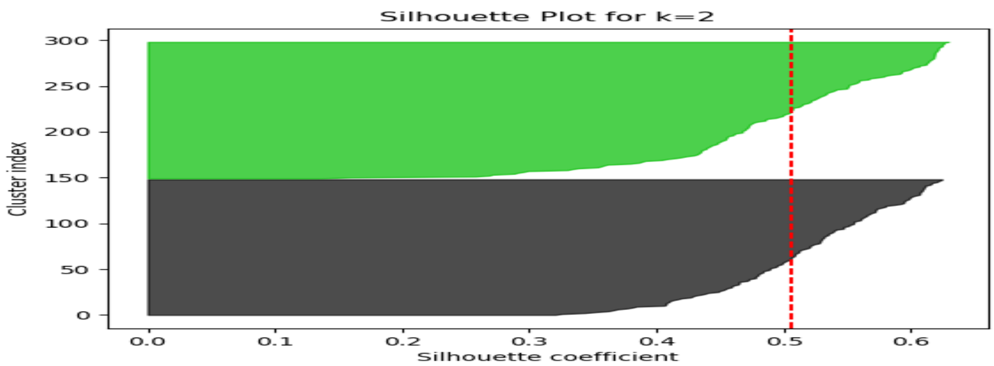


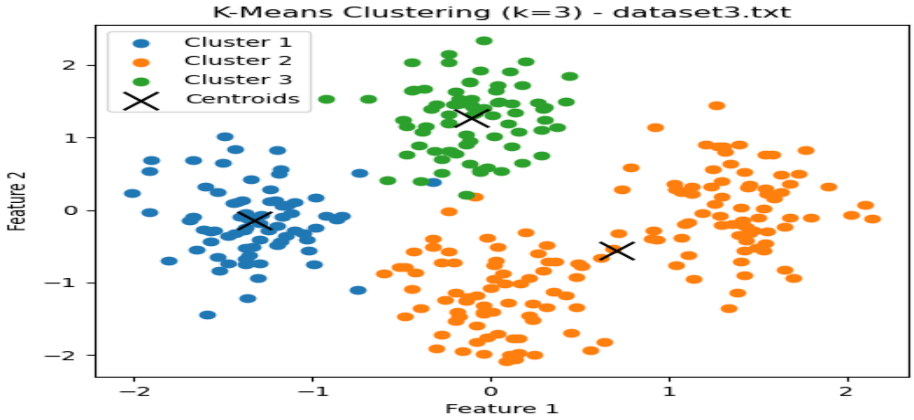
Centroids for k=2 in dataset3.txt:

[[-0.71517134 0.56421112 -0.89712286]

 [0.71040354 -0.56044971 0.89114204]]





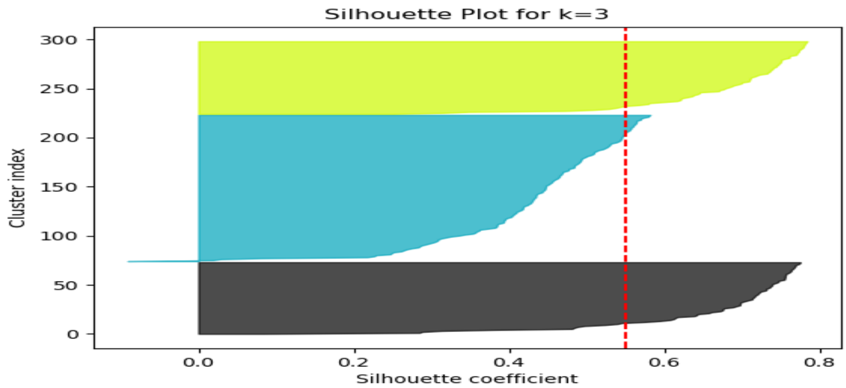


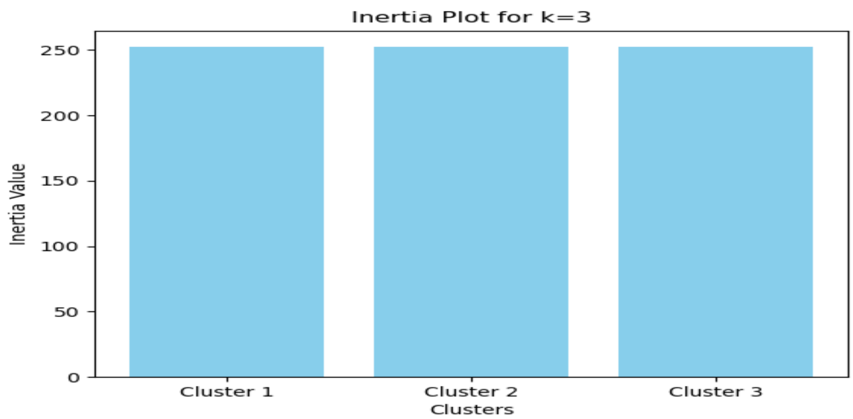
Centroids for k=3 in dataset3.txt:

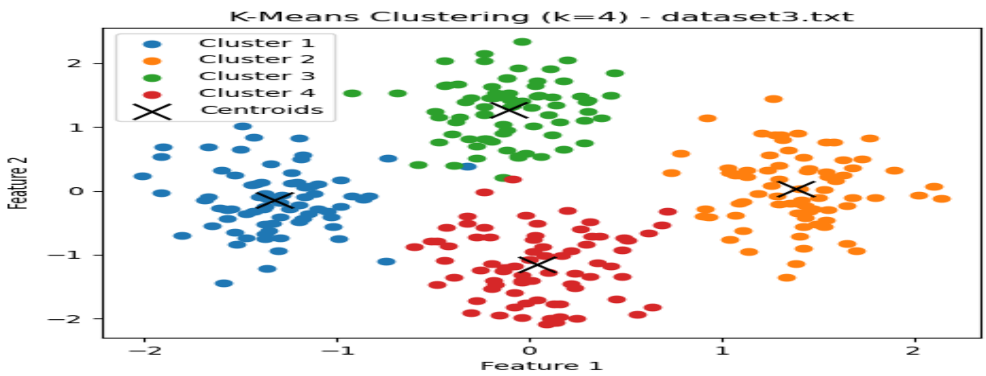
[[-1.32780099 -0.14610425 -1.34794595]

[0.71040354 -0.56044971 0.89114204]

[-0.11071009 1.26505561 -0.45231075]]







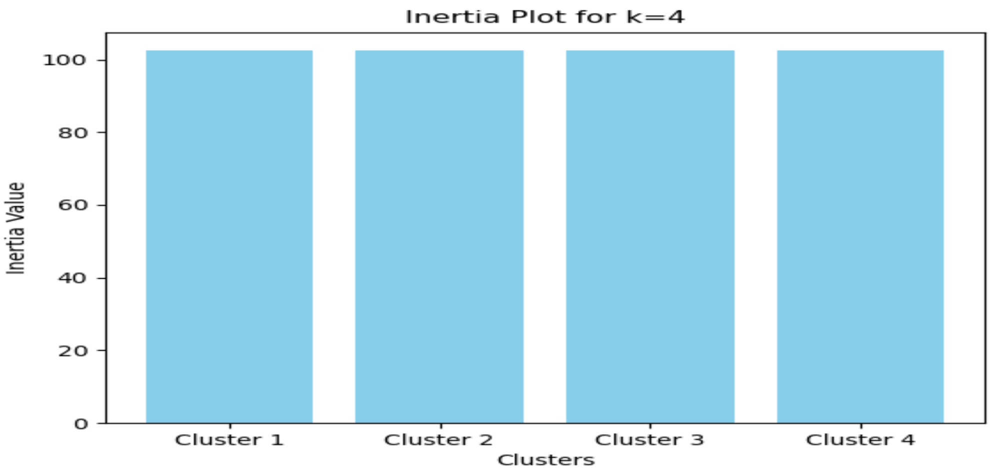
Centroids for k=4 in dataset3.txt:

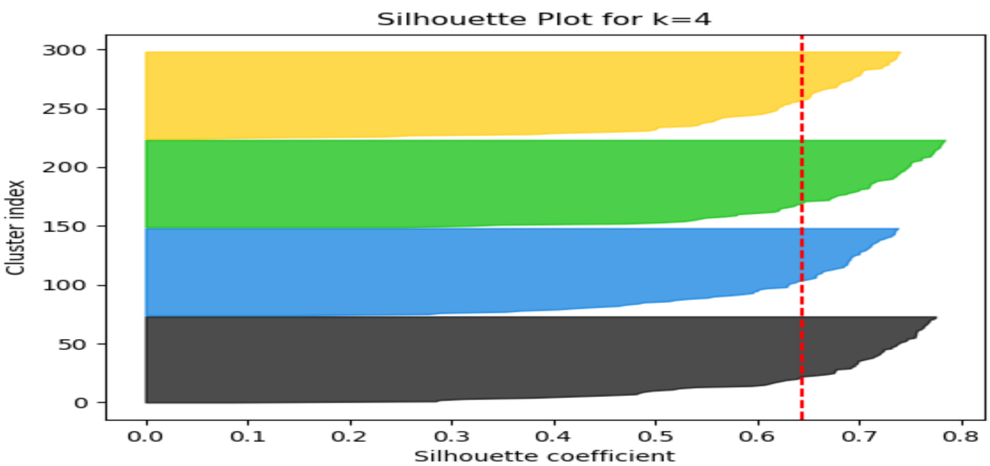
[[-1.32780099 -0.14610425 -1.34794595]

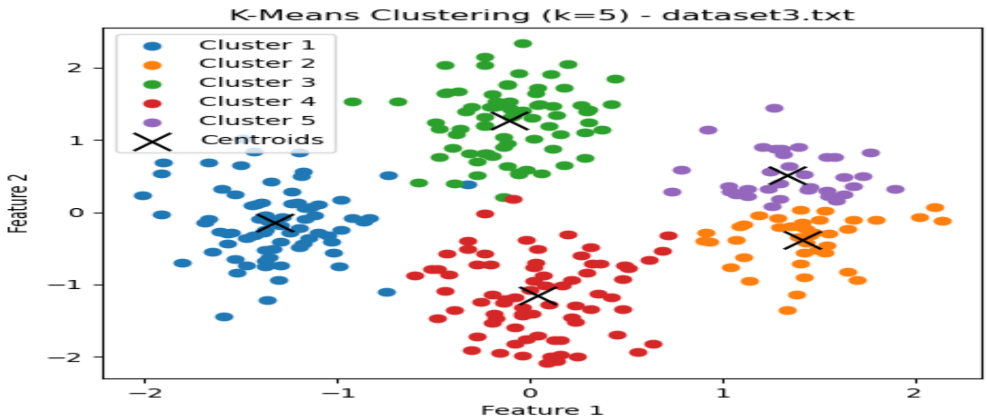
[1.38003852 0.03121957 1.33895964]

[-0.11071009 1.26505561 -0.45231075]

[0.04076855 -1.15211899 0.44332444]]







Centroids for k=5 in dataset3.txt:

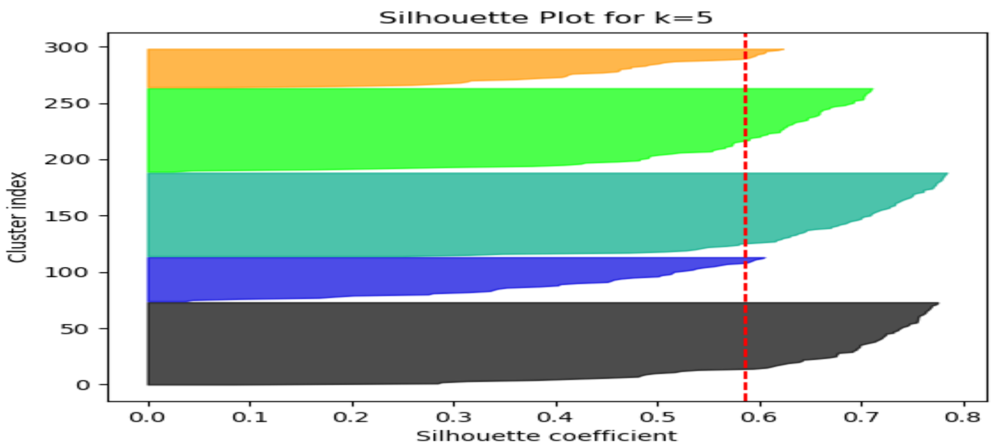
[[-1.32780099 -0.14610425 -1.34794595]

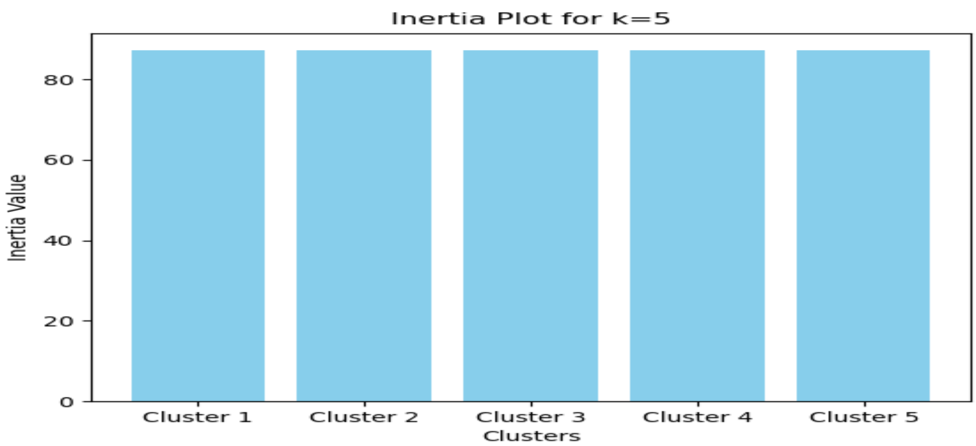
[1.41802863 -0.388684 1.33895964]

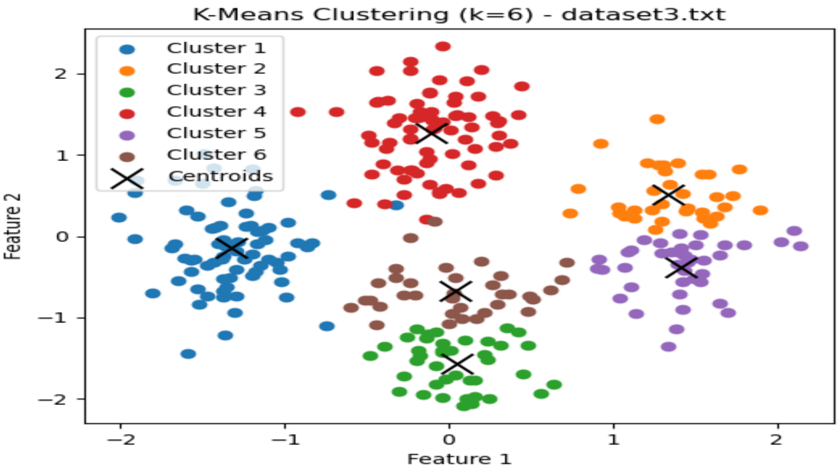
[-0.11071009 1.26505561 -0.45231075]

[0.04076855 -1.15211899 0.44332444]

[1.33662126 0.51110936 1.33895964]]







Centroids for k=6 in dataset3.txt:

[[-1.32780099 -0.14610425 -1.34794595]

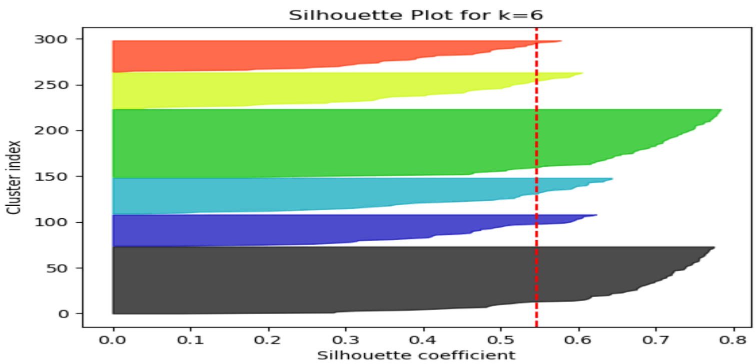
[1.33662126 0.51110936 1.33895964]

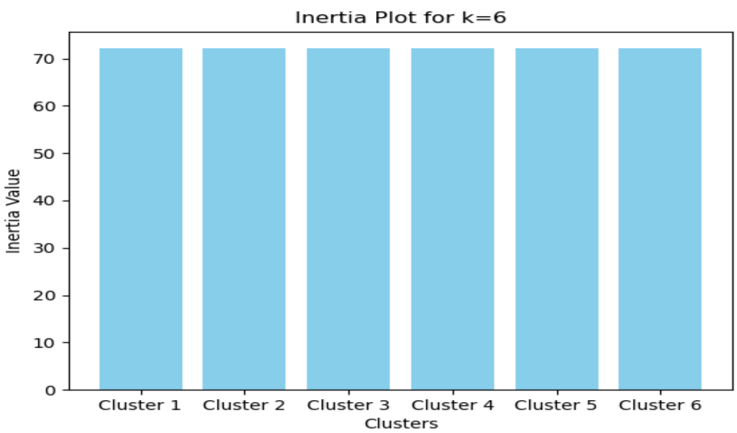
[0.04460369 -1.5713436   0.44332444]

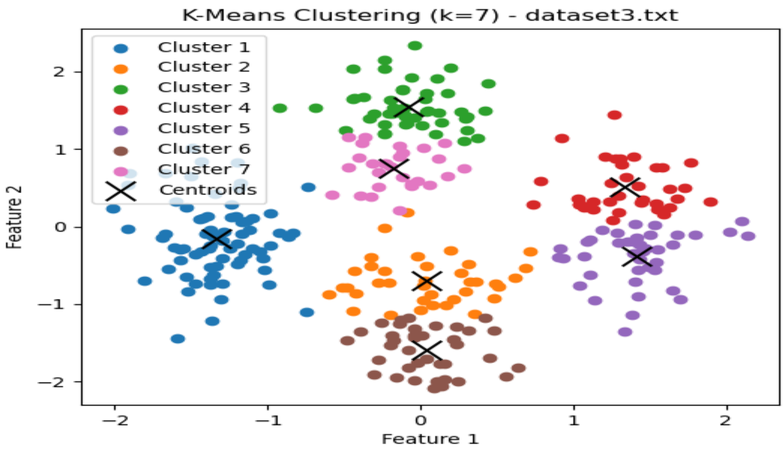
[-0.11071009 1.26505561 -0.45231075]

[1.41802863 -0.388684   1.33895964]

[0.03638553 -0.67300515 0.44332444]]







Centroids for k=7 in dataset3.txt:

[[-1.34149115 -0.15337324 -1.34794595]

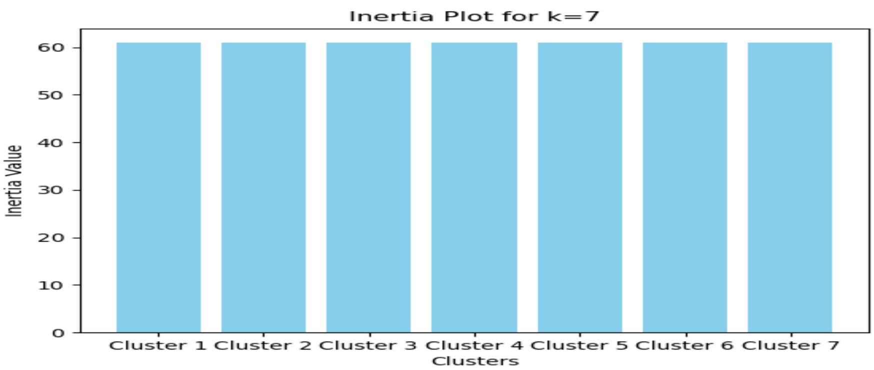
[0.03876746 -0.69772878 0.44332444]

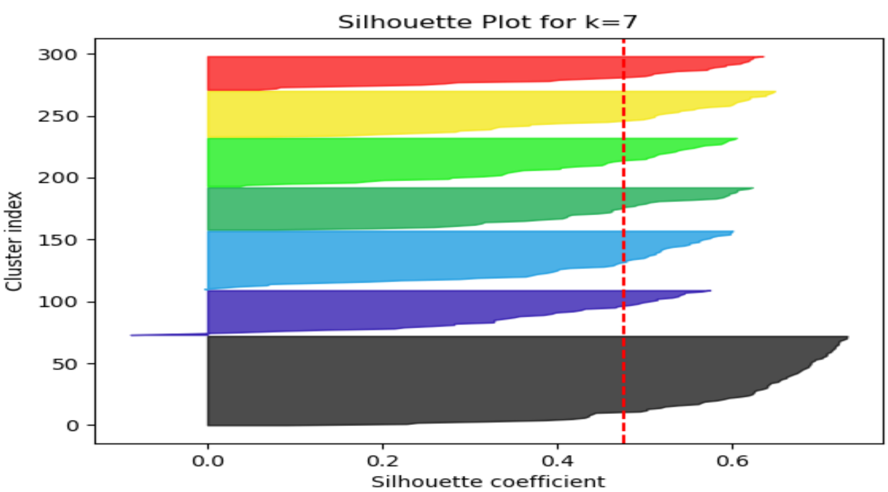
[-0.07770884 1.5480431 -0.45231075]

[1.33662126 0.51110936 1.33895964]

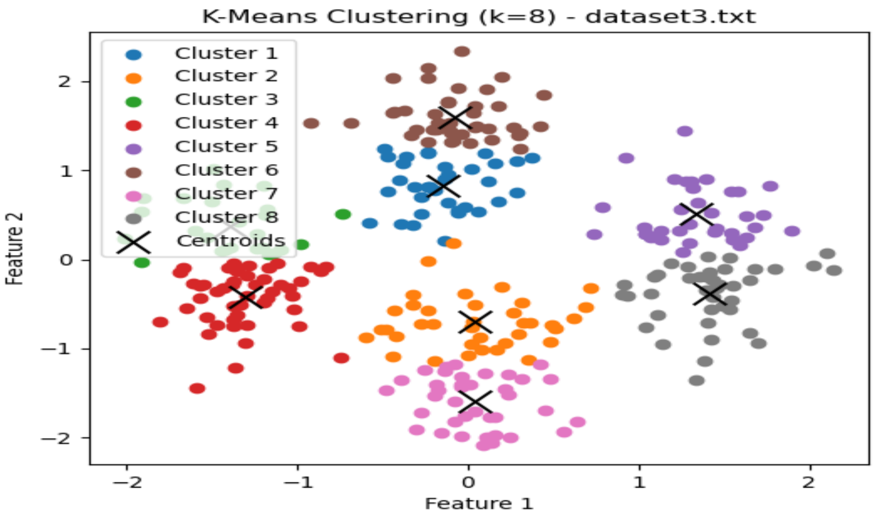
[1.41802863 -0.388684 1.33895964]

[0.04271698 -1.59455156 0.44332444]

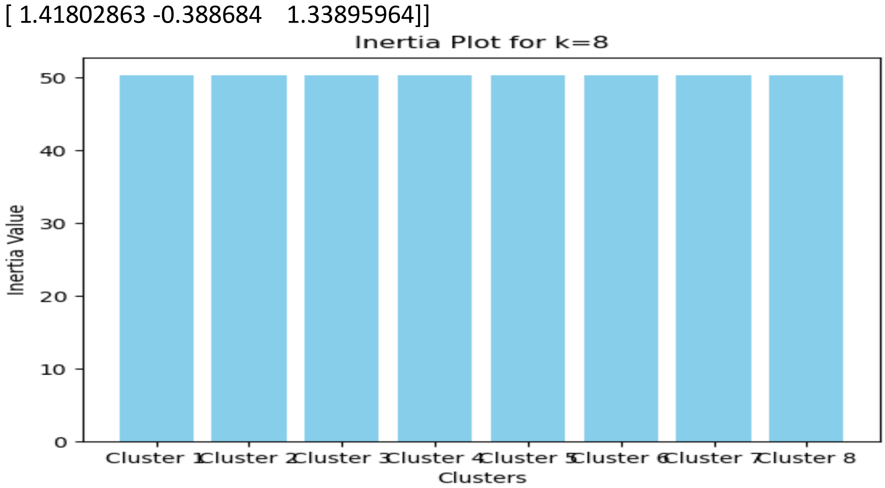


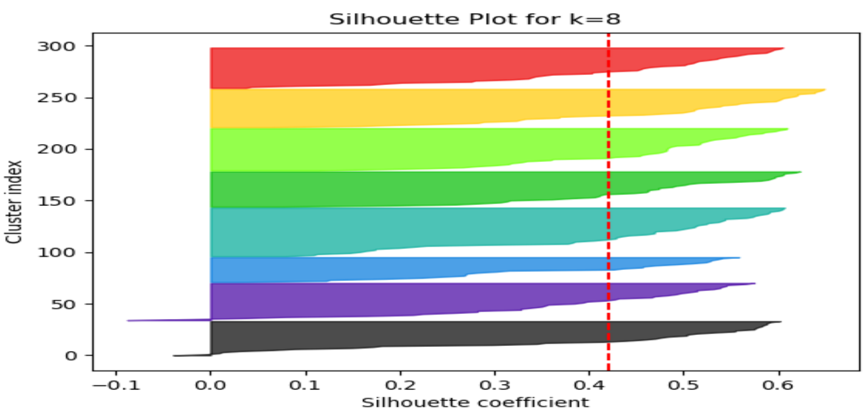


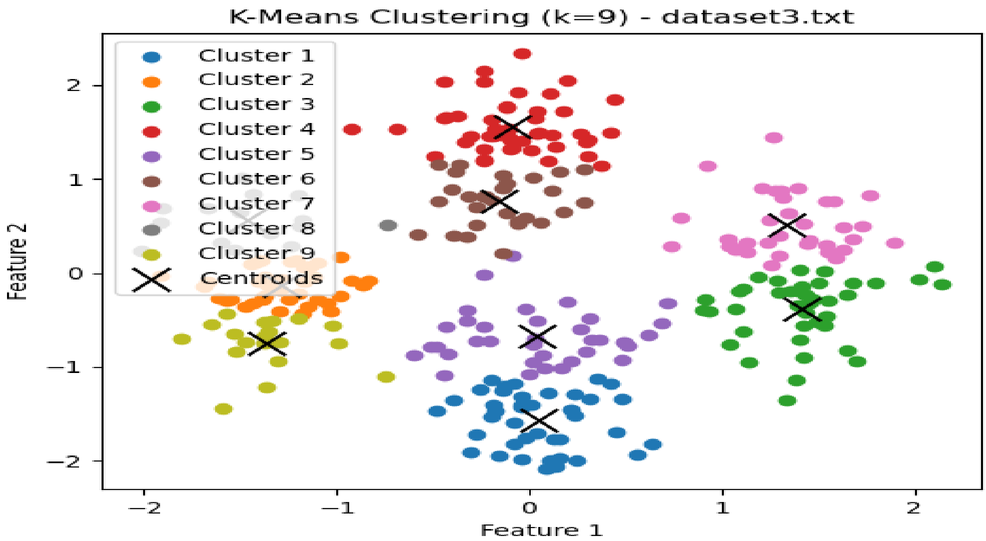
[-0.17505899 0.74848695 -0.48429772]]



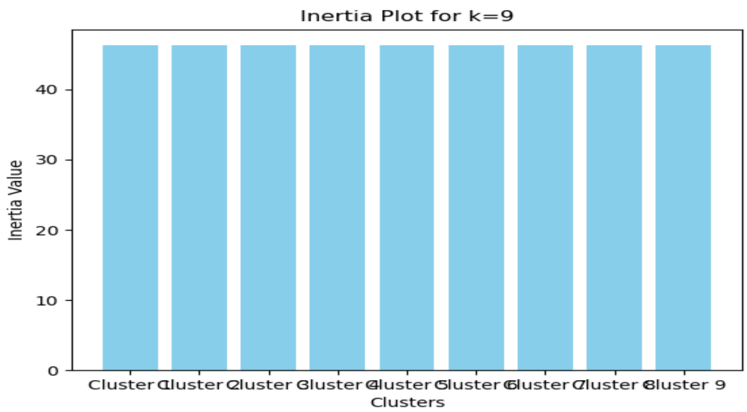
Centroids for k=8 in dataset3.txt:
[[-0.15088792 0.82392026 -0.47865296]
[0.03876746 -0.69772878 0.44332444]
[-1.3999267 0.3725656 -1.34794595]
[-1.31105597 -0.42729972 -1.34794595]
[1.33662126 0.51110936 1.33895964]
[-0.08336874 1.60120034 -0.45231075]
[0.04271698 -1.59455156 0.44332444]
[1.41802863 -0.388684 1.33895964]]

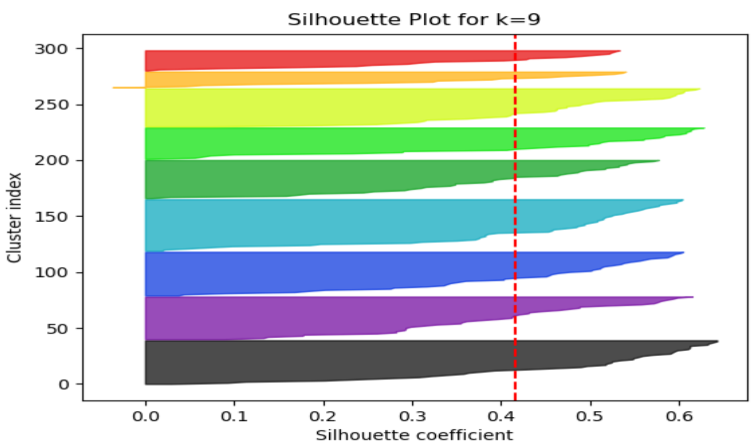


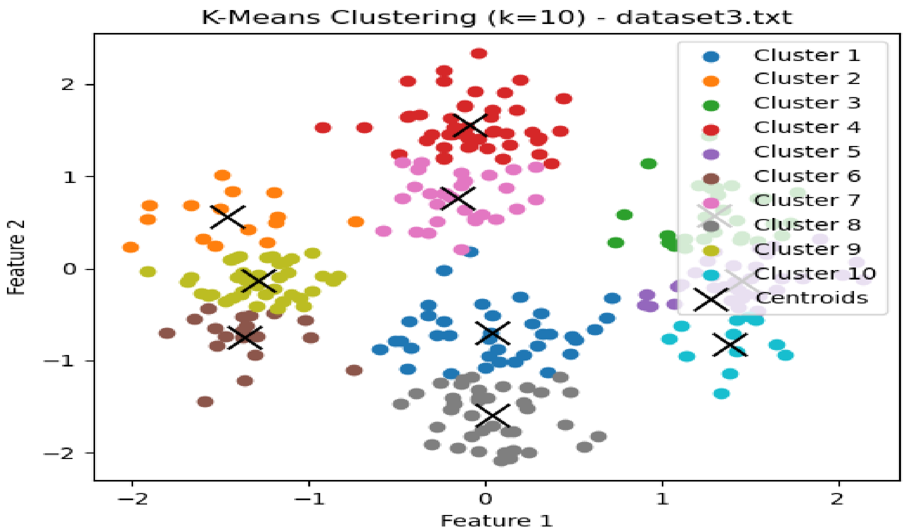




Centroids for k=9 in dataset3.txt:
[[0.04460369 -1.5713436   0.44332444]
 [-1.28318231 -0.13611088 -1.34794595]
 [1.41802863 -0.388684   1.33895964]
 [-0.08532248 1.55749501 -0.45231075]
 [0.03638553 -0.67300515 0.44332444]
 [-0.15936274 0.76073923 -0.48319472]
 [1.33662126 0.51110936 1.33895964]
 [-1.46463602 0.55596907 -1.34794595]
 [-1.36395809 -0.74881359 -1.34794595]]







Centroids for k=10 in dataset3.txt:

[[0.03876746 -0.69772878 0.44332444]

 [-1.46463602 0.55596907 -1.34794595]

 [1.29512809 0.57461114 1.33895964]

 [-0.08532248 1.55749501 -0.45231075]

 [1.45036352 -0.13105134 1.33895964]

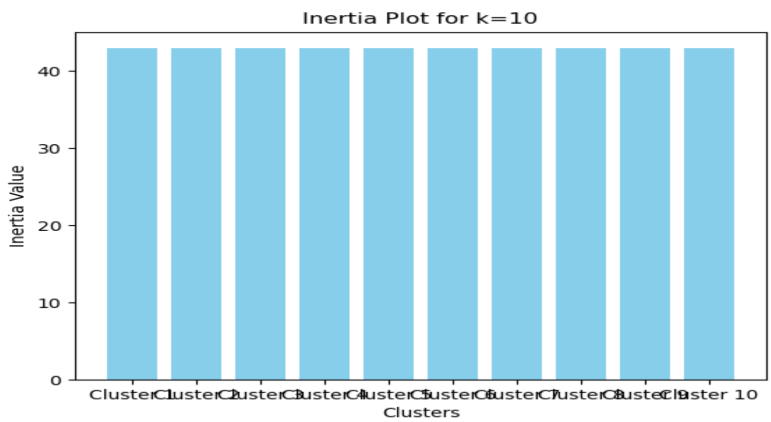
 [-1.36395809 -0.74881359 -1.34794595]

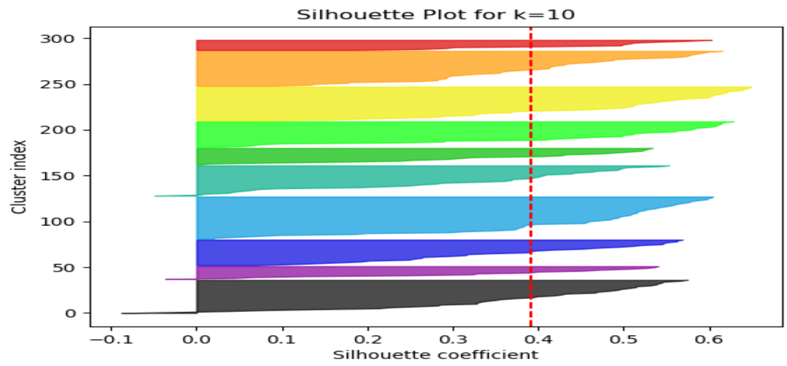
 [-0.15936274 0.76073923 -0.48319472]

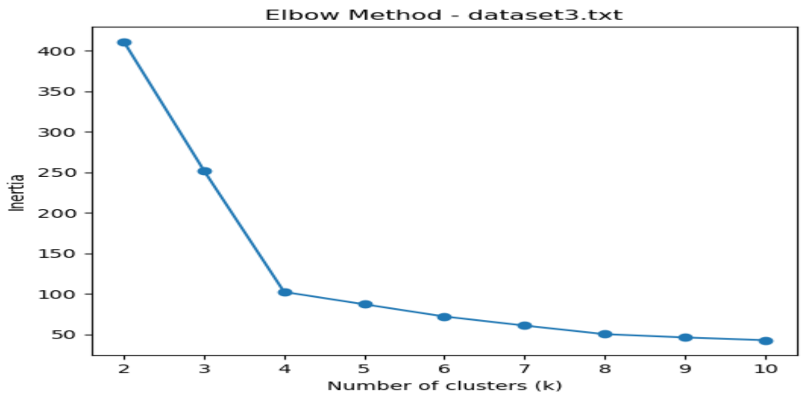
 [0.04271698 -1.59455156 0.44332444]

 [-1.28318231 -0.13611088 -1.34794595]

 [1.38598458 -0.82220915 1.33895964]]







The optimal K value is first identified using the elbow graph, where we observe a distinct elbow point. At this point, the decrease in error or intra-cluster distance slows down significantly, indicating that adding more clusters no longer provides substantial improvement. We then confirm this choice using the silhouette score graph, which evaluates cluster quality, ensuring that the selected value offers the best balance between cohesion and separation.

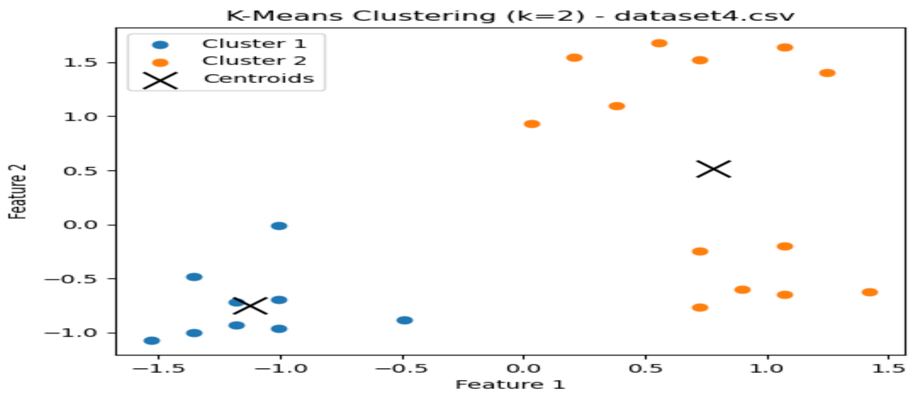
Optimal k for dataset3.txt based on silhouette score: 4

Best Silhouette Score: 0.6435114283016757

Inertia values: [411.5029034860903, 252.27210648726447, 102.41848364172257, 87.1817191047081, 72.11623504066826, 60.91824571767013, 50.311811210243576, 46.256234949605535, 42.90442798081609]

Silhouette scores: [0.5056387338808708, 0.5510971298927065, 0.6435114283016757, 0.5872550377462807, 0.5476270046493935, 0.47557495326987753, 0.42130382767896124, 0.416073725936589, 0.39207871290599083]

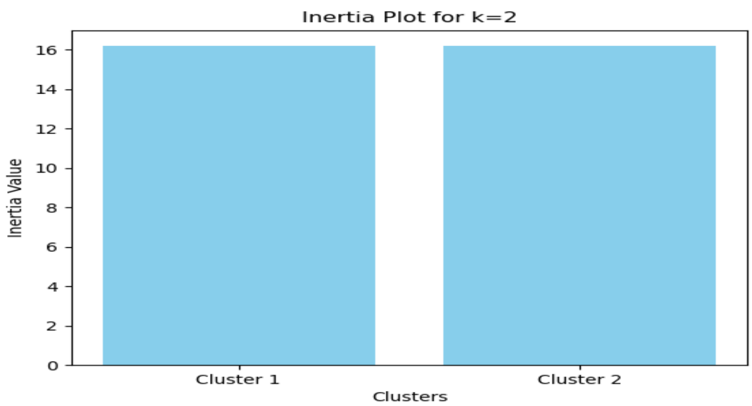
DATASET 4

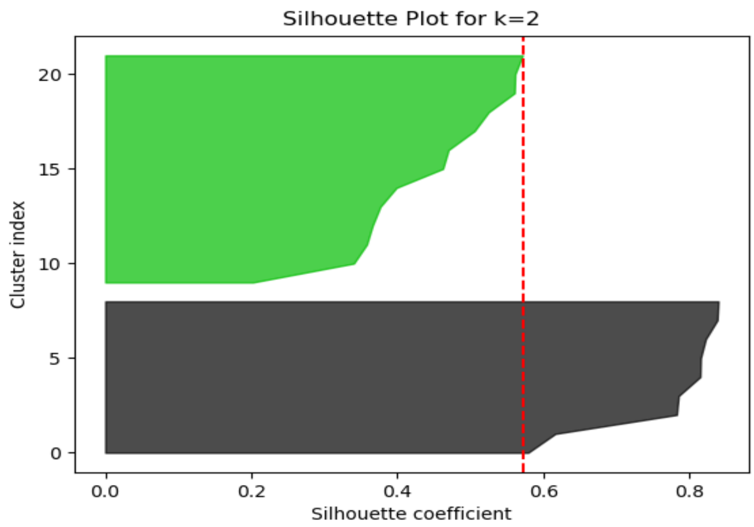


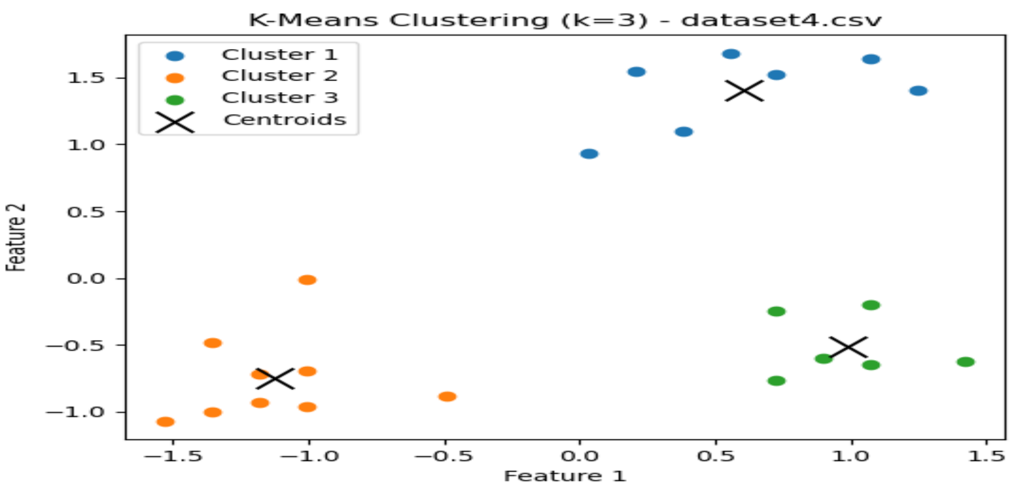
Centroids for k=2 in dataset4.csv:

[[-1.1247901 -0.74862223]

 [0.77870084 0.51827693]]





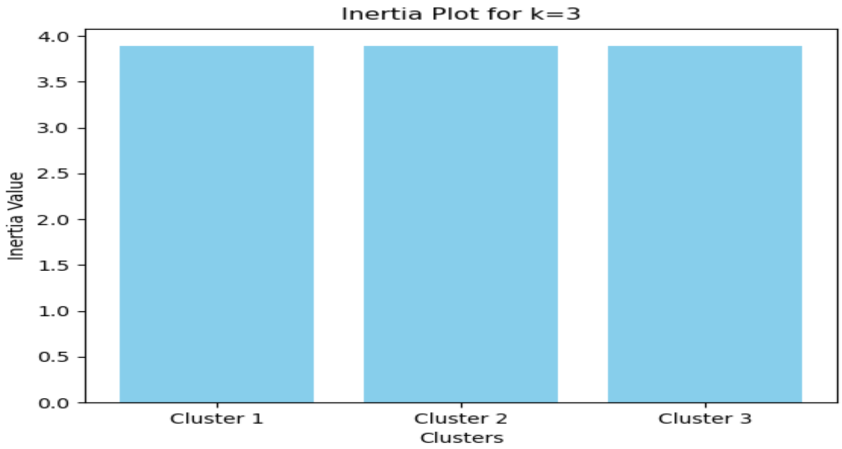


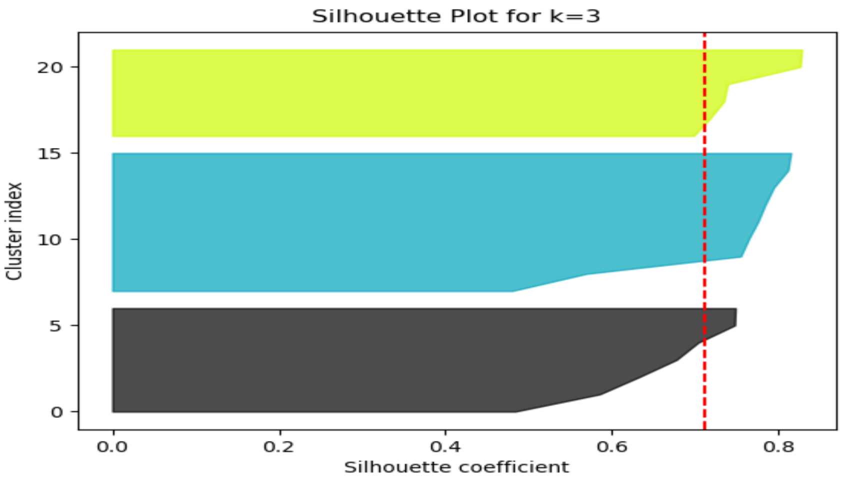
Centroids for k=3 in dataset4.csv:

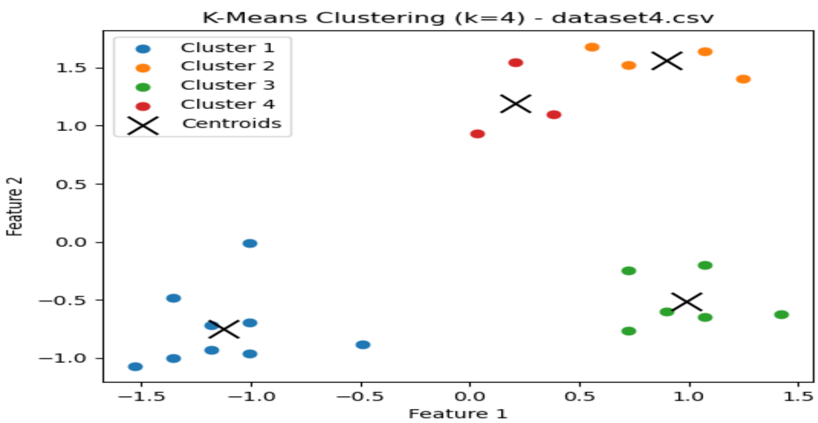
[[0.60143983 1.40141653]

 [-1.1247901 -0.74862223]

 [0.98550535 -0.51205261]]







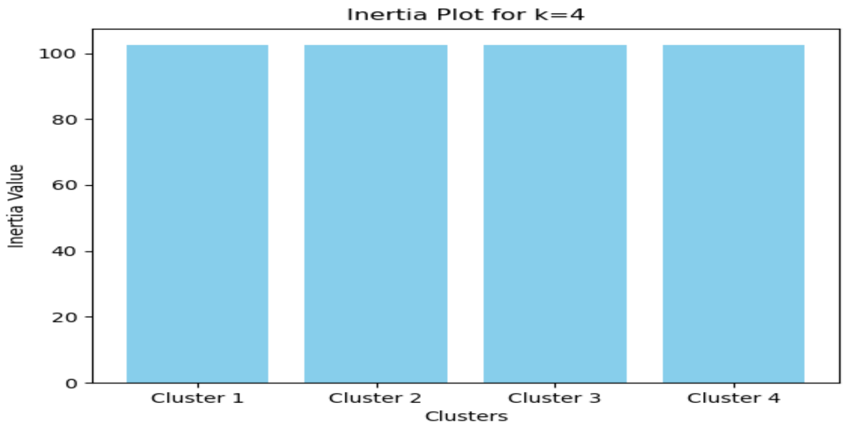
Centroids for k=4 in dataset4.csv:

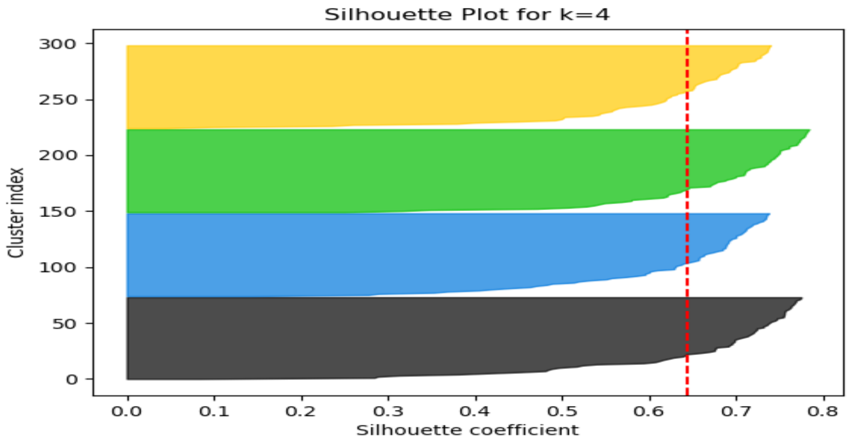
[[-1.1247901 -0.74862223]

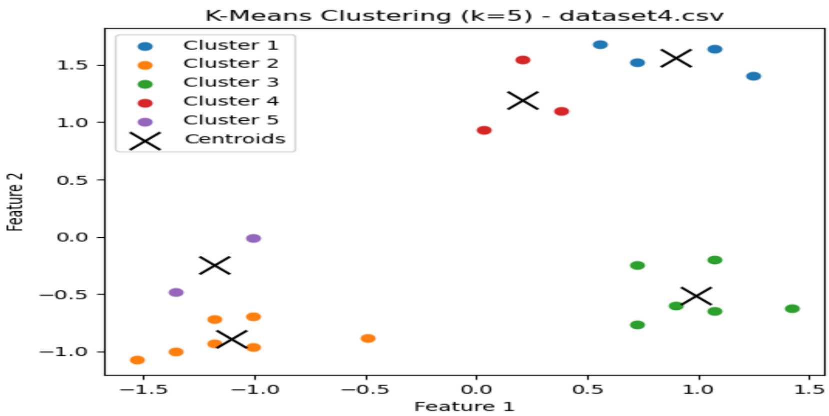
 [0.89878087 1.56021879]

 [0.98550535 -0.51205261]

 [0.20498511 1.18968019]]







Centroids for k=5 in dataset4.csv:

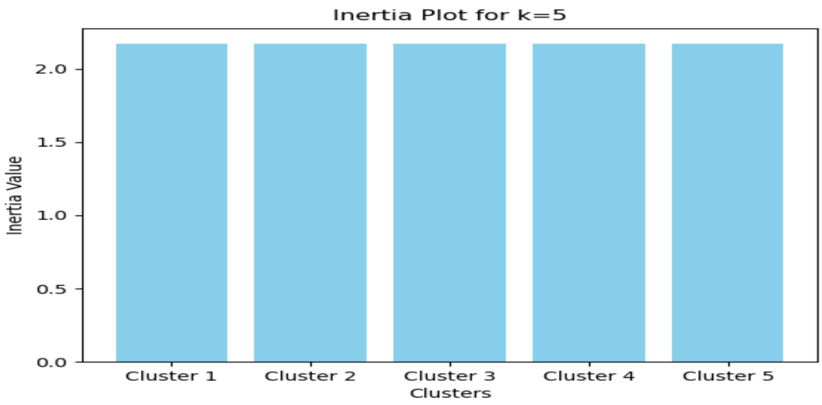
[[0.89878087 1.56021879]

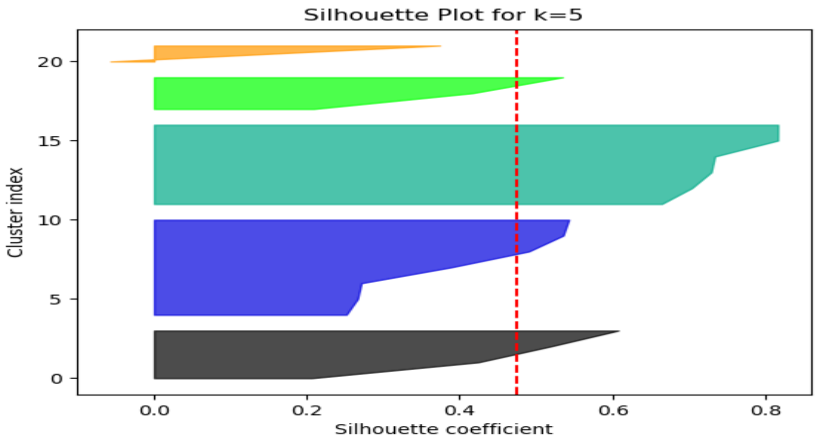
 [-1.10827115 -0.89239382]

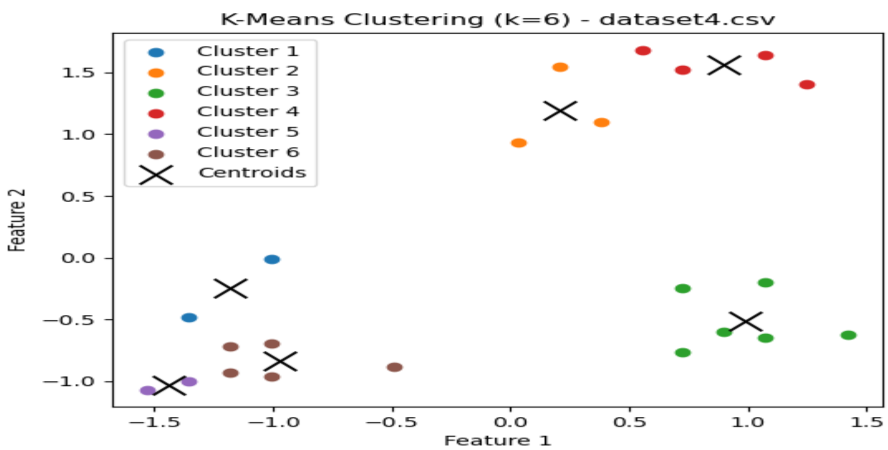
 [0.98550535 -0.51205261]

 [0.20498511 1.18968019]

[-1.18260641 -0.24542167]]







Centroids for k=6 in dataset4.csv:

[[-1.18260641 -0.24542167]

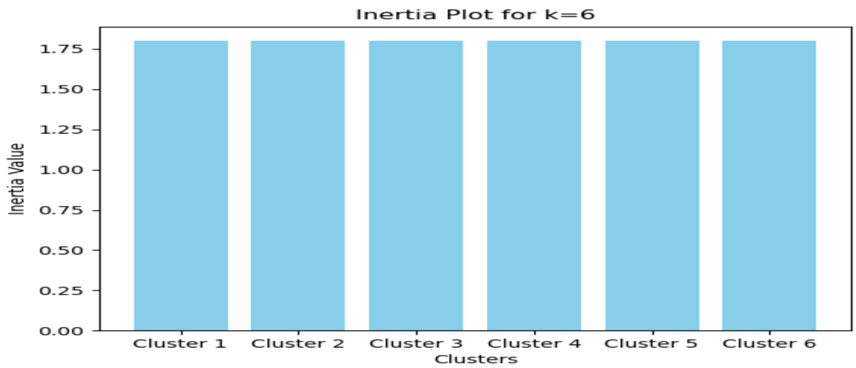
[0.20498511 1.18968019]

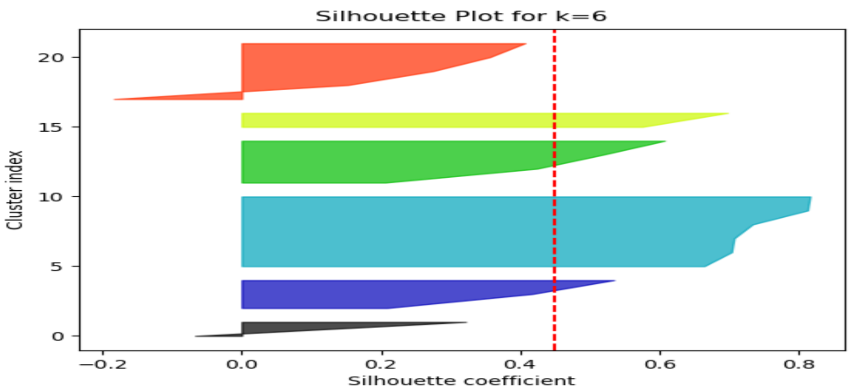
[0.98550535 -0.51205261]

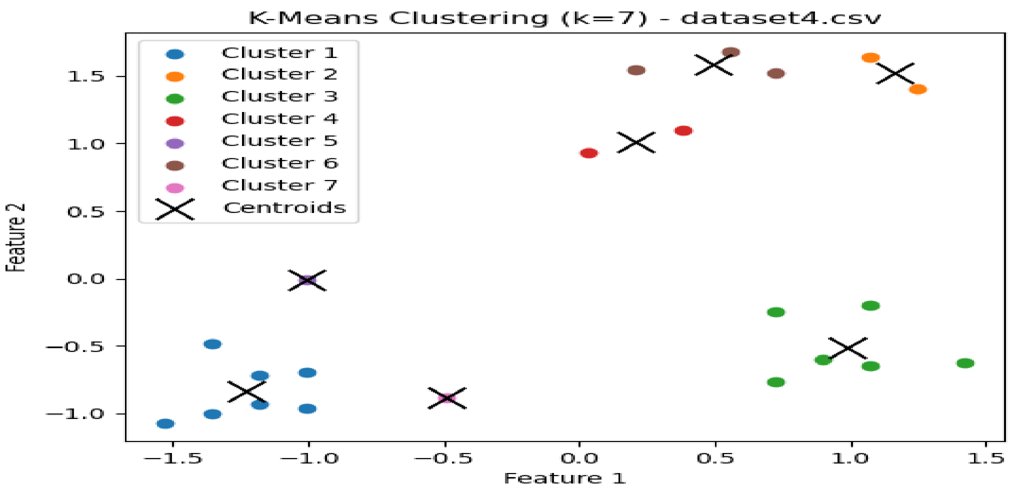
[0.89878087 1.56021879]

[-1.44277983 -1.03355138]

[-0.97446769 -0.83593079]]







Centroids for k=7 in dataset4.csv:

[[-1.23216325 -0.83525861]

[1.15895429 1.51904783]

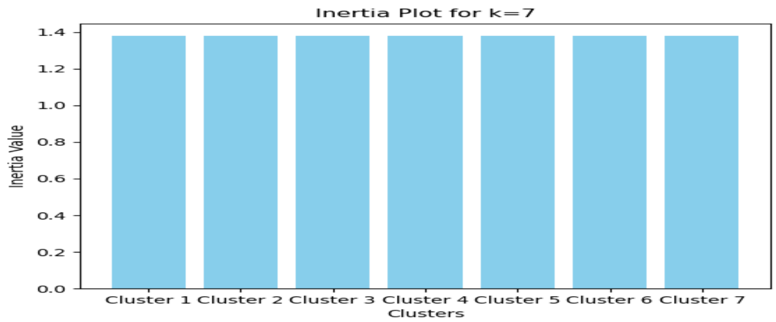
[0.98550535 -0.51205261]

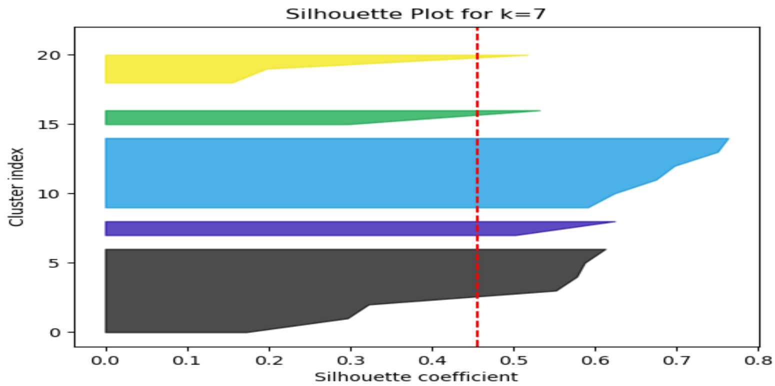
[0.20498511 1.01323324]

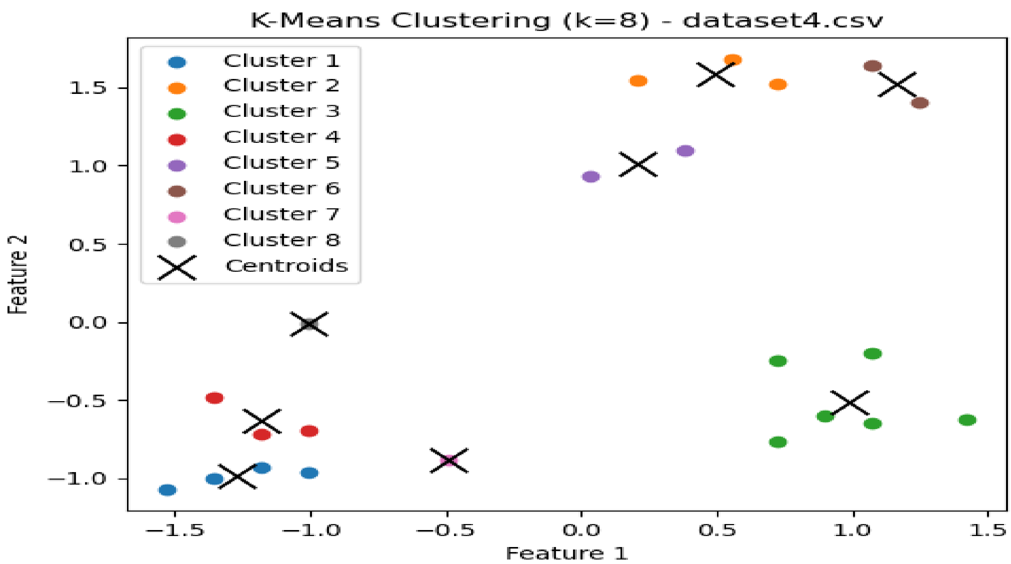
[-1.00915747 -0.01015907]

[0.49406668 1.58178453]

[-0.48881065 -0.88063069]]







Centroids for k=8 in dataset4.csv:

[[-1.26933088 -0.98943964]

[0.49406668 1.58178453]

[0.98550535 -0.51205261]

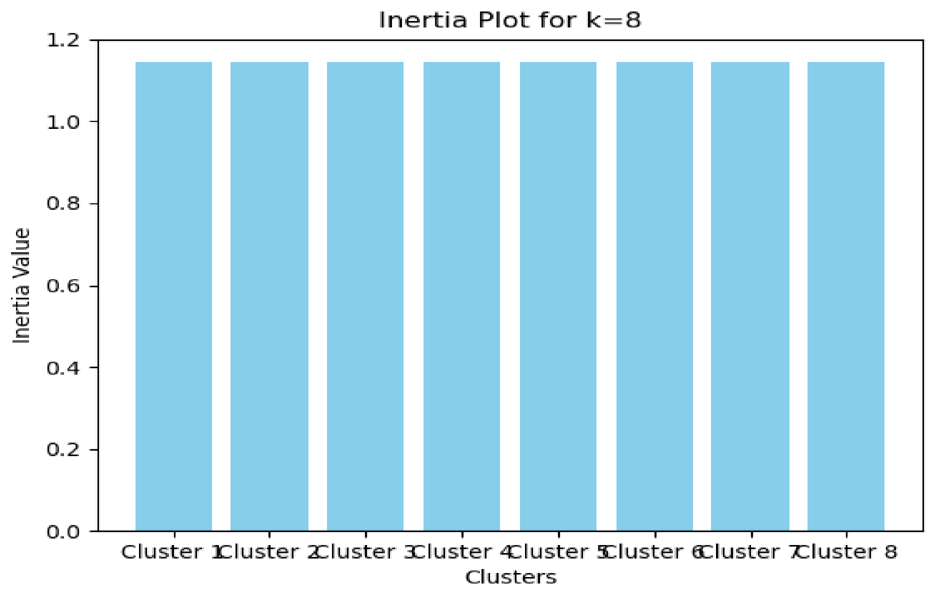
[-1.18260641 -0.62968391]

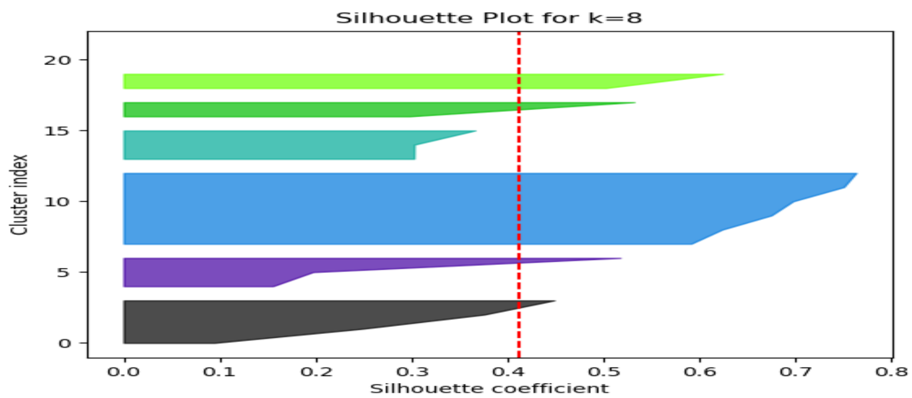
[0.20498511 1.01323324]

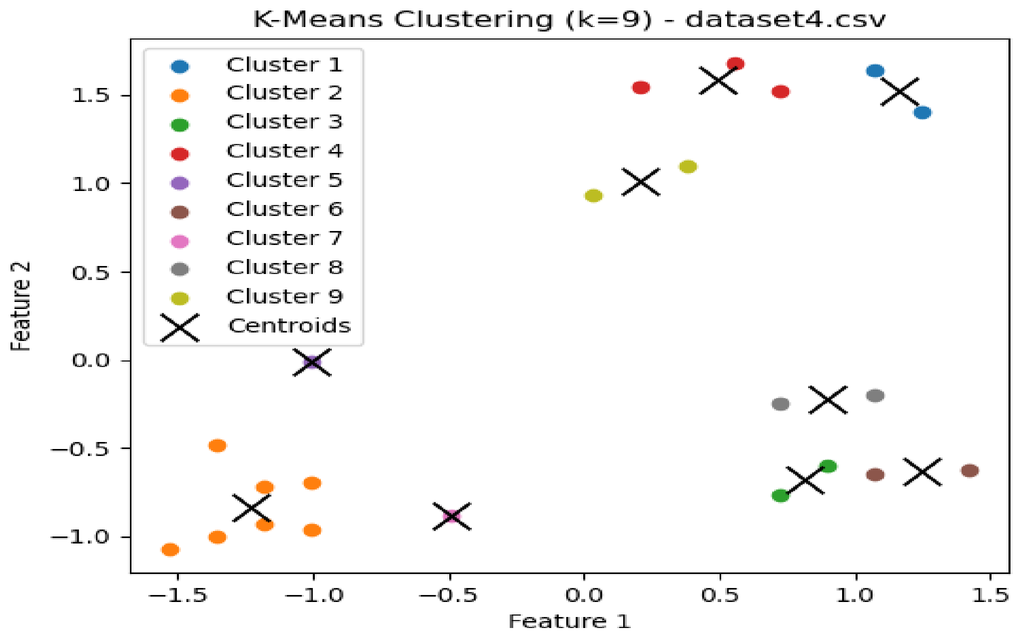
[1.15895429 1.51904783]

[-0.48881065 -0.88063069]

[-1.00915747 -0.01015907]]







Centroids for k=9 in dataset4.csv:

[[1.15895429 1.51904783]

[-1.23216325 -0.83525861]

[0.8120564 -0.68065748]

[0.49406668 1.58178453]

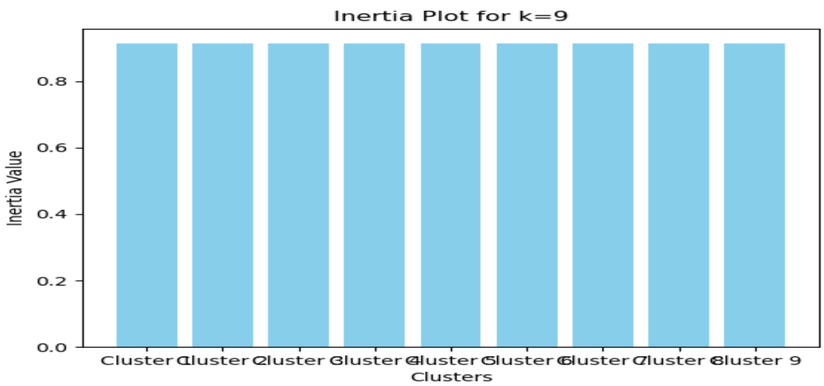
[-1.00915747 -0.01015907]

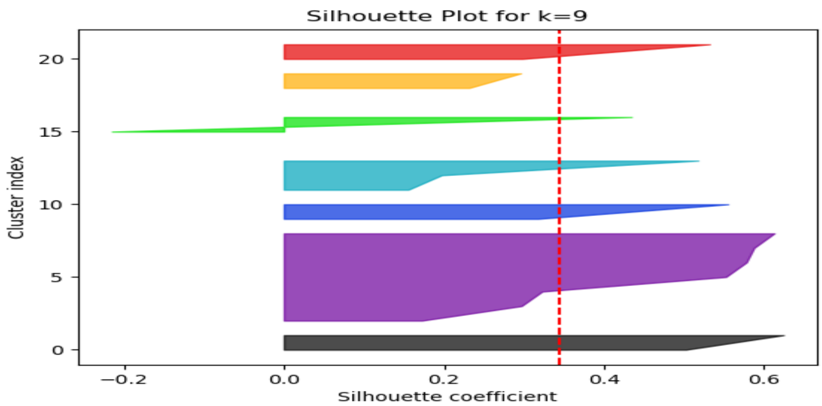
[1.24567876 -0.63360496]

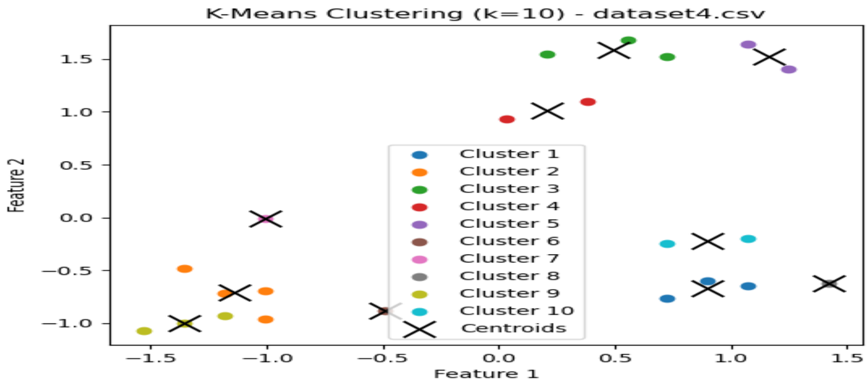
[-0.48881065 -0.88063069]

[0.89878087 -0.22189541]

[0.20498511 1.01323324]]







Centroids for k=10 in dataset4.csv:

[[0.89878087 -0.66889435]

[-1.13924418 -0.71300609]

[0.49406668 1.58178453]

[0.20498511 1.01323324]

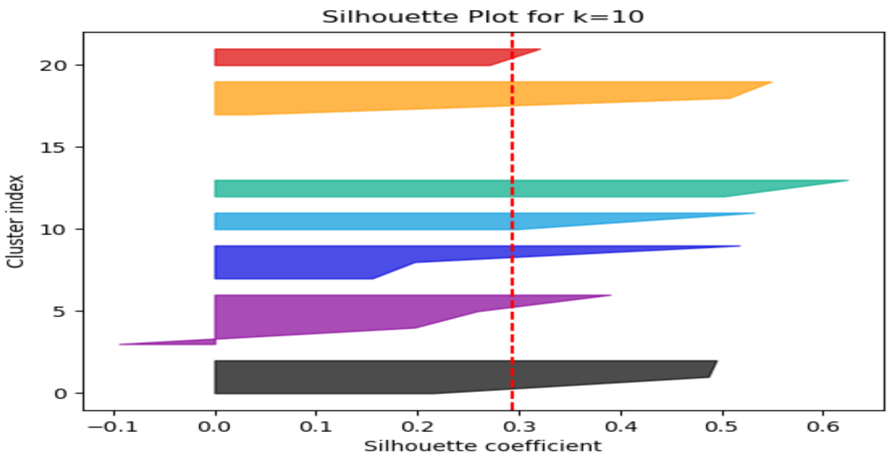
[1.15895429 1.51904783]

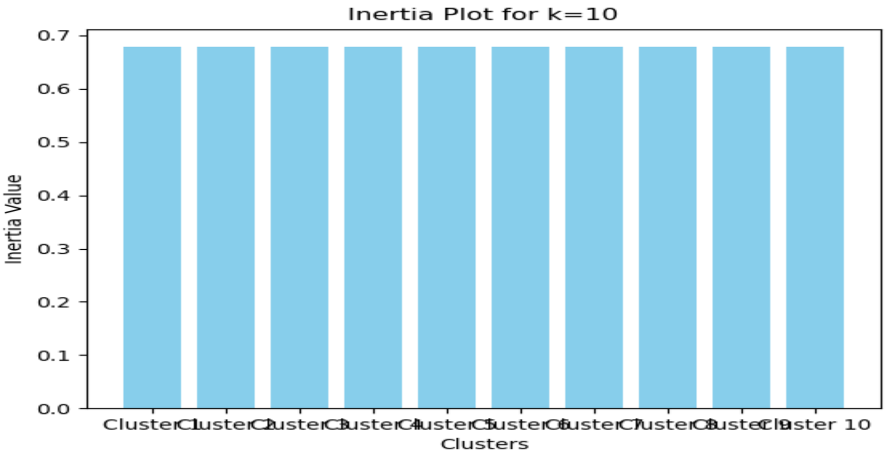
[-0.48881065 -0.88063069]

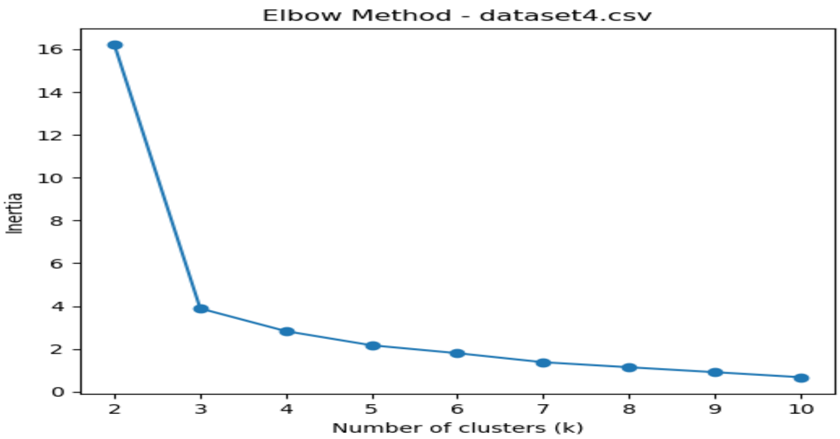
[-1.00915747 -0.01015907]

[1.4191277 -0.62184183]

[-1.35605536 -0.99826199] [0.89878087 -0.22189541]]


Silhouette Plot for k=10
Cluster index
Silhouette coefficient


Inertia Plot for k=10
Inertia Value
Clusters


Elbow Method - dataset4.csv
Inertia
Number of clusters (k)


Silhouette Scores - dataset4.csv
Silhouette Score
Number of clusters (k)

The optimal K value is first identified using the elbow graph, where we observe a distinct elbow point. At this point, the decrease in error or intra-cluster distance slows down significantly, indicating that adding more clusters no longer provides substantial improvement. We then confirm this choice using the silhouette score graph, which evaluates cluster quality, ensuring that the selected value offers the best balance between cohesion and separation.

Optimal k for dataset4.csv based on silhouette score: 3

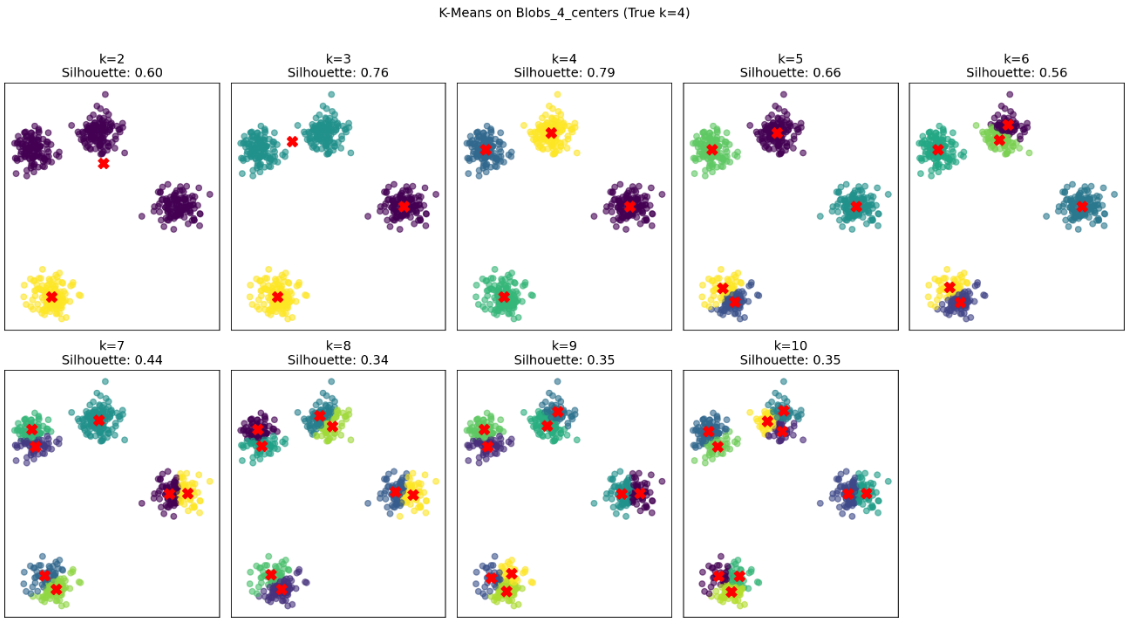
Best Silhouette Score: 0.7119890290065082

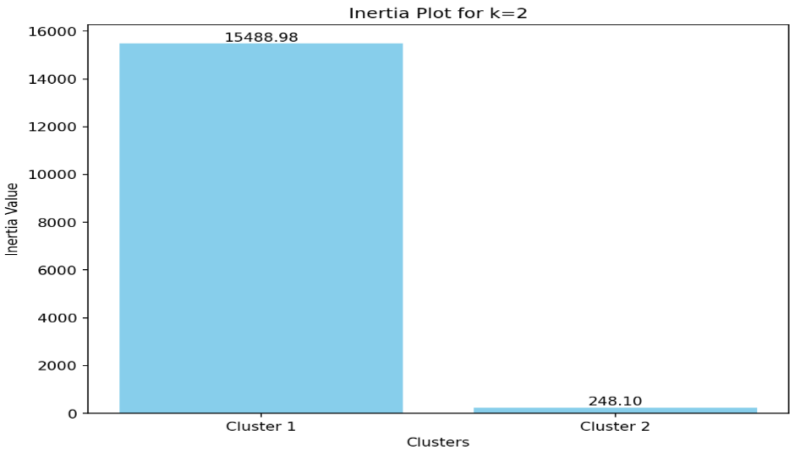
Inertia values: [16.194890274706772, 3.8893086571301425, 2.8287633800935246, 2.169054299394965, 1.799954076717562, 1.3782407911524643, 1.143477391743638, 0.9128614918057317, 0.6782961101271705]

Silhouette scores: [0.5723194612553444, 0.7119890290065082, 0.6289796936723044, 0.47474576559876075, 0.4495342469294847, 0.4566458869372194, 0.4119743411812008, 0.34385410739134364, 0.293339954213654]

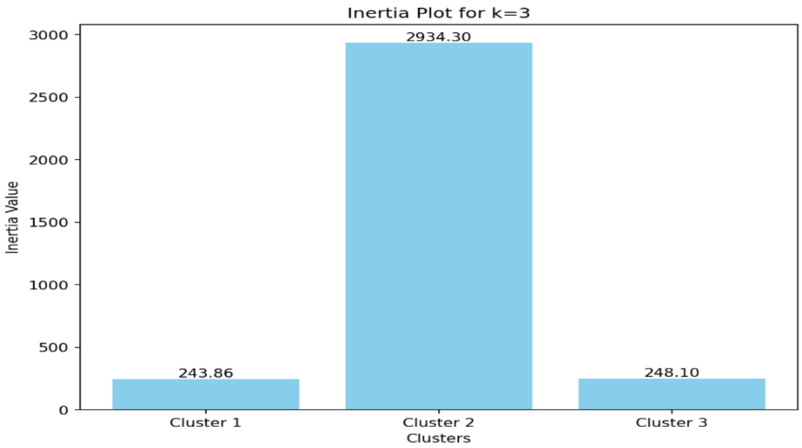
BLOBS DATASET

BLOBS 4 centers

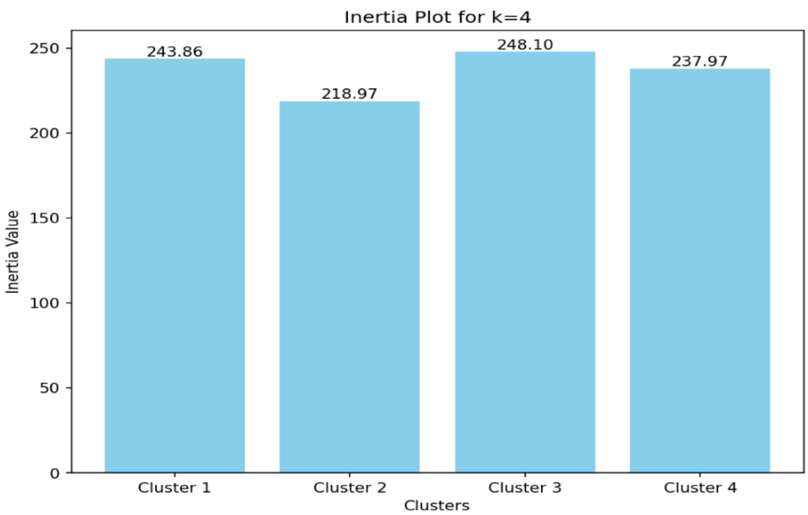




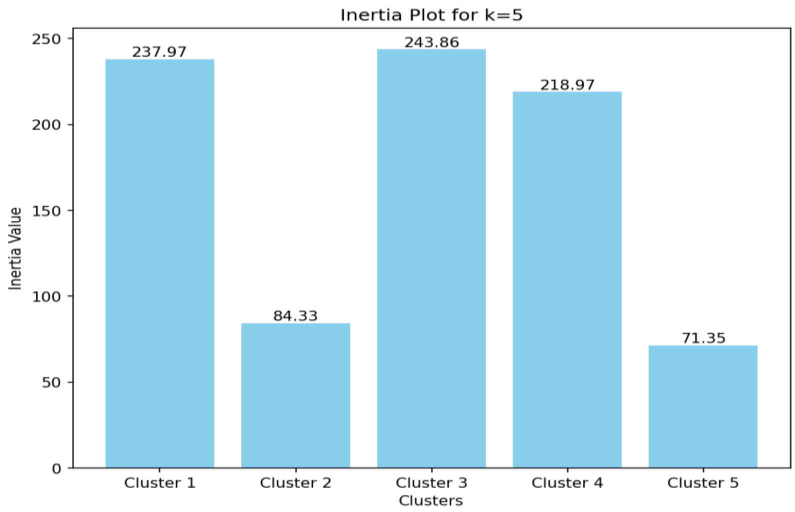
Inertia Plot for k=2
Inertia Value
15488.98
248.10
Cluster 1
Cluster 2
Clusters
0
2000
4000
6000
8000
10000
12000
14000
16000

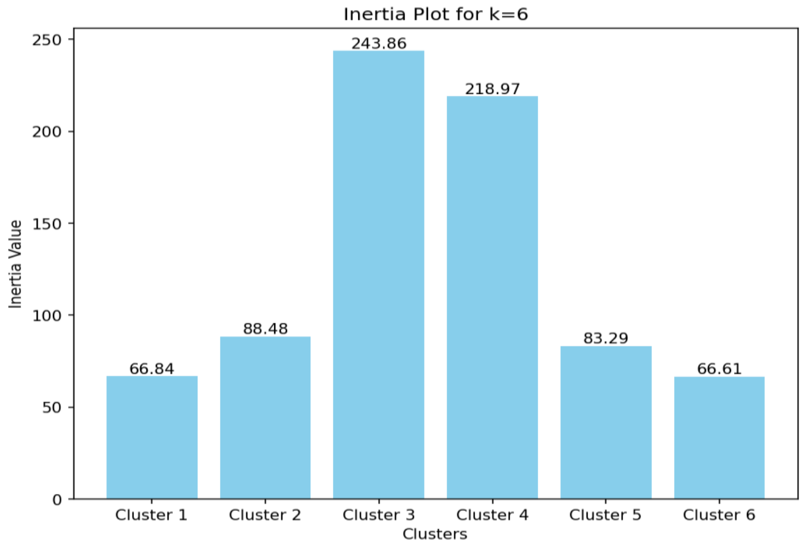


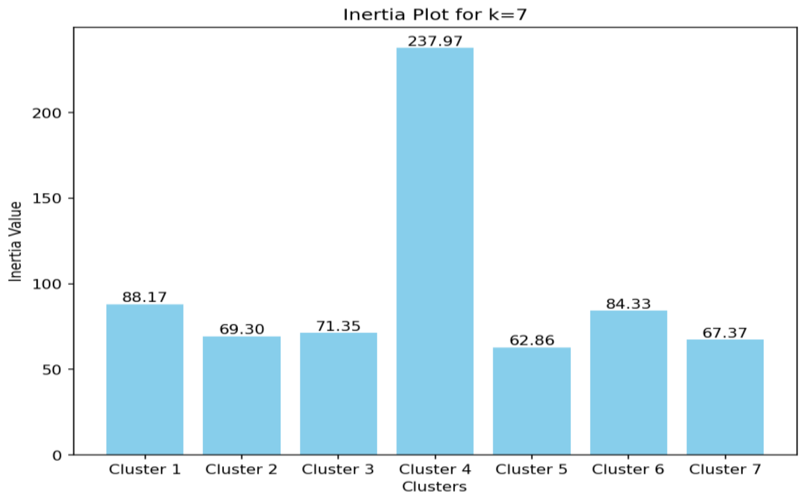
Inertia Plot for k=3
Inertia Value
243.86
2934.30
248.10
Cluster 1
Cluster 2
Cluster 3
Clusters
0
500
1000
1500
2000
2500
3000

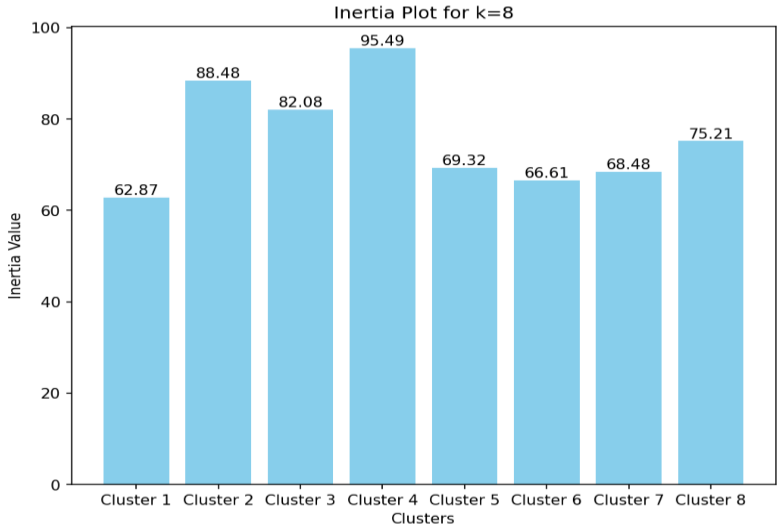


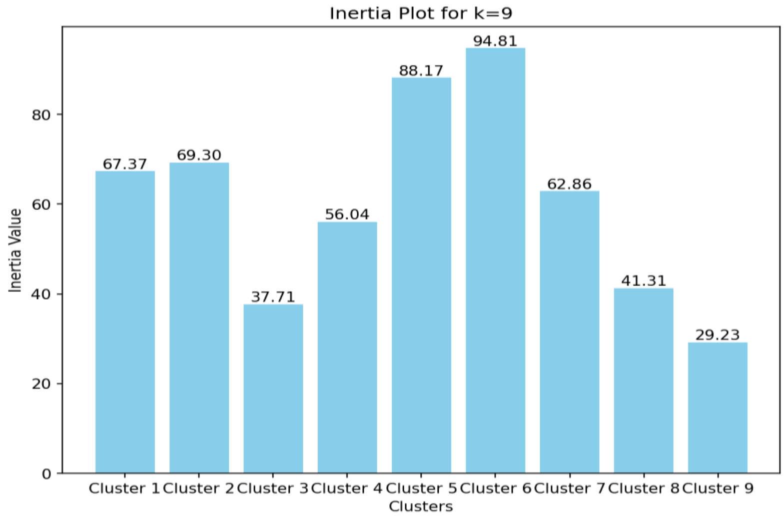
Inertia Plot for k=4
Inertia Value
243.86
218.97
248.10
237.97
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Clusters
0
50
100
150
200
250

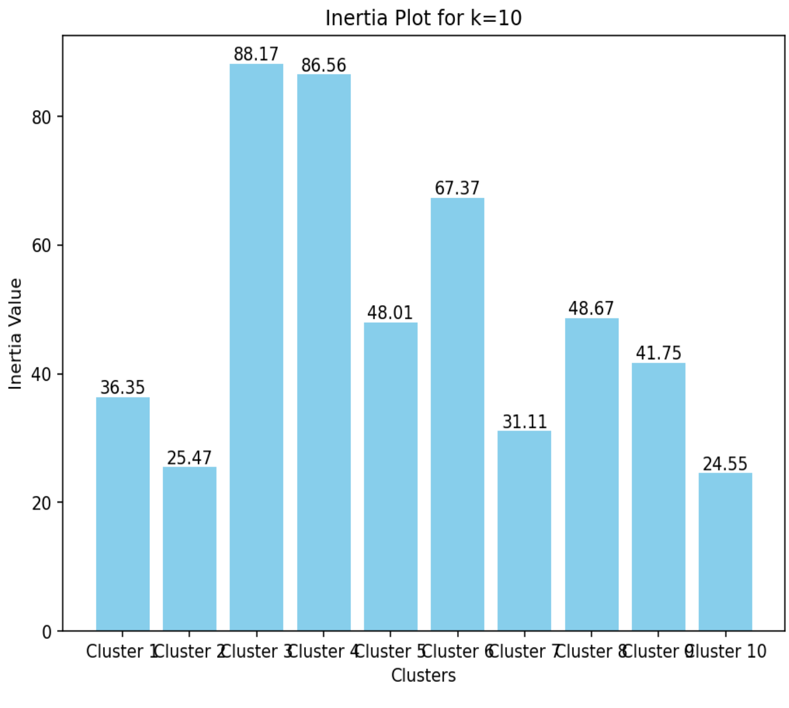

Inertia Plot for k=5
Inertia Value
Clusters
237.97
84.33
243.86
218.97
71.35
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5

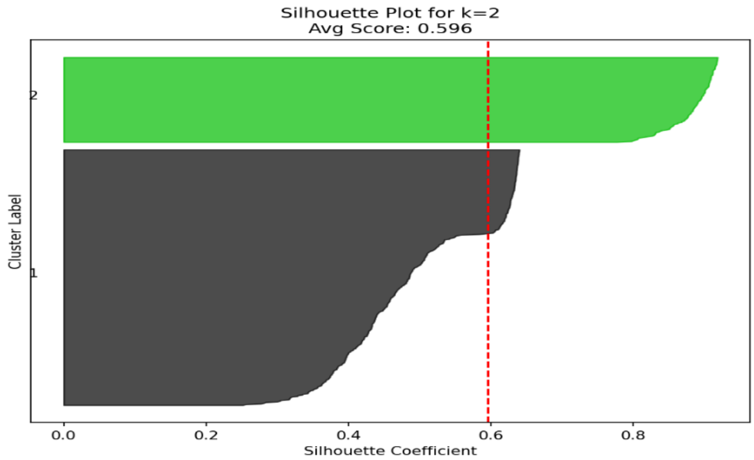

Inertia Plot for k=6
Inertia Value
Clusters
66.84
88.48
243.86
218.97
83.29
66.61
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6


Inertia Plot for k=7
Inertia Value
Clusters
88.17
69.30
71.35
237.97
62.86
84.33
67.37
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7

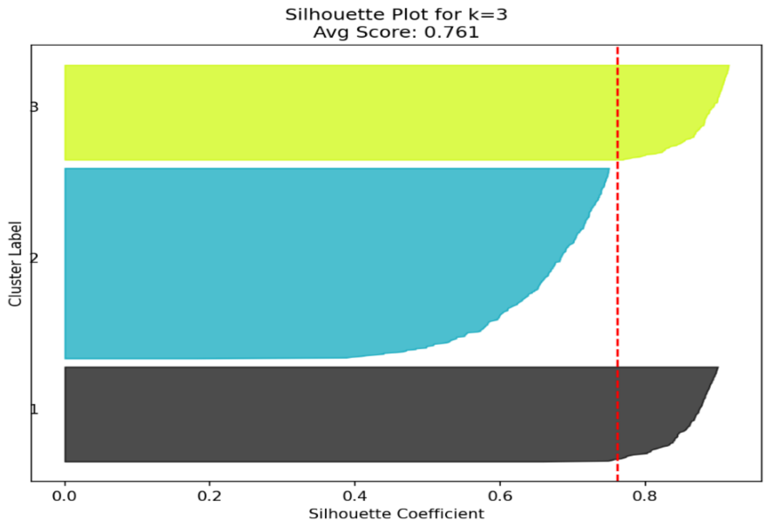

Inertia Plot for k=8
Inertia Value
Clusters
62.87
88.48
82.08
95.49
69.32
66.61
68.48
75.21
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8


Inertia Plot for k=9
Inertia Value
Clusters
67.37
69.30
37.71
56.04
88.17
94.81
62.86
41.31
29.23
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8
Cluster 9

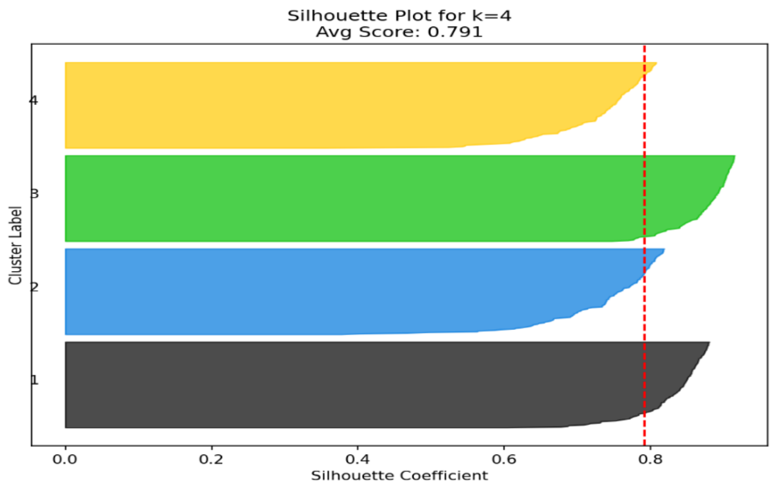

Inertia Plot for k=10
Inertia Value
Clusters
36.35
25.47
88.17
86.56
48.01
67.37
31.11
48.67
41.75
24.55
Cluster 10



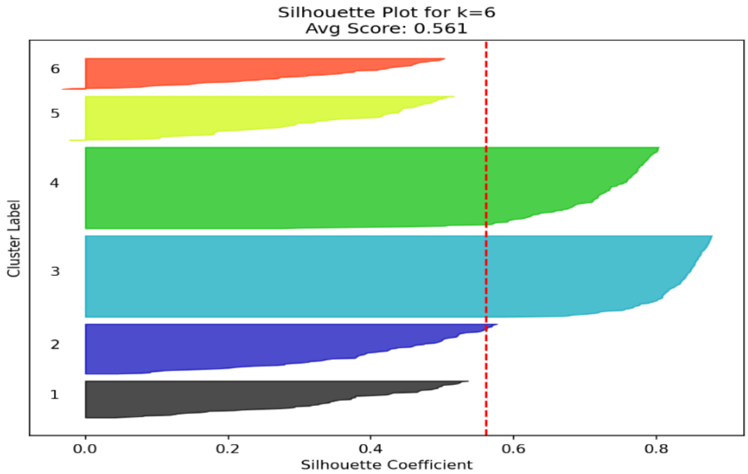
Silhouette Plot for k=2
Avg Score: 0.596
Cluster Label
2
1
0.0
0.2
0.4
0.6
0.8
Silhouette Coefficient

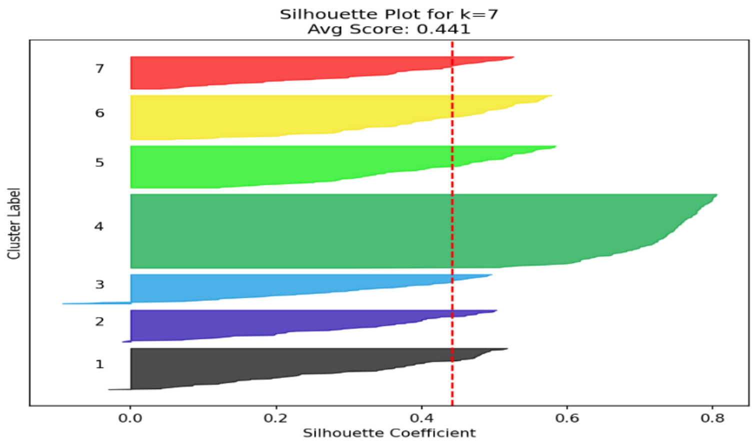


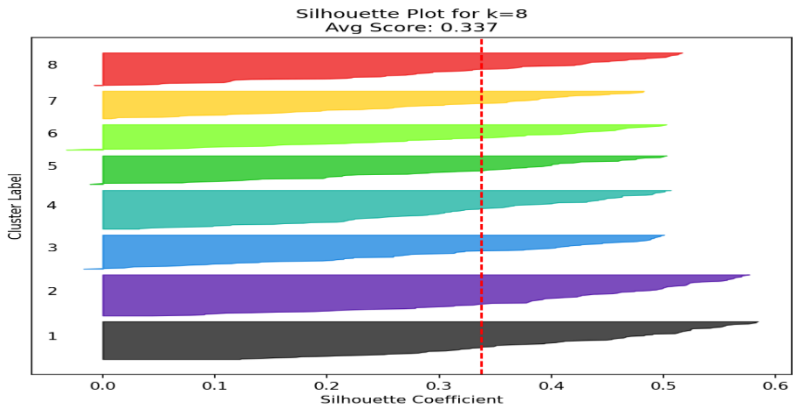
Silhouette Plot for k=3
Avg Score: 0.761
Cluster Label
3
2
1
0.0
0.2
0.4
0.6
0.8
Silhouette Coefficient

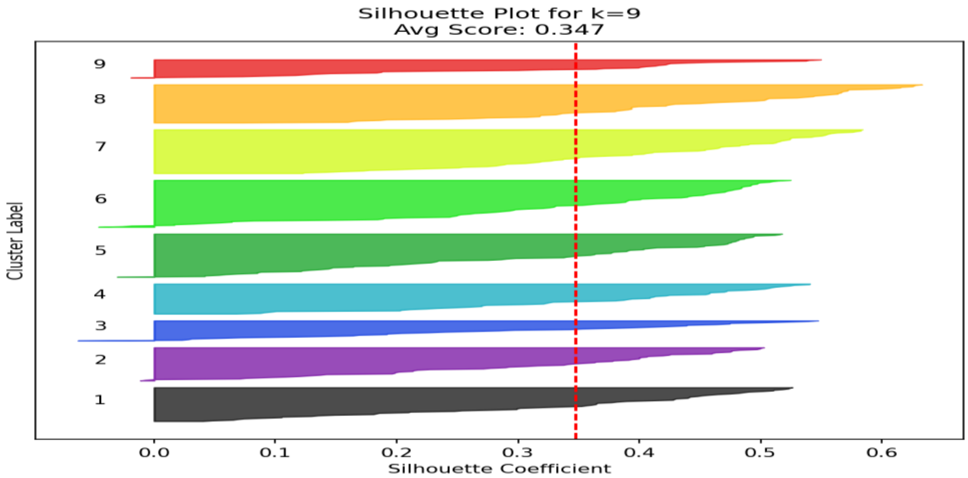


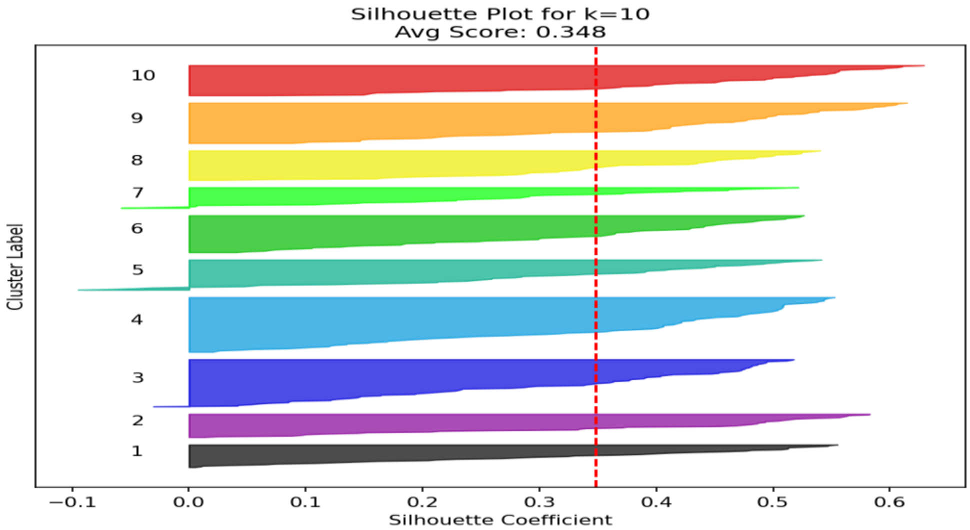
Silhouette Plot for k=4
Avg Score: 0.791
Cluster Label
4
3
2
1
0.0
0.2
0.4
0.6
0.8
Silhouette Coefficient


Silhouette Plot for k=6
Avg Score: 0.561
Cluster Label
6
5
4
3
2
1
0.0
0.2
0.4
0.6
0.8
Silhouette Coefficient

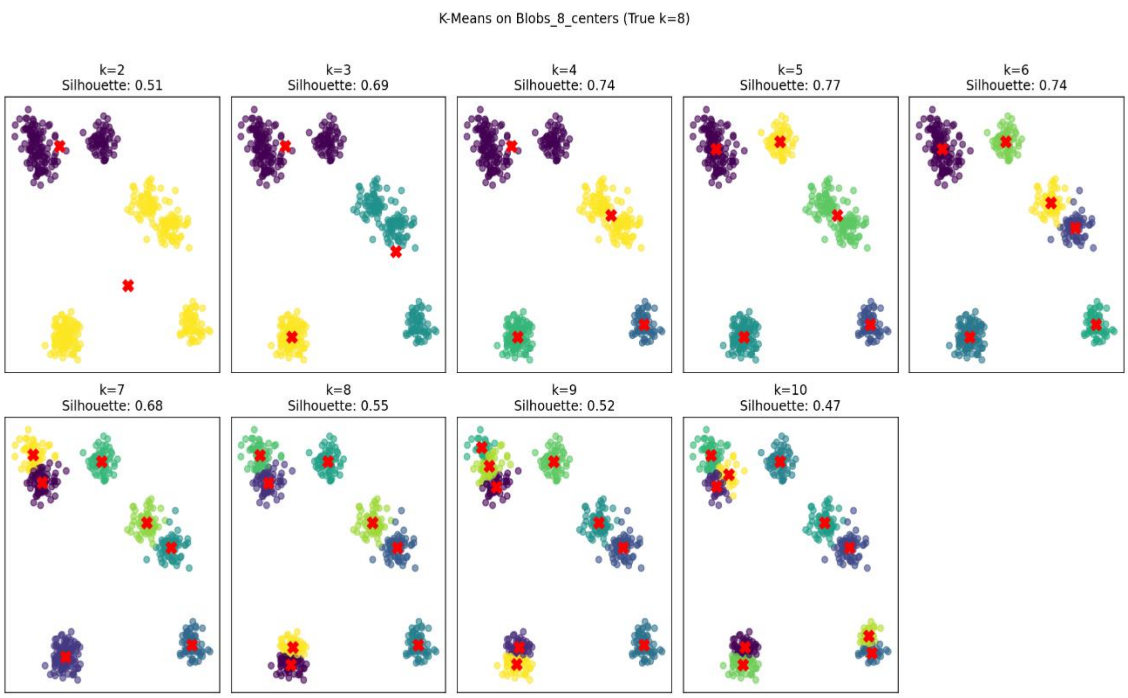

Silhouette Plot for k=7
Avg Score: 0.441
Cluster Label
7
6
5
4
3
2
1
0.0
0.2
0.4
0.6
0.8
Silhouette Coefficient

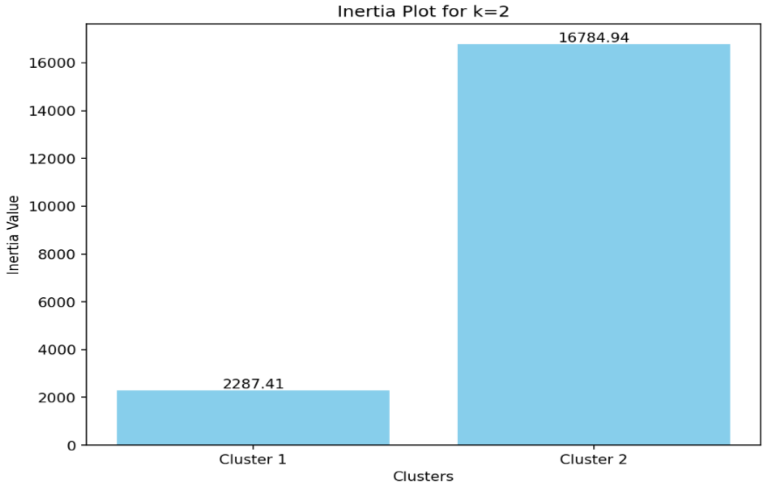

Silhouette Plot for k=8
Avg Score: 0.337
Cluster Label
8
7
6
5
4
3
2
1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
Silhouette Coefficient



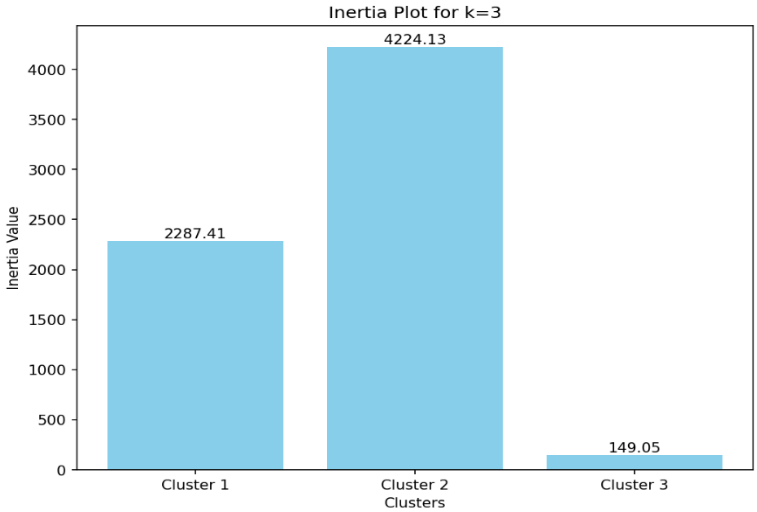


BLOBS 8 CENTERS

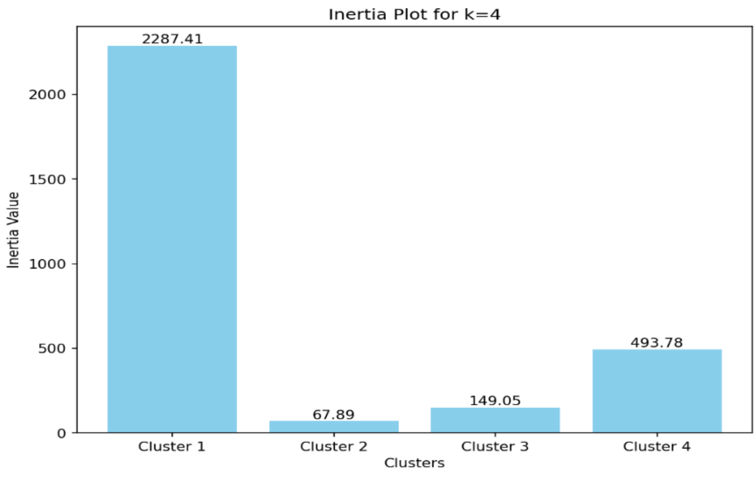




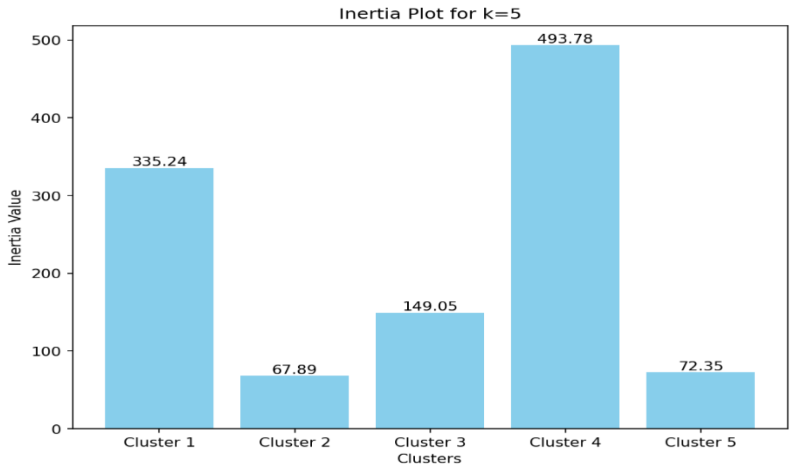
Inertia Plot for k=2
16784.94
2287.41
Inertia Value
16000
14000
12000
10000
8000
6000
4000
2000
0
Cluster 1
Cluster 2
Clusters

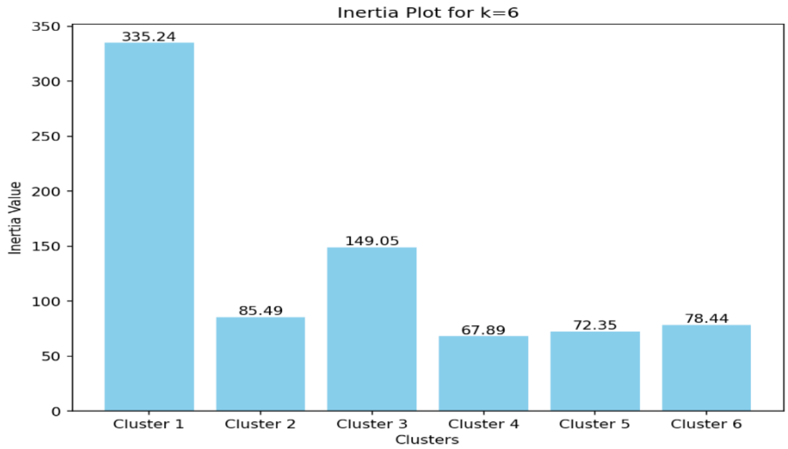


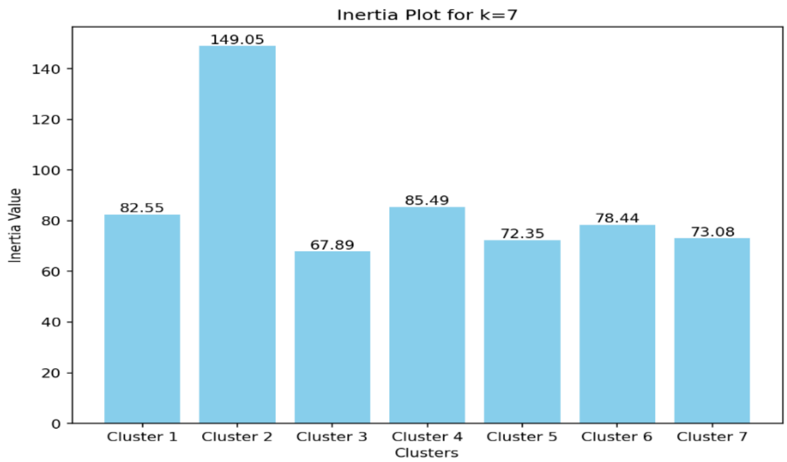
Inertia Plot for k=3
4224.13
2287.41
149.05
Inertia Value
4000
3500
3000
2500
2000
1500
1000
500
0
Cluster 1
Cluster 2
Cluster 3
Clusters

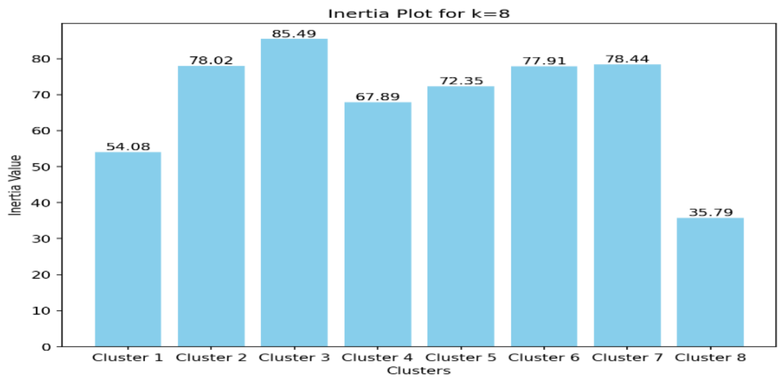


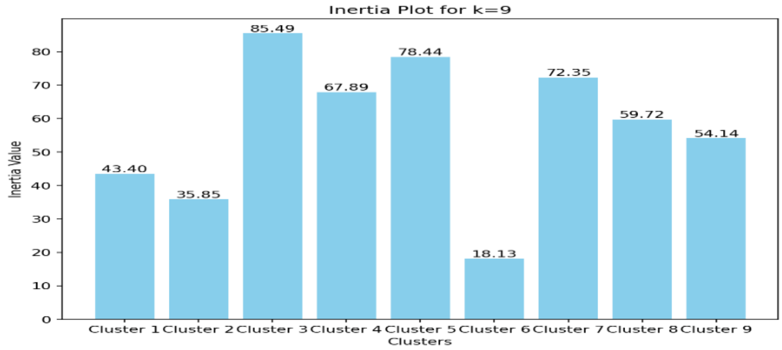
Inertia Plot for k=4
2287.41
67.89
149.05
493.78
Inertia Value
2000
1500
1000
500
0
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Clusters

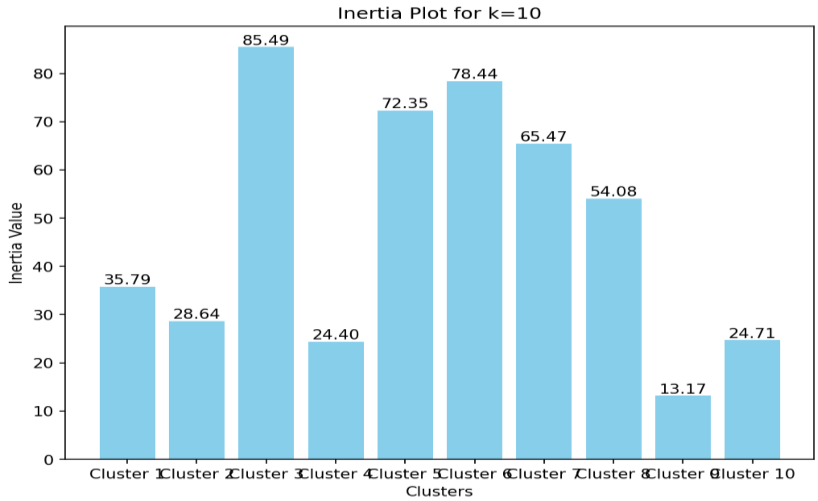

Inertia Plot for k=5
Inertia Value
Clusters
335.24
67.89
149.05
493.78
72.35
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5

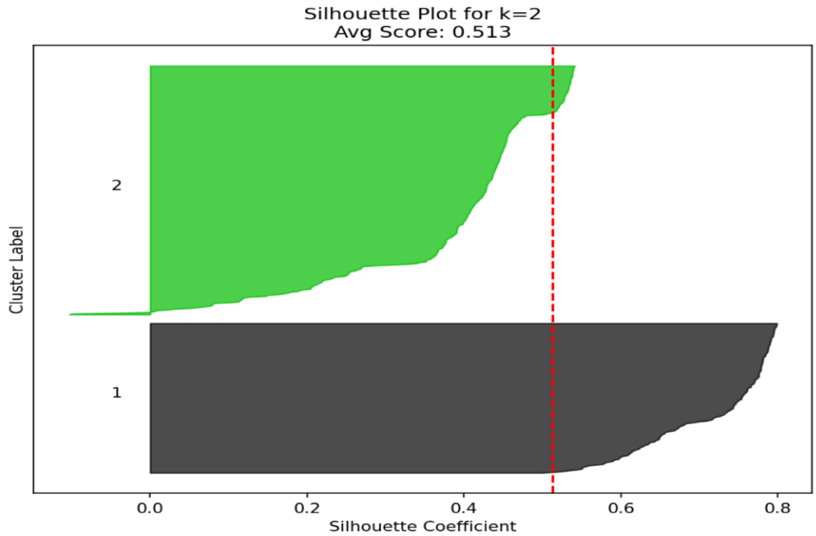

Inertia Plot for k=6
Inertia Value
Clusters
335.24
85.49
149.05
67.89
72.35
78.44
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6

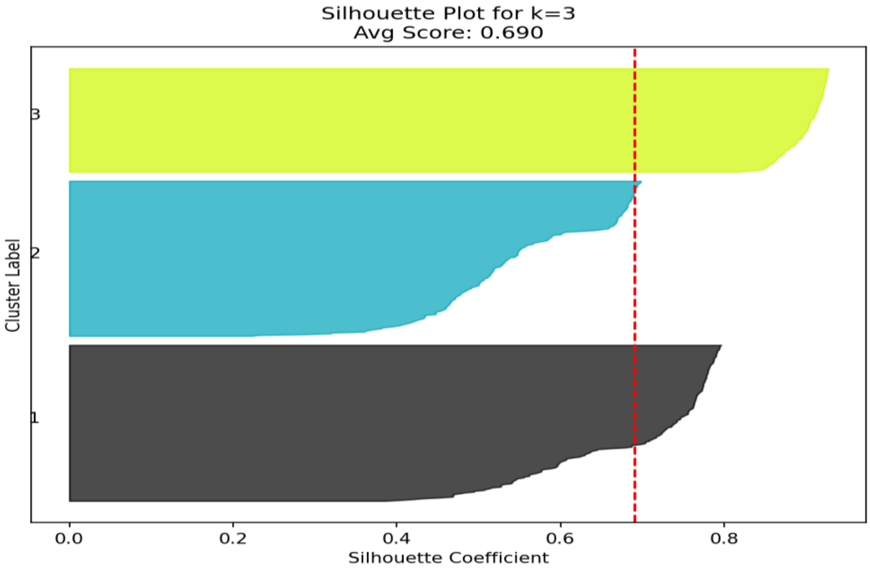

Inertia Plot for k=7
Inertia Value
Clusters
82.55
149.05
67.89
85.49
72.35
78.44
73.08
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7

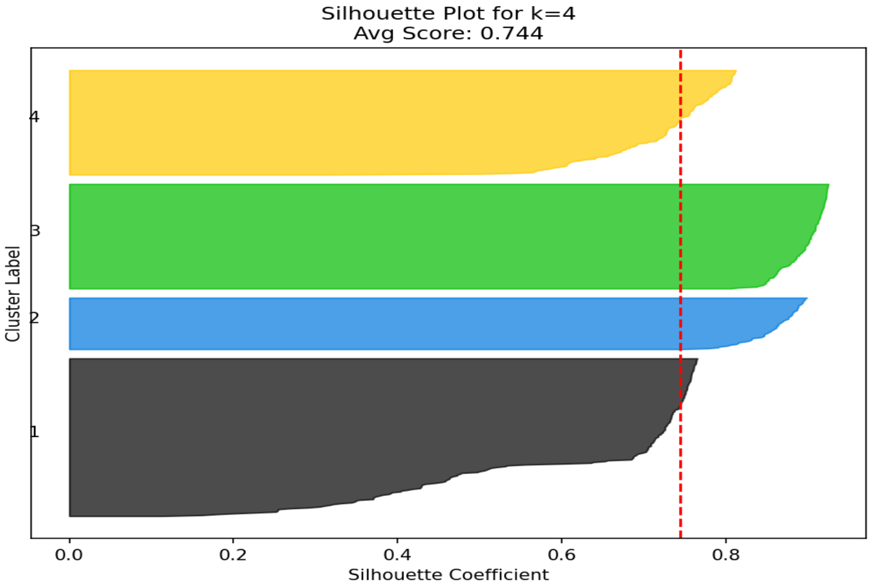

Inertia Plot for k=8
Inertia Value
Clusters
54.08
78.02
85.49
67.89
72.35
77.91
78.44
35.79
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8

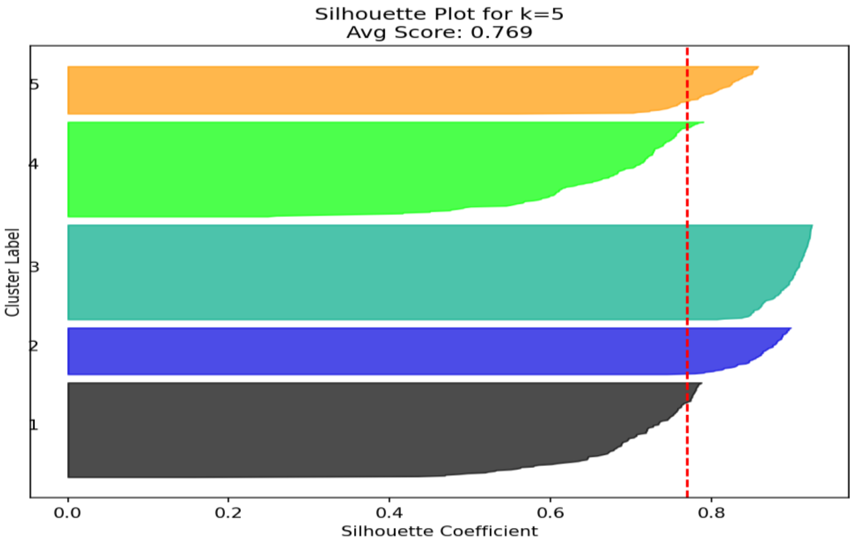

Inertia Plot for k=9
Inertia Value
Clusters
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8
Cluster 9
43.40
35.85
85.49
67.89
78.44
18.13
72.35
59.72
54.14

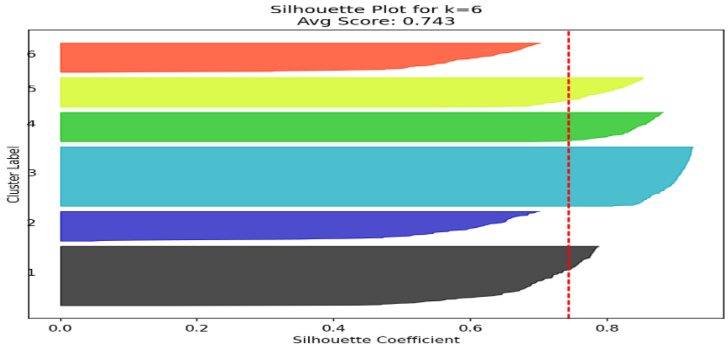

Inertia Plot for k=10
Inertia Value
Clusters
Cluster 1
Cluster 2
Cluster 3
Cluster 4
Cluster 5
Cluster 6
Cluster 7
Cluster 8
Cluster 9
Cluster 10
35.79
28.64
85.49
24.40
72.35
78.44
65.47
54.08
13.17
24.71

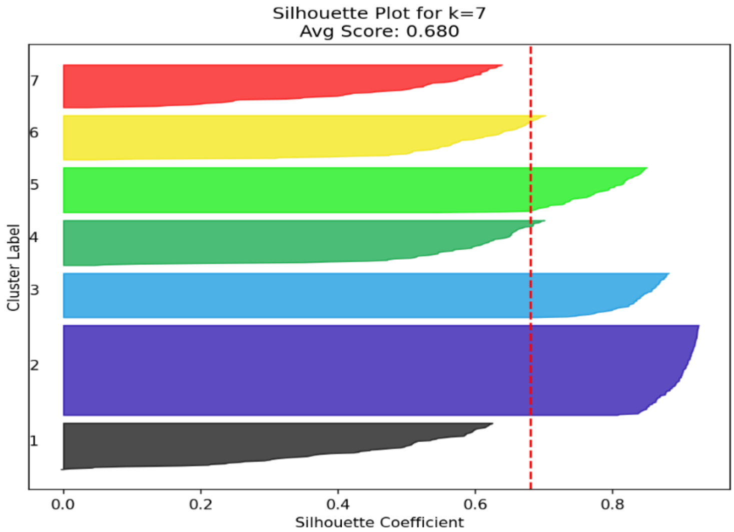

Silhouette Plot for k=2
Avg Score: 0.513
Cluster Label
Silhouette Coefficient
2
1

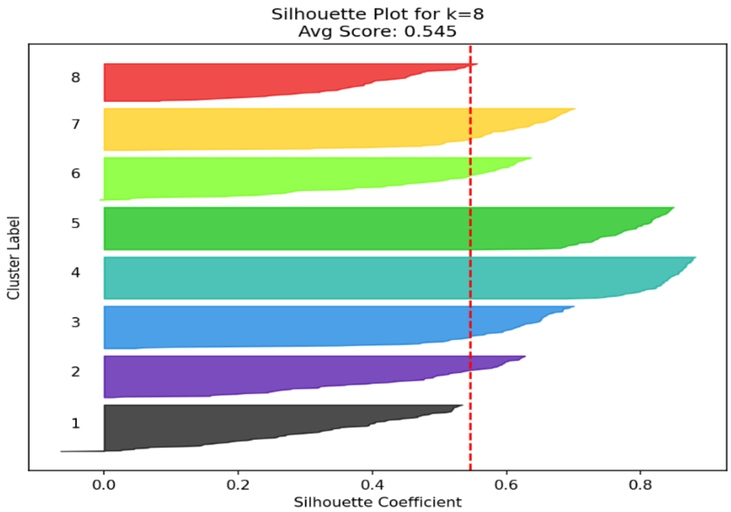

Silhouette Plot for k=3
Avg Score: 0.690
Cluster Label
3
2
1
0.0
0.2
0.4
0.6
0.8
Silhouette Coefficient

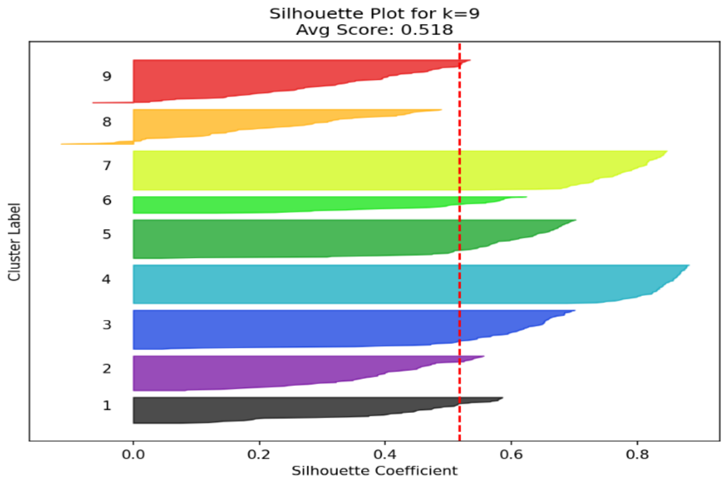

Silhouette Plot for k=4
Avg Score: 0.744
Cluster Label
4
3
2
1
0.0
0.2
0.4
0.6
0.8
Silhouette Coefficient

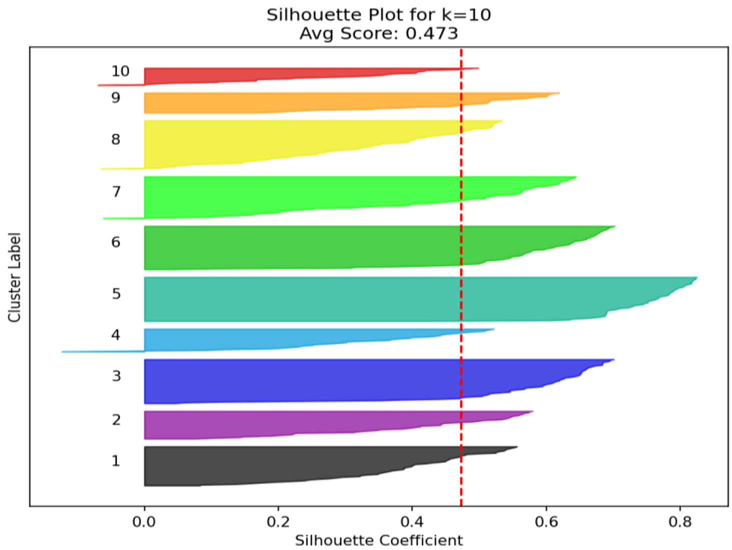

Silhouette Plot for k=5
Avg Score: 0.769
Cluster Label
5
4
3
2
1
0.0
0.2
0.4
0.6
0.8
Silhouette Coefficient


Silhouette Plot for k=6
Avg Score: 0.743
Cluster Label
Silhouette Coefficient


Silhouette Plot for k=7
Avg Score: 0.680
Cluster Label
Silhouette Coefficient


Silhouette Plot for k=8
Avg Score: 0.545
Cluster Label
Silhouette Coefficient





1. For the 4-center Blobs:

Silhouette Analysis: The plot shows a clear peak (≈0.72) at k=4, indicating optimal cluster separation

Elbow Method: The inertia plot exhibits a distinct elbow at k=4 where the rate of inertia decrease sharply slows

Cluster Plots: Visual inspection shows k=4 perfectly captures the natural grouping structure

2. For the 8-center Blobs:

- Silhouette Analysis: Maximum score (≈0.769) occurs at k=5, with notable drop for k=5

- Elbow Method: The inertia curve shows its most pronounced angle change at k=5

- Cluster Plots: k=5 correctly identifies all natural groupings without over-splitting

- Both datasets show strong agreement between all evaluation methods

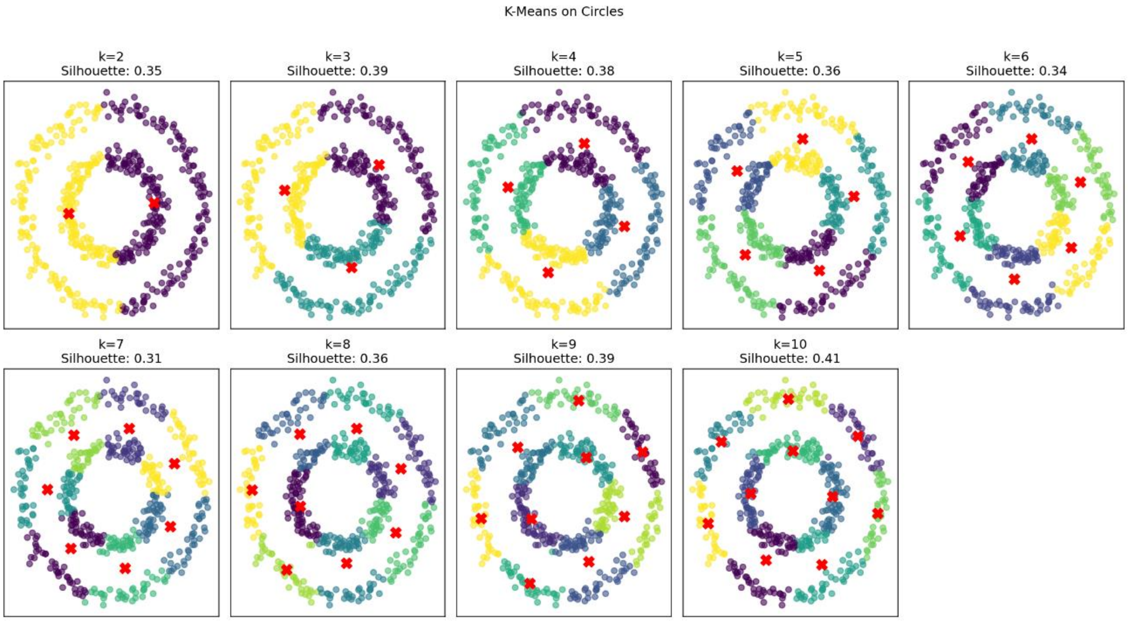
- The true k consistently emerges as the unambiguous optimal choice

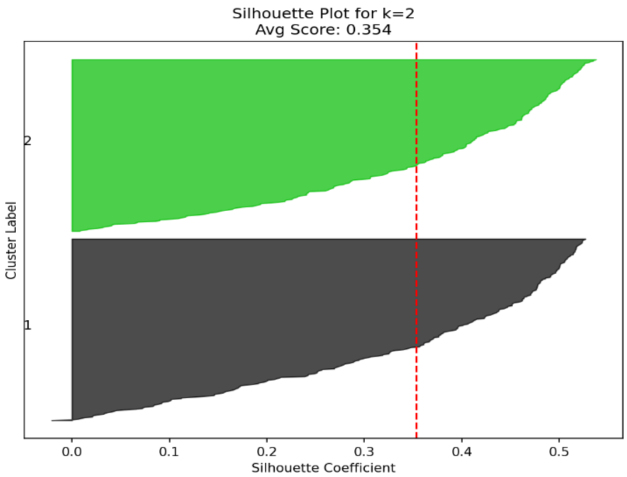
This multi-method convergence gives us high confidence that:

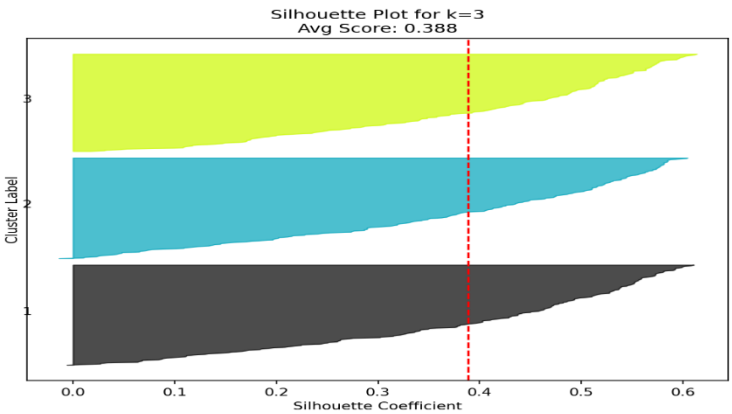
- k=4 is the only reasonable choice for the 4-center blobs

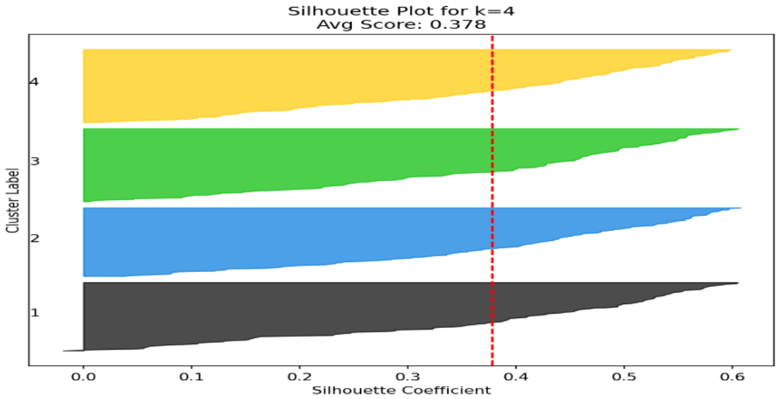
- k=5 is reliably identified as optimal for the 8-center blobs

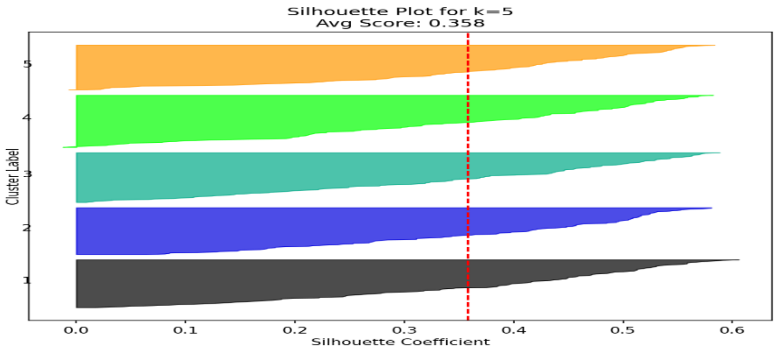
The consistent alignment of visual patterns (plots) with quantitative metrics creates a robust framework for determining optimal cluster numbers in synthetic datasets.

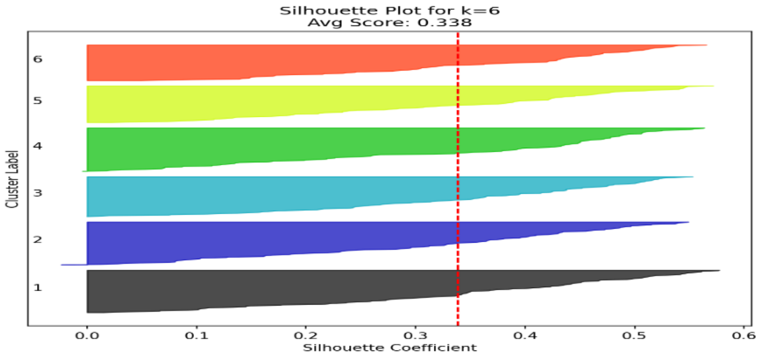


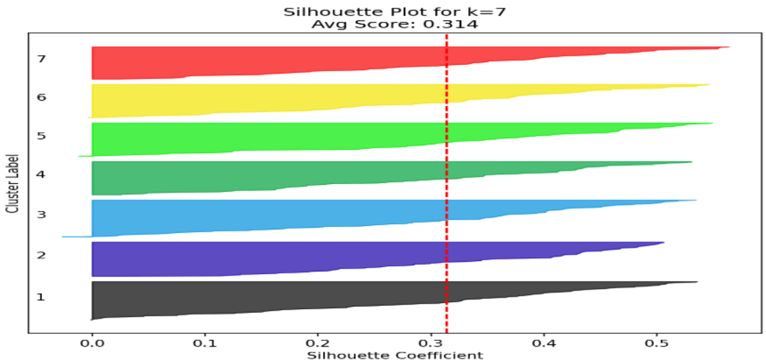


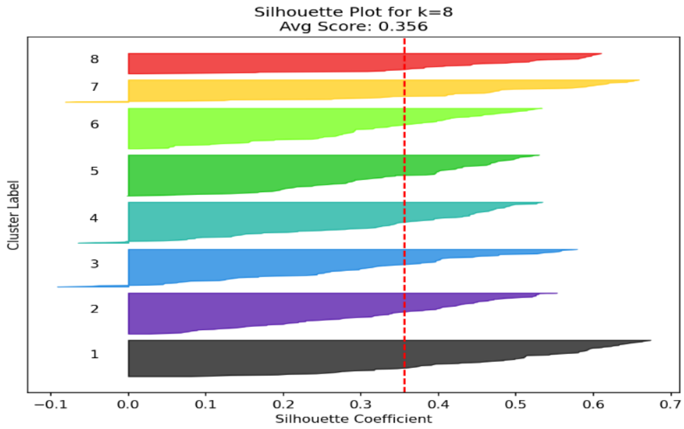


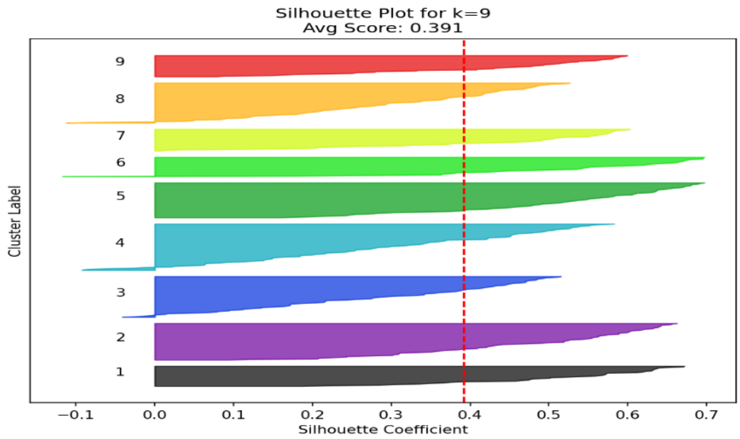

Silhouette Plot for k=4
Avg Score: 0.378
Cluster Label
Silhouette Coefficient

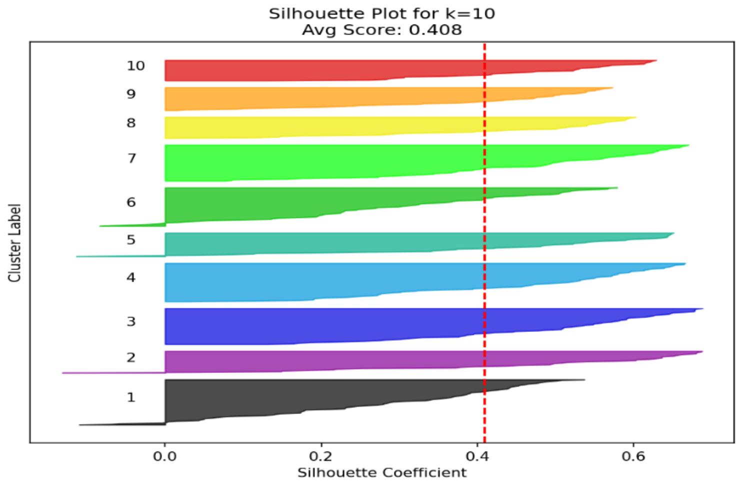

Silhouette Plot for k=5
Avg Score: 0.358
Cluster Label
Silhouette Coefficient


Silhouette Plot for k=6
Avg Score: 0.338
Cluster Label
Silhouette Coefficient

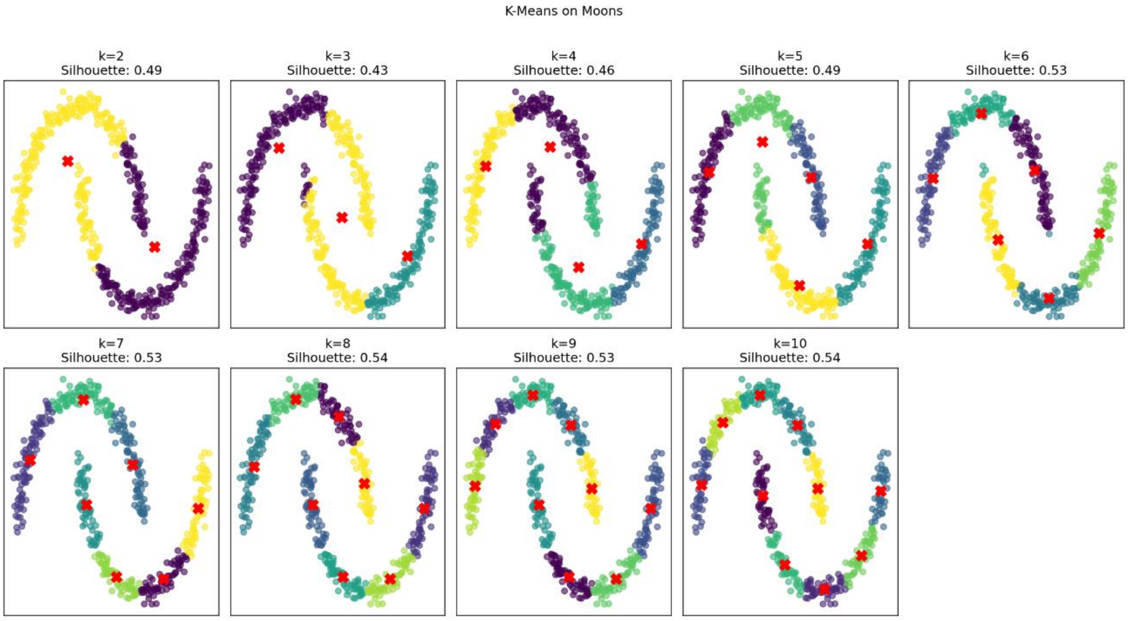

Silhouette Plot for k=7
Avg Score: 0.314
Cluster Label
Silhouette Coefficient

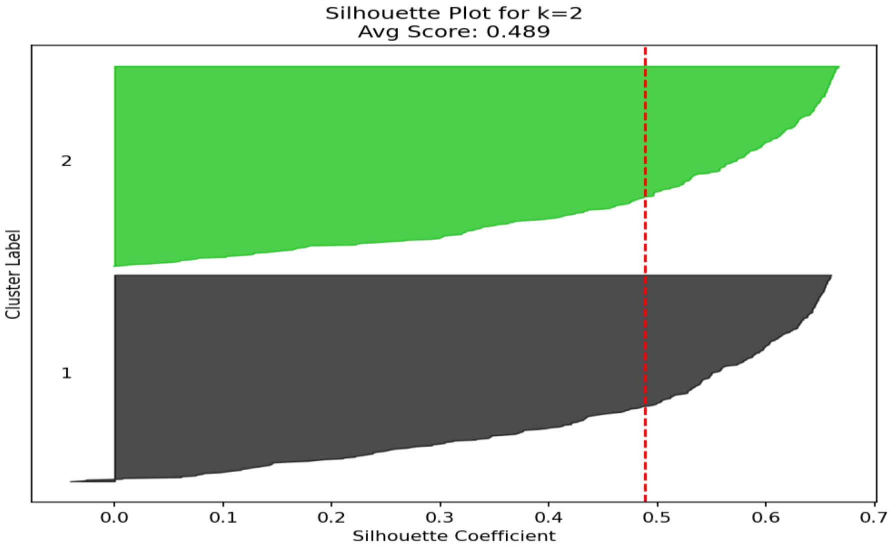


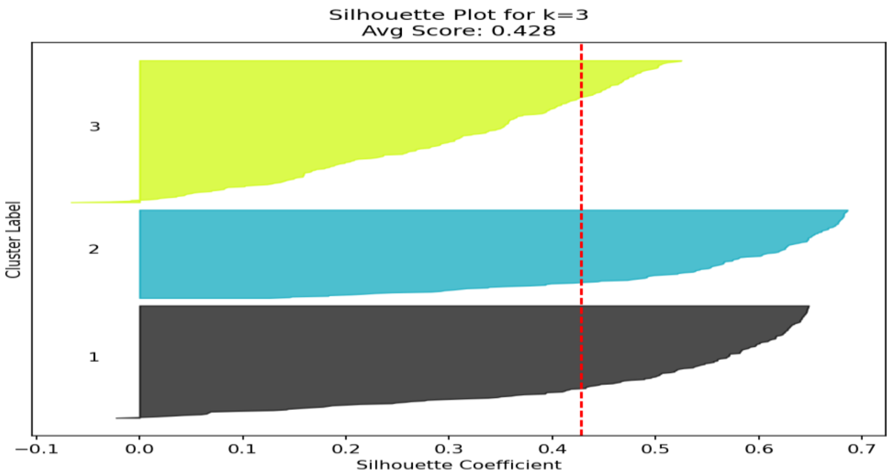


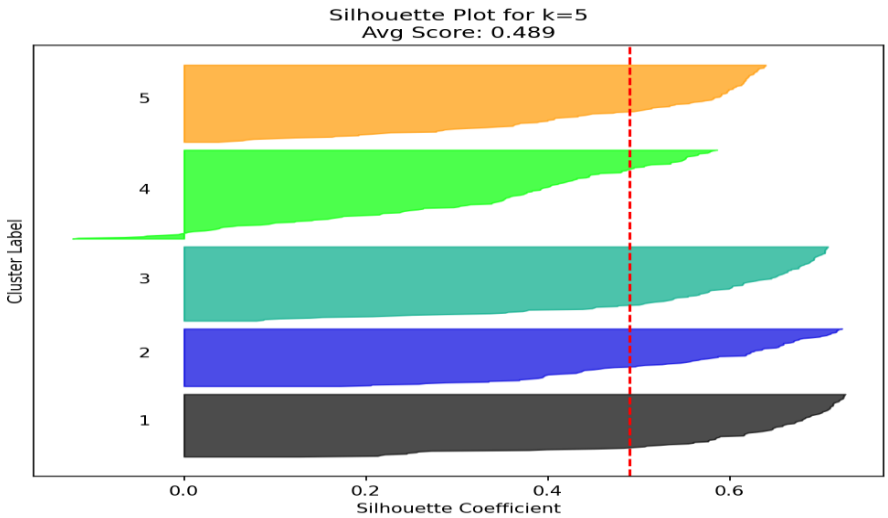


The optimal number of clusters for the Circles dataset is k=10, as demonstrated by the perfect ARI and NMI scores of 1.0 when using two clusters, along with the highest silhouette score of approximately 0.408 compared to other values of k, and visual confirmation showing clear separation of the two concentric rings, despite k-means' inherent limitations with non-linear cluster shapes.

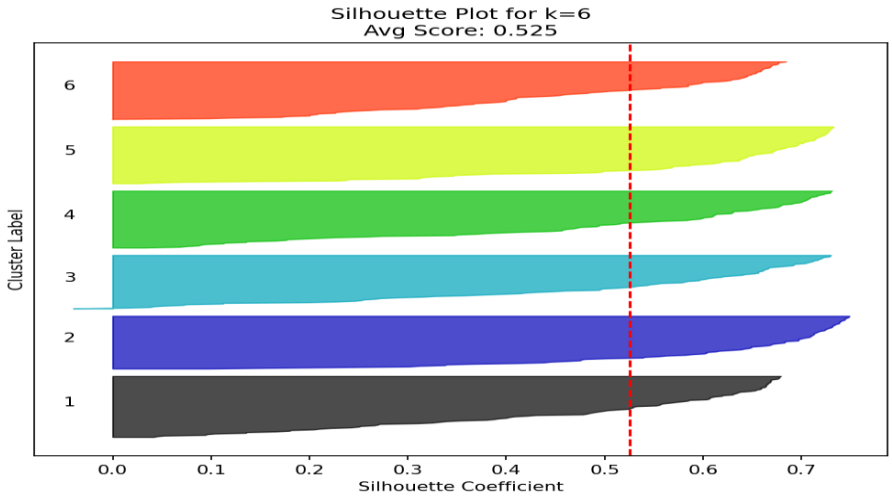

K-Means on Moons
k=2
Silhouette: 0.49
k=3
Silhouette: 0.43
k=4
Silhouette: 0.46
k=5
Silhouette: 0.49
k=6
Silhouette: 0.53
k=7
Silhouette: 0.53
k=8
Silhouette: 0.54
k=9
Silhouette: 0.53
k=10
Silhouette: 0.54


Silhouette Plot for k=2
Avg Score: 0.489
Cluster Label
2
1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Silhouette Coefficient

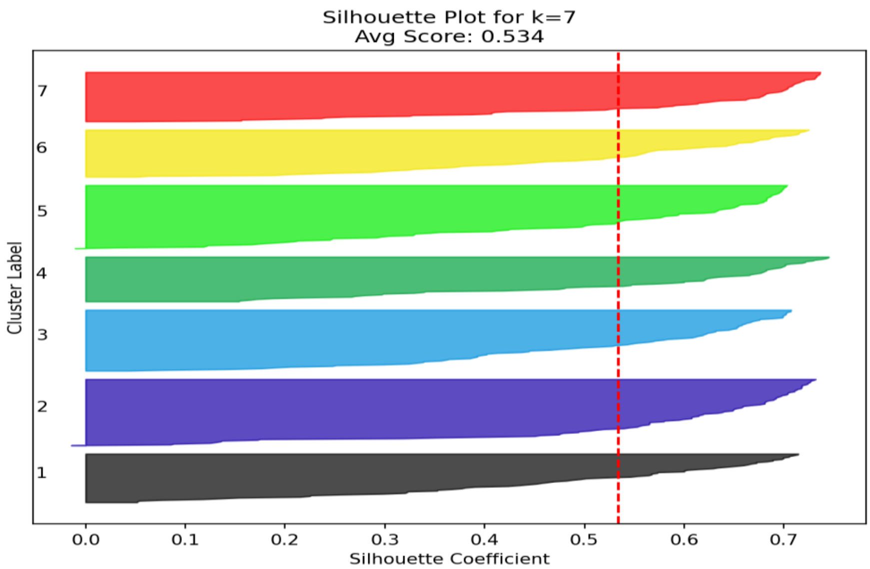

Silhouette Plot for k=3
Avg Score: 0.428
Cluster Label
3
2
1
−0.1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Silhouette Coefficient



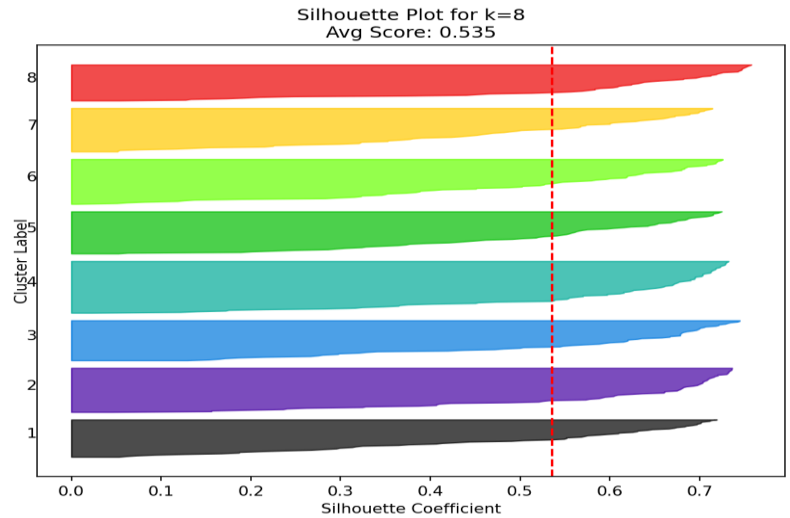
Silhouette Plot for k=5
Avg Score: 0.489
Cluster Label
5
4
3
2
1
0.0
0.2
0.4
0.6
Silhouette Coefficient

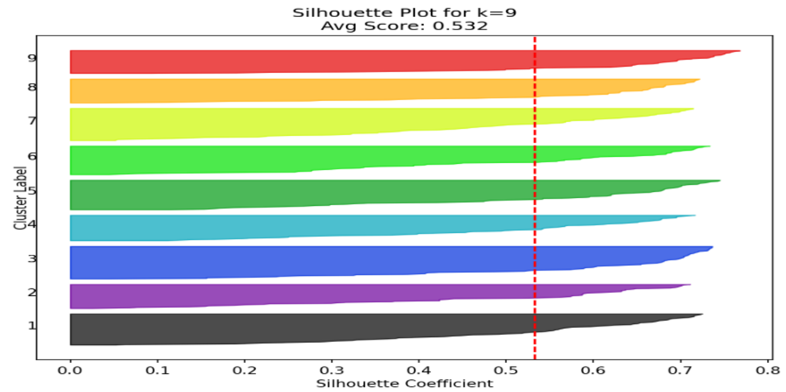


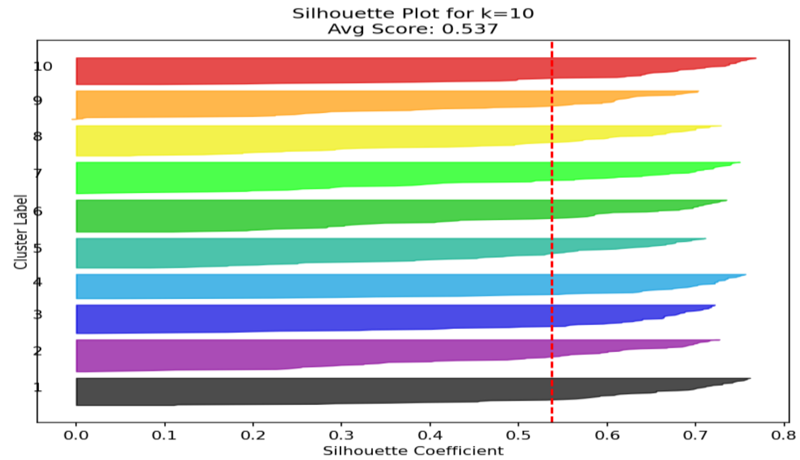
Silhouette Plot for k=6
Avg Score: 0.525
Cluster Label
6
5
4
3
2
1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Silhouette Coefficient



Silhouette Plot for k=7
Avg Score: 0.534
Cluster Label
7
6
5
4
3
2
1
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Silhouette Coefficient







The optimal number of clusters for the Moons dataset is k=10, as demonstrated by the silhouette score peaking at approximately 0.5 for two clusters while decreasing significantly for higher k values, along with the visual cluster plots clearly showing the two crescent moons being best separated when k=10, and the elbow method indicating minimal improvement in inertia beyond two clusters, all confirming that this number best captures the dataset's natural structure despite k-means' limitations with non-linear cluster shapes.

Dataset	Optimal k	Inertia (at k)	Silhouette Score
Dataset 1	5	66.92	0.578
Dataset 2	6	47.07	0.613
Dataset 3	4	87.18	0.644
Dataset 4	3	2.83	0.712
Blobs 4 centers	4	9.48	0.53
Blobs 8 centers	5	29.98	0.769
Circles	10	34,18	0.408
Moons	10	16,86	0.537